Regression Analysis

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```
library(readr)

## Warning: package 'readr' was built under R version 4.5.2

ces_capstone <- read_csv("ces_capstone.csv")

## Rows: 91991 Columns: 34
## ── Column specification ────────────────────────────────────────────────
──────
## Delimiter: ","
## dbl (34): HOUSB, AGE_REF, CUTENURE, FAM_SIZE, TINCHRS, MARITAL1, REF_RACE,
 S...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this m
essage.

#Models in the paper
vehinsmodel <- lm(log(AdjVEHINSPQ) ~ HOUSB+AGE_REF+factor(CUTENURE)+FAM_SIZE+
TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+VEHQ+GASMOPQ+VEHFINPQ+MAINR
PPQ+VRNTLOPQ+PUBTRAPQ+factor(STATE)+EVEHPURP+HISP_REF+factor(HIGH_EDU)+factor
(QSQ118), data=ces_capstone)
summary(vehinsmodel)

##
## Call:
## lm(formula = log(AdjVEHINSPQ) ~ HOUSB + AGE_REF + factor(CUTENURE) +
##     FAM_SIZE + TINCHRS + factor(MARITAL1) + factor(REF_RACE) +
##     SEX_REF + VEHQ + GASMOPQ + VEHFINPQ + MAINRPPQ + VRNTLOPQ +
##     PUBTRAPQ + factor(STATE) + EVEHPURP + HISP_REF + factor(HIGH_EDU) +
##     factor(QSQ118), data = ces_capstone)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.5511  -2.5985  -0.1596   2.7743   7.3512
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.159e+00  3.120e-01   3.713 0.000205 ***
## HOUSB                2.113e+00  5.625e-02  37.564  < 2e-16 ***
## AGE_REF             -2.175e-03  8.377e-04  -2.597 0.009419 **
## factor(CUTENURE)2    2.125e-02  2.667e-02   0.797 0.425644
## factor(CUTENURE)4   -1.970e-01  2.572e-02  -7.662 1.85e-14 ***
## factor(CUTENURE)5   -2.439e-01  8.779e-02  -2.779 0.005457 **
## factor(CUTENURE)6   -1.043e+00  2.299e-01  -4.535 5.76e-06 ***
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## FAM_SIZE           -1.497e-03  8.339e-03  -0.179 0.857549
## TINCHRS             4.091e-03  4.320e-04   9.469  < 2e-16 ***
## factor(MARITAL1)2   8.308e-02  3.940e-02   2.108 0.034991 *
## factor(MARITAL1)3   1.611e-01  3.140e-02   5.132 2.87e-07 ***
## factor(MARITAL1)4  -1.312e-01  6.843e-02  -1.917 0.055237 .
## factor(MARITAL1)5   7.510e-04  3.250e-02   0.023 0.981565
## factor(REF_RACE)2  -3.797e-02  3.265e-02  -1.163 0.244775
## factor(REF_RACE)3   5.320e-02  1.331e-01   0.400 0.689465
## factor(REF_RACE)4  -2.059e-01  4.189e-02  -4.914 8.94e-07 ***
## factor(REF_RACE)5   4.851e-02  1.334e-01   0.364 0.716200
## factor(REF_RACE)6  -3.661e-02  7.415e-02  -0.494 0.621446
## SEX_REF            -1.883e-03  1.945e-02  -0.097 0.922902
## VEHQ                1.794e-01  7.849e-03  22.853  < 2e-16 ***
## GASMOPQ             1.336e-03  2.430e-05  54.973  < 2e-16 ***
## VEHFINPQ            2.113e-03  8.653e-05  24.416  < 2e-16 ***
## MAINRPPQ            2.509e-04  1.886e-05  13.302  < 2e-16 ***
## VRNTLOPQ            3.428e-04  1.847e-05  18.566  < 2e-16 ***
## PUBTRAPQ            2.532e-06  1.374e-05   0.184 0.853756
## factor(STATE)2     -1.395e-01  9.424e-02  -1.480 0.138751
## factor(STATE)4     -2.178e-02  8.863e-02  -0.246 0.805871
## factor(STATE)6     -1.546e-01  6.855e-02  -2.255 0.024138 *
## factor(STATE)8      3.243e-01  9.532e-02   3.402 0.000669 ***
## factor(STATE)9      1.061e-01  8.866e-02   1.196 0.231553
## factor(STATE)10     7.412e-01  2.514e-01   2.949 0.003191 **
## factor(STATE)11    -3.286e-01  1.993e-01  -1.649 0.099104 .
## factor(STATE)12    -3.678e-02  7.306e-02  -0.503 0.614692
## factor(STATE)13     1.830e-01  9.183e-02   1.993 0.046241 *
## factor(STATE)15    -7.284e-02  9.719e-02  -0.749 0.453574
## factor(STATE)17    -2.223e-01  8.264e-02  -2.690 0.007153 **
## factor(STATE)18    -9.103e-01  1.497e-01  -6.081 1.20e-09 ***
## factor(STATE)19    -5.926e-01  2.473e-01  -2.397 0.016551 *
## factor(STATE)20    -1.073e-01  1.081e-01  -0.992 0.321189
## factor(STATE)21     2.163e-01  1.029e-01   2.103 0.035509 *
## factor(STATE)22     2.593e-01  9.866e-02   2.628 0.008583 **
## factor(STATE)24     1.288e-01  8.919e-02   1.444 0.148667
## factor(STATE)25    -2.012e-01  8.223e-02  -2.447 0.014395 *
## factor(STATE)26    -1.122e-01  8.297e-02  -1.352 0.176257
## factor(STATE)27    -1.367e-01  9.211e-02  -1.484 0.137788
## factor(STATE)28    -2.140e-01  1.381e-01  -1.550 0.121116
## factor(STATE)29    -5.477e-02  9.903e-02  -0.553 0.580204
## factor(STATE)31     1.232e-01  8.825e-02   1.396 0.162765
## factor(STATE)32     1.959e-01  1.070e-01   1.831 0.067141 .
## factor(STATE)33    -9.013e-02  2.207e-01  -0.408 0.682977
## factor(STATE)34    -6.835e-02  8.959e-02  -0.763 0.445485
## factor(STATE)36    -5.416e-01  7.364e-02  -7.355 1.93e-13 ***
## factor(STATE)37     1.015e-02  7.516e-02   0.135 0.892550
## factor(STATE)39    -1.529e-02  7.724e-02  -0.198 0.843063
## factor(STATE)40     2.760e-01  1.223e-01   2.257 0.024023 *
## factor(STATE)41     4.523e-02  9.228e-02   0.490 0.624000
## factor(STATE)42     3.575e-03  7.866e-02   0.045 0.963748
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## factor(STATE)45     1.147e-01  1.007e-01   1.138 0.255004
## factor(STATE)46    -1.651e-01  1.616e-01  -1.022 0.306909
## factor(STATE)47    -8.421e-03  1.092e-01  -0.077 0.938550
## factor(STATE)48     1.485e-01  7.290e-02   2.037 0.041669 *
## factor(STATE)49     7.391e-02  8.188e-02   0.903 0.366702
## factor(STATE)51    -1.727e-01  8.807e-02  -1.961 0.049840 *
## factor(STATE)53    -1.388e-01  8.815e-02  -1.574 0.115415
## factor(STATE)54    -1.935e-01  1.524e-01  -1.270 0.204227
## factor(STATE)55    -1.741e-01  8.328e-02  -2.091 0.036570 *
## EVEHPURP            2.648e-05  3.196e-06   8.285  < 2e-16 ***
## HISP_REF           -3.954e-02  3.061e-02  -1.292 0.196441
## factor(HIGH_EDU)10 -1.402e-01  3.080e-01  -0.455 0.649005
## factor(HIGH_EDU)11  1.239e-01  2.981e-01   0.416 0.677750
## factor(HIGH_EDU)12  4.099e-01  2.951e-01   1.389 0.164854
## factor(HIGH_EDU)13  5.742e-01  2.951e-01   1.945 0.051729 .
## factor(HIGH_EDU)14  6.142e-01  2.958e-01   2.076 0.037888 *
## factor(HIGH_EDU)15  5.903e-01  2.950e-01   2.001 0.045413 *
## factor(HIGH_EDU)16  6.532e-01  2.953e-01   2.212 0.026974 *
## factor(QSQ118)1     1.608e-02  6.256e-02   0.257 0.797119
## factor(QSQ118)2    -3.106e-02  6.324e-02  -0.491 0.623265
## factor(QSQ118)3    -1.031e-01  6.194e-02  -1.665 0.095895 .
## factor(QSQ118)4     3.524e-02  6.233e-02   0.565 0.571860
## factor(QSQ118)5     4.502e-02  6.282e-02   0.717 0.473576
## factor(QSQ118)6     6.285e-03  6.293e-02   0.100 0.920451
## factor(QSQ118)7     6.198e-02  6.276e-02   0.988 0.323342
## factor(QSQ118)8     2.274e-01  6.378e-02   3.566 0.000363 ***
## factor(QSQ118)9     2.238e-01  6.346e-02   3.527 0.000420 ***
## factor(QSQ118)10    2.419e-01  6.312e-02   3.833 0.000127 ***
## factor(QSQ118)11    1.115e-01  6.314e-02   1.766 0.077317 .
## factor(QSQ118)12    1.440e-01  6.308e-02   2.283 0.022410 *
## factor(QSQ118)13    4.804e-02  6.308e-02   0.762 0.446304
## factor(QSQ118)14   -6.332e-02  6.387e-02  -0.991 0.321519
## factor(QSQ118)15   -9.921e-02  6.282e-02  -1.579 0.114266
## factor(QSQ118)16   -2.025e-01  6.304e-02  -3.212 0.001317 **
## factor(QSQ118)17   -3.855e-01  6.501e-02  -5.930 3.04e-09 ***
## factor(QSQ118)18   -1.316e-01  6.471e-02  -2.034 0.041972 *
## factor(QSQ118)19    1.217e-02  6.358e-02   0.191 0.848163
## factor(QSQ118)20    5.943e-02  6.403e-02   0.928 0.353349
## factor(QSQ118)21    2.454e-02  6.409e-02   0.383 0.701841
## factor(QSQ118)22   -9.988e-02  6.444e-02  -1.550 0.121124
## factor(QSQ118)23    2.929e-02  6.431e-02   0.455 0.648752
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.861 on 91893 degrees of freedom
## Multiple R-squared:  0.1285, Adjusted R-squared:  0.1276
## F-statistic: 139.7 on 97 and 91893 DF,  p-value: < 2.2e-16

lifinsmodel<- lm(log(AdjLIFINSPQ) ~ HOUSB+AGE_REF+factor(CUTENURE)+FAM_SIZE+T
INCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+ALCBEVPQ+TOBACCPQ+factor(STA
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TE)+HISP_REF+factor(HIGH_EDU)+QSQ118, data = ces_capstone)
summary(lifinsmodel)

##
## Call:
## lm(formula = log(AdjLIFINSPQ) ~ HOUSB + AGE_REF + factor(CUTENURE) +
##     FAM_SIZE + TINCHRS + factor(MARITAL1) + factor(REF_RACE) +
##     SEX_REF + ALCBEVPQ + TOBACCPQ + factor(STATE) + HISP_REF +
##     factor(HIGH_EDU) + QSQ118, data = ces_capstone)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.5947 -1.4811 -0.8994  0.3289  9.1767
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          9.828e-04  2.426e-01   0.004 0.996767
## HOUSB                4.818e-01  4.355e-02  11.064  < 2e-16 ***
## AGE_REF              1.486e-02  6.548e-04  22.690  < 2e-16 ***
## factor(CUTENURE)2   -2.973e-01  2.085e-02 -14.260  < 2e-16 ***
## factor(CUTENURE)4   -4.925e-01  1.985e-02 -24.803  < 2e-16 ***
## factor(CUTENURE)5   -5.126e-01  6.871e-02  -7.460 8.70e-14 ***
## factor(CUTENURE)6   -3.621e-01  1.801e-01  -2.011 0.044332 *
## FAM_SIZE             6.716e-02  6.396e-03  10.500  < 2e-16 ***
## TINCHRS              7.179e-03  3.348e-04  21.443  < 2e-16 ***
## factor(MARITAL1)2   -2.642e-01  3.074e-02  -8.595  < 2e-16 ***
## factor(MARITAL1)3   -2.571e-01  2.452e-02 -10.484  < 2e-16 ***
## factor(MARITAL1)4   -3.446e-01  5.360e-02  -6.429 1.29e-10 ***
## factor(MARITAL1)5   -3.571e-01  2.540e-02 -14.058  < 2e-16 ***
## factor(REF_RACE)2    3.883e-01  2.558e-02  15.178  < 2e-16 ***
## factor(REF_RACE)3    2.723e-01  1.043e-01   2.611 0.009031 **
## factor(REF_RACE)4   -1.542e-01  3.281e-02  -4.701 2.59e-06 ***
## factor(REF_RACE)5    5.346e-02  1.046e-01   0.511 0.609205
## factor(REF_RACE)6    2.166e-01  5.810e-02   3.729 0.000192 ***
## SEX_REF              1.081e-01  1.523e-02   7.097 1.29e-12 ***
## ALCBEVPQ             8.548e-04  3.333e-05  25.643  < 2e-16 ***
## TOBACCPQ             3.594e-05  3.879e-05   0.926 0.354194
## factor(STATE)2      -6.071e-01  7.379e-02  -8.228  < 2e-16 ***
## factor(STATE)4      -6.784e-01  6.943e-02  -9.772  < 2e-16 ***
## factor(STATE)6      -5.829e-01  5.371e-02 -10.854  < 2e-16 ***
## factor(STATE)8      -7.646e-01  7.468e-02 -10.238  < 2e-16 ***
## factor(STATE)9      -3.362e-01  6.945e-02  -4.842 1.29e-06 ***
## factor(STATE)10     -4.615e-01  1.970e-01  -2.343 0.019114 *
## factor(STATE)11     -1.425e-01  1.560e-01  -0.913 0.361014
## factor(STATE)12     -7.711e-01  5.719e-02 -13.482  < 2e-16 ***
## factor(STATE)13     -4.242e-01  7.195e-02  -5.896 3.73e-09 ***
## factor(STATE)15     -7.416e-01  7.615e-02  -9.739  < 2e-16 ***
## factor(STATE)17     -4.071e-01  6.473e-02  -6.290 3.20e-10 ***
## factor(STATE)18     -8.916e-01  1.173e-01  -7.602 2.95e-14 ***
## factor(STATE)19     -3.813e-01  1.938e-01  -1.968 0.049044 *
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## factor(STATE)20    -7.119e-02  8.470e-02  -0.841 0.400589
## factor(STATE)21    -2.803e-01  8.062e-02  -3.477 0.000507 ***
## factor(STATE)22    -9.271e-01  7.726e-02 -12.000  < 2e-16 ***
## factor(STATE)24    -4.187e-01  6.987e-02  -5.993 2.07e-09 ***
## factor(STATE)25    -4.591e-01  6.437e-02  -7.131 1.00e-12 ***
## factor(STATE)26    -3.383e-01  6.492e-02  -5.211 1.88e-07 ***
## factor(STATE)27    -2.386e-01  7.219e-02  -3.305 0.000950 ***
## factor(STATE)28     4.928e-03  1.082e-01   0.046 0.963673
## factor(STATE)29    -2.429e-02  7.759e-02  -0.313 0.754288
## factor(STATE)31    -2.481e-01  6.913e-02  -3.589 0.000332 ***
## factor(STATE)32    -3.509e-01  8.383e-02  -4.186 2.84e-05 ***
## factor(STATE)33     1.947e-01  1.729e-01   1.126 0.260217
## factor(STATE)34    -4.346e-01  7.007e-02  -6.202 5.59e-10 ***
## factor(STATE)36    -6.248e-01  5.747e-02 -10.872  < 2e-16 ***
## factor(STATE)37    -4.682e-01  5.889e-02  -7.950 1.89e-15 ***
## factor(STATE)39    -3.827e-01  6.052e-02  -6.324 2.56e-10 ***
## factor(STATE)40    -6.675e-02  9.585e-02  -0.696 0.486202
## factor(STATE)41    -4.746e-01  7.231e-02  -6.563 5.31e-11 ***
## factor(STATE)42    -4.045e-01  6.158e-02  -6.568 5.13e-11 ***
## factor(STATE)45    -2.381e-01  7.894e-02  -3.017 0.002556 **
## factor(STATE)46    -7.338e-01  1.266e-01  -5.797 6.75e-09 ***
## factor(STATE)47     1.481e-01  8.560e-02   1.730 0.083680 .
## factor(STATE)48    -5.236e-01  5.710e-02  -9.171  < 2e-16 ***
## factor(STATE)49    -5.196e-01  6.421e-02  -8.093 5.90e-16 ***
## factor(STATE)51    -2.405e-01  6.901e-02  -3.484 0.000494 ***
## factor(STATE)53    -4.415e-01  6.909e-02  -6.390 1.66e-10 ***
## factor(STATE)54    -1.331e-01  1.194e-01  -1.115 0.264884
## factor(STATE)55    -2.106e-01  6.525e-02  -3.227 0.001250 **
## HISP_REF            2.600e-01  2.405e-02  10.812  < 2e-16 ***
## factor(HIGH_EDU)10 -1.583e-02  2.414e-01  -0.066 0.947723
## factor(HIGH_EDU)11  1.309e-01  2.336e-01   0.560 0.575256
## factor(HIGH_EDU)12  1.848e-01  2.312e-01   0.799 0.424194
## factor(HIGH_EDU)13  3.204e-01  2.313e-01   1.386 0.165850
## factor(HIGH_EDU)14  4.667e-01  2.318e-01   2.013 0.044074 *
## factor(HIGH_EDU)15  4.489e-01  2.312e-01   1.942 0.052179 .
## factor(HIGH_EDU)16  6.856e-01  2.314e-01   2.963 0.003047 **
## QSQ118             -7.837e-04  1.074e-03  -0.730 0.465502
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.242 on 91920 degrees of freedom
## Multiple R-squared:  0.08256,    Adjusted R-squared:  0.08186
## F-statistic: 118.2 on 70 and 91920 DF,  p-value: < 2.2e-16

perinsmodel <- lm(log(AdjPERINSPQ) ~ HOUSB+AGE_REF+factor(CUTENURE)+FAM_SIZE+
TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+factor(STATE)+HISP_REF+fact
or(HIGH_EDU)+QSQ118, data= ces_capstone)
summary(perinsmodel)
```

```
##
## Call:
## lm(formula = log(AdjPERINSPQ) ~ HOUSB + AGE_REF + factor(CUTENURE) +
##     FAM_SIZE + TINCHRS + factor(MARITAL1) + factor(REF_RACE) +
##     SEX_REF + factor(STATE) + HISP_REF + factor(HIGH_EDU) + QSQ118,
##     data = ces_capstone)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.4863  -1.2403   0.1586   1.2763   7.8486
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         3.1080099  0.2087466  14.889  < 2e-16 ***
## HOUSB              -0.7420821  0.0374366 -19.822  < 2e-16 ***
## AGE_REF            -0.0210482  0.0005633 -37.367  < 2e-16 ***
## factor(CUTENURE)2  -0.4151274  0.0179398 -23.140  < 2e-16 ***
## factor(CUTENURE)4  -0.4948586  0.0170725 -28.986  < 2e-16 ***
## factor(CUTENURE)5  -0.9941565  0.0591242 -16.815  < 2e-16 ***
## factor(CUTENURE)6  -2.3220130  0.1549424 -14.986  < 2e-16 ***
## FAM_SIZE            0.1957230  0.0054922  35.636  < 2e-16 ***
## TINCHRS             0.0482072  0.0002871 167.913  < 2e-16 ***
## factor(MARITAL1)2   0.1152064  0.0264507   4.356 1.33e-05 ***
## factor(MARITAL1)3   0.7116680  0.0210929  33.740  < 2e-16 ***
## factor(MARITAL1)4   0.8215740  0.0461183  17.815  < 2e-16 ***
## factor(MARITAL1)5   0.7871297  0.0218588  36.010  < 2e-16 ***
## factor(REF_RACE)2   0.0943971  0.0219725   4.296 1.74e-05 ***
## factor(REF_RACE)3  -0.1406422  0.0897560  -1.567 0.117132
## factor(REF_RACE)4   0.1584203  0.0281666   5.624 1.87e-08 ***
## factor(REF_RACE)5   0.2213115  0.0899839   2.459 0.013917 *
## factor(REF_RACE)6   0.0894879  0.0499954   1.790 0.073470 .
## SEX_REF            -0.0898466  0.0130918  -6.863 6.79e-12 ***
## factor(STATE)2      0.2755926  0.0634735   4.342 1.41e-05 ***
## factor(STATE)4     -0.1271020  0.0597375  -2.128 0.033367 *
## factor(STATE)6      0.2480725  0.0461843   5.371 7.83e-08 ***
## factor(STATE)8      0.3031561  0.0642323   4.720 2.37e-06 ***
## factor(STATE)9      0.1865623  0.0597575   3.122 0.001797 **
## factor(STATE)10     0.5827064  0.1694856   3.438 0.000586 ***
## factor(STATE)11     0.3377939  0.1342457   2.516 0.011863 *
## factor(STATE)12    -0.1196450  0.0492112  -2.431 0.015048 *
## factor(STATE)13     0.1287351  0.0619083   2.079 0.037579 *
## factor(STATE)15     0.2767958  0.0655143   4.225 2.39e-05 ***
## factor(STATE)17     0.1125345  0.0556884   2.021 0.043304 *
## factor(STATE)18    -0.0199866  0.1009395  -0.198 0.843041
## factor(STATE)19     0.0478881  0.1667351   0.287 0.773952
## factor(STATE)20    -0.0356988  0.0728868  -0.490 0.624288
## factor(STATE)21     0.0441877  0.0693810   0.637 0.524202
## factor(STATE)22    -0.2029583  0.0664815  -3.053 0.002267 **
## factor(STATE)24     0.3985820  0.0601132   6.631 3.36e-11 ***
## factor(STATE)25     0.1481301  0.0553827   2.675 0.007482 **
```

```
## factor(STATE)26     0.0052042  0.0558695   0.093 0.925785
## factor(STATE)27     0.1286430  0.0620906   2.072 0.038281 *
## factor(STATE)28    -0.1863923  0.0931166  -2.002 0.045319 *
## factor(STATE)29     0.1311735  0.0667608   1.965 0.049437 *
## factor(STATE)31     0.0134347  0.0594899   0.226 0.821333
## factor(STATE)32     0.1274223  0.0721358   1.766 0.077328 .
## factor(STATE)33     0.3529553  0.1488043   2.372 0.017697 *
## factor(STATE)34     0.2472620  0.0602895   4.101 4.11e-05 ***
## factor(STATE)36     0.1583259  0.0494294   3.203 0.001360 **
## factor(STATE)37    -0.0042755  0.0506772  -0.084 0.932764
## factor(STATE)39     0.1426257  0.0520687   2.739 0.006160 **
## factor(STATE)40     0.2694878  0.0824826   3.267 0.001087 **
## factor(STATE)41     0.1679256  0.0622033   2.700 0.006943 **
## factor(STATE)42     0.1221207  0.0529871   2.305 0.021184 *
## factor(STATE)45    -0.1043786  0.0679266  -1.537 0.124385
## factor(STATE)46    -0.2779029  0.1089252  -2.551 0.010733 *
## factor(STATE)47     0.1363567  0.0736613   1.851 0.064154 .
## factor(STATE)48     0.1061567  0.0491273   2.161 0.030710 *
## factor(STATE)49     0.0270768  0.0552126   0.490 0.623845
## factor(STATE)51     0.2716345  0.0593795   4.575 4.78e-06 ***
## factor(STATE)53     0.3626853  0.0594126   6.105 1.04e-09 ***
## factor(STATE)54     0.0822190  0.1027135   0.800 0.423441
## factor(STATE)55     0.1142731  0.0561501   2.035 0.041840 *
## HISP_REF           -0.1686496  0.0206125  -8.182 2.83e-16 ***
## factor(HIGH_EDU)10 -0.0304802  0.2077245  -0.147 0.883343
## factor(HIGH_EDU)11  0.3797817  0.2010080   1.889 0.058843 .
## factor(HIGH_EDU)12  0.8403727  0.1989832   4.223 2.41e-05 ***
## factor(HIGH_EDU)13  1.0954499  0.1990039   5.505 3.71e-08 ***
## factor(HIGH_EDU)14  1.3666159  0.1994763   6.851 7.38e-12 ***
## factor(HIGH_EDU)15  1.5700899  0.1989370   7.892 3.00e-15 ***
## factor(HIGH_EDU)16  1.7133042  0.1991029   8.605  < 2e-16 ***
## QSQ118              0.0066676  0.0009234   7.220 5.22e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.929 on 91922 degrees of freedom
## Multiple R-squared:  0.5424, Adjusted R-squared:  0.542
## F-statistic:  1602 on 68 and 91922 DF,  p-value: < 2.2e-16
```

```
hlthinsmodel <-lm(log(AdjHLTHINPQ) ~ HOUSB+AGE_REF+factor(CUTENURE)+FAM_SIZE+
TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+FOODPQ+ALCBEVPQ+MEDSRVPQ+PR
EDRGPQ+MEDSUPPQ+TOBACCPQ+factor(STATE)+HISP_REF+factor(HIGH_EDU)+QSQ118,data=
ces_capstone)
summary(hlthinsmodel)
```

```
##
## Call:
## lm(formula = log(AdjHLTHINPQ) ~ HOUSB + AGE_REF + factor(CUTENURE) +
##     FAM_SIZE + TINCHRS + factor(MARITAL1) + factor(REF_RACE) +
##     SEX_REF + FOODPQ + ALCBEVPQ + MEDSRVPQ + PREDRGPQ + MEDSUPPQ +
```

```
##     TOBACCPQ + factor(STATE) + HISP_REF + factor(HIGH_EDU) +
##     QSQ118, data = ces_capstone)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.2022 -0.8847  0.5650  1.5560  7.0215
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.538e+00  2.529e-01   6.079 1.21e-09 ***
## HOUSB               -5.378e-01  4.591e-02 -11.712  < 2e-16 ***
## AGE_REF              5.209e-02  6.835e-04  76.217  < 2e-16 ***
## factor(CUTENURE)2   -2.605e-01  2.174e-02 -11.980  < 2e-16 ***
## factor(CUTENURE)4   -6.774e-01  2.073e-02 -32.675  < 2e-16 ***
## factor(CUTENURE)5   -1.317e+00  7.164e-02 -18.387  < 2e-16 ***
## factor(CUTENURE)6   -2.353e+00  1.877e-01 -12.536  < 2e-16 ***
## FAM_SIZE            -1.910e-02  6.843e-03  -2.791 0.005256 **
## TINCHRS              5.041e-04  3.509e-04   1.437 0.150839
## factor(MARITAL1)2   -3.006e-01  3.207e-02  -9.375  < 2e-16 ***
## factor(MARITAL1)3   -5.045e-01  2.558e-02 -19.721  < 2e-16 ***
## factor(MARITAL1)4   -8.254e-01  5.589e-02 -14.770  < 2e-16 ***
## factor(MARITAL1)5   -4.773e-01  2.650e-02 -18.014  < 2e-16 ***
## factor(REF_RACE)2   -1.413e-01  2.669e-02  -5.296 1.19e-07 ***
## factor(REF_RACE)3   -2.353e-01  1.087e-01  -2.164 0.030458 *
## factor(REF_RACE)4   -3.603e-02  3.421e-02  -1.053 0.292334
## factor(REF_RACE)5    1.605e-01  1.090e-01   1.472 0.140995
## factor(REF_RACE)6    8.134e-03  6.058e-02   0.134 0.893185
## SEX_REF              6.817e-02  1.589e-02   4.291 1.78e-05 ***
## FOODPQ               1.767e-04  6.128e-06  28.831  < 2e-16 ***
## ALCBEVPQ             1.824e-04  3.739e-05   4.877 1.08e-06 ***
## MEDSRVPQ             1.079e-04  9.729e-06  11.086  < 2e-16 ***
## PREDRGPQ            -5.364e-02  1.875e-01  -0.286 0.774778
## MEDSUPPQ             2.431e-04  4.210e-05   5.775 7.72e-09 ***
## TOBACCPQ            -2.447e-05  4.048e-05  -0.604 0.545568
## factor(STATE)2      -2.145e-01  7.694e-02  -2.787 0.005313 **
## factor(STATE)4      -2.687e-01  7.239e-02  -3.712 0.000206 ***
## factor(STATE)6      -4.314e-01  5.600e-02  -7.703 1.34e-14 ***
## factor(STATE)8      -4.129e-02  7.787e-02  -0.530 0.595976
## factor(STATE)9       1.562e-01  7.242e-02   2.157 0.030991 *
## factor(STATE)10     -4.666e-01  2.054e-01  -2.272 0.023068 *
## factor(STATE)11     -6.279e-02  1.627e-01  -0.386 0.699557
## factor(STATE)12     -4.134e-02  5.964e-02  -0.693 0.488223
## factor(STATE)13      2.313e-01  7.502e-02   3.083 0.002047 **
## factor(STATE)15     -6.268e-01  7.942e-02  -7.892 3.00e-15 ***
## factor(STATE)17      1.438e-01  6.750e-02   2.130 0.033172 *
## factor(STATE)18     -2.721e-01  1.223e-01  -2.225 0.026075 *
## factor(STATE)19     -2.270e-01  2.020e-01  -1.124 0.261221
## factor(STATE)20     -6.148e-02  8.832e-02  -0.696 0.486385
## factor(STATE)21     -1.571e-01  8.406e-02  -1.868 0.061698 .
## factor(STATE)22     -3.878e-01  8.056e-02  -4.814 1.48e-06 ***
```

```
## factor(STATE)24     1.941e-01  7.285e-02   2.664 0.007720 **
## factor(STATE)25     1.066e-01  6.712e-02   1.589 0.112108
## factor(STATE)26    -3.030e-01  6.770e-02  -4.476 7.62e-06 ***
## factor(STATE)27     1.590e-01  7.528e-02   2.113 0.034626 *
## factor(STATE)28    -1.190e-02  1.128e-01  -0.105 0.916019
## factor(STATE)29     1.546e-01  8.091e-02   1.911 0.055964 .
## factor(STATE)31     1.500e-01  7.211e-02   2.080 0.037511 *
## factor(STATE)32    -3.544e-01  8.741e-02  -4.055 5.02e-05 ***
## factor(STATE)33     4.032e-01  1.803e-01   2.236 0.025336 *
## factor(STATE)34    -1.906e-01  7.306e-02  -2.608 0.009102 **
## factor(STATE)36    -5.239e-01  5.992e-02  -8.743  < 2e-16 ***
## factor(STATE)37    -8.534e-02  6.140e-02  -1.390 0.164597
## factor(STATE)39    -5.102e-03  6.311e-02  -0.081 0.935568
## factor(STATE)40    -7.877e-02  9.994e-02  -0.788 0.430588
## factor(STATE)41    -3.149e-01  7.541e-02  -4.176 2.97e-05 ***
## factor(STATE)42    -9.337e-03  6.423e-02  -0.145 0.884419
## factor(STATE)45     1.931e-02  8.231e-02   0.235 0.814491
## factor(STATE)46     1.048e-01  1.320e-01   0.794 0.427389
## factor(STATE)47    -3.961e-02  8.925e-02  -0.444 0.657194
## factor(STATE)48    -3.695e-02  5.954e-02  -0.621 0.534810
## factor(STATE)49    -2.794e-01  6.697e-02  -4.172 3.02e-05 ***
## factor(STATE)51     1.201e-01  7.196e-02   1.669 0.095193 .
## factor(STATE)53    -5.212e-01  7.204e-02  -7.235 4.70e-13 ***
## factor(STATE)54    -2.560e-01  1.245e-01  -2.056 0.039767 *
## factor(STATE)55     4.374e-01  6.805e-02   6.428 1.30e-10 ***
## HISP_REF            6.574e-01  2.508e-02  26.213  < 2e-16 ***
## factor(HIGH_EDU)10 -4.398e-02  2.517e-01  -0.175 0.861269
## factor(HIGH_EDU)11 -5.114e-01  2.435e-01  -2.100 0.035749 *
## factor(HIGH_EDU)12  6.054e-02  2.411e-01   0.251 0.801739
## factor(HIGH_EDU)13  3.074e-01  2.411e-01   1.275 0.202412
## factor(HIGH_EDU)14  6.007e-01  2.417e-01   2.486 0.012939 *
## factor(HIGH_EDU)15  8.372e-01  2.411e-01   3.473 0.000515 ***
## factor(HIGH_EDU)16  9.862e-01  2.413e-01   4.087 4.36e-05 ***
## QSQ118              8.043e-04  1.131e-03   0.711 0.477088
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.337 on 91916 degrees of freedom
## Multiple R-squared:  0.248,  Adjusted R-squared:  0.2474
## F-statistic: 409.6 on 74 and 91916 DF,  p-value: < 2.2e-16

#Robust Standard Errors

library(lmtest)

## Warning: package 'lmtest' was built under R version 4.5.3

## Loading required package: zoo
##
## Attaching package: 'zoo'
```

```
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
library(sandwich)

coeftest(vehinsmodel,vcov. = vcovHC(vehinsmodel, type = "HC3"))
```

```
##
## t test of coefficients:
##
##                        Estimate  Std. Error t value  Pr(>|t|)
## (Intercept)         1.1586e+00  2.8148e-01  4.1161 3.857e-05 ***
## HOUSB               2.1129e+00  5.7216e-02 36.9290 < 2.2e-16 ***
## AGE_REF            -2.1750e-03  8.3021e-04 -2.6199 0.0087979 **
## factor(CUTENURE)2   2.1246e-02  2.7226e-02  0.7803 0.4351897
## factor(CUTENURE)4  -1.9704e-01  2.5764e-02 -7.6479 2.062e-14 ***
## factor(CUTENURE)5  -2.4394e-01  8.4168e-02 -2.8983 0.0037528 **
## factor(CUTENURE)6  -1.0428e+00  1.5092e-01 -6.9095 4.897e-12 ***
## FAM_SIZE           -1.4968e-03  8.6095e-03 -0.1738 0.8619851
## TINCHRS             4.0907e-03  4.4628e-04  9.1662 < 2.2e-16 ***
## factor(MARITAL1)2   8.3078e-02  3.8669e-02  2.1484 0.0316809 *
## factor(MARITAL1)3   1.6114e-01  3.1226e-02  5.1605 2.468e-07 ***
## factor(MARITAL1)4  -1.3118e-01  6.6771e-02 -1.9646 0.0494661 *
## factor(MARITAL1)5   7.5095e-04  3.2299e-02  0.0232 0.9814511
## factor(REF_RACE)2  -3.7974e-02  3.2484e-02 -1.1690 0.2423967
## factor(REF_RACE)3   5.3197e-02  1.2979e-01  0.4099 0.6819158
## factor(REF_RACE)4  -2.0586e-01  4.1753e-02 -4.9306 8.214e-07 ***
## factor(REF_RACE)5   4.8513e-02  1.3765e-01  0.3524 0.7245016
## factor(REF_RACE)6  -3.6614e-02  7.2057e-02 -0.5081 0.6113654
## SEX_REF            -1.8826e-03  1.9529e-02 -0.0964 0.9232020
## VEHQ                1.7936e-01  8.8711e-03 20.2186 < 2.2e-16 ***
## GASMOPQ             1.3359e-03  4.2913e-05 31.1313 < 2.2e-16 ***
## VEHFINPQ            2.1127e-03  9.5501e-05 22.1222 < 2.2e-16 ***
## MAINRPPQ            2.5091e-04  2.8362e-05  8.8469 < 2.2e-16 ***
## VRNTLOPQ            3.4284e-04  4.3383e-05  7.9026 2.761e-15 ***
## PUBTRAPQ            2.5319e-06  1.4750e-05  0.1717 0.8637106
## factor(STATE)2     -1.3951e-01  9.4069e-02 -1.4831 0.1380510
## factor(STATE)4     -2.1780e-02  8.8856e-02 -0.2451 0.8063635
## factor(STATE)6     -1.5459e-01  6.8645e-02 -2.2520 0.0243254 *
## factor(STATE)8      3.2429e-01  9.8181e-02  3.3030 0.0009570 ***
## factor(STATE)9      1.0607e-01  8.8871e-02  1.1935 0.2326621
## factor(STATE)10     7.4122e-01  2.5512e-01  2.9054 0.0036686 **
## factor(STATE)11    -3.2862e-01  1.8594e-01 -1.7673 0.0771750 .
## factor(STATE)12    -3.6779e-02  7.3668e-02 -0.4992 0.6176048
## factor(STATE)13     1.8304e-01  9.4394e-02  1.9391 0.0524910 .
## factor(STATE)15    -7.2843e-02  9.4936e-02 -0.7673 0.4429182
## factor(STATE)17    -2.2229e-01  8.1434e-02 -2.7297 0.0063409 **
## factor(STATE)18    -9.1035e-01  1.4297e-01 -6.3674 1.932e-10 ***
```

```
## factor(STATE)19    -5.9258e-01  2.4512e-01 -2.4175 0.0156279 *
## factor(STATE)20    -1.0726e-01  1.0776e-01 -0.9954 0.3195633
## factor(STATE)21     2.1633e-01  1.0306e-01  2.0990 0.0358189 *
## factor(STATE)22     2.5931e-01  1.0152e-01  2.5544 0.0106390 *
## factor(STATE)24     1.2882e-01  9.0088e-02  1.4299 0.1527423
## factor(STATE)25    -2.0124e-01  8.1538e-02 -2.4681 0.0135864 *
## factor(STATE)26    -1.1221e-01  8.5140e-02 -1.3179 0.1875389
## factor(STATE)27    -1.3670e-01  9.3277e-02 -1.4655 0.1427940
## factor(STATE)28    -2.1405e-01  1.3528e-01 -1.5823 0.1135876
## factor(STATE)29    -5.4770e-02  9.8997e-02 -0.5533 0.5800929
## factor(STATE)31     1.2318e-01  8.7453e-02  1.4085 0.1589735
## factor(STATE)32     1.9592e-01  1.0886e-01  1.7998 0.0718954 .
## factor(STATE)33    -9.0133e-02  2.1407e-01 -0.4210 0.6737296
## factor(STATE)34    -6.8353e-02  9.1152e-02 -0.7499 0.4533260
## factor(STATE)36    -5.4164e-01  7.3293e-02 -7.3901 1.479e-13 ***
## factor(STATE)37     1.0152e-02  7.4749e-02  0.1358 0.8919655
## factor(STATE)39    -1.5292e-02  7.6336e-02 -0.2003 0.8412252
## factor(STATE)40     2.7605e-01  1.2090e-01  2.2833 0.0224148 *
## factor(STATE)41     4.5233e-02  9.1936e-02  0.4920 0.6227158
## factor(STATE)42     3.5752e-03  7.8211e-02  0.0457 0.9635397
## factor(STATE)45     1.1467e-01  1.0115e-01  1.1336 0.2569668
## factor(STATE)46    -1.6512e-01  1.6036e-01 -1.0297 0.3031715
## factor(STATE)47    -8.4211e-03  1.0906e-01 -0.0772 0.9384519
## factor(STATE)48     1.4849e-01  7.3465e-02  2.0212 0.0432627 *
## factor(STATE)49     7.3913e-02  8.1692e-02  0.9048 0.3655816
## factor(STATE)51    -1.7274e-01  8.8461e-02 -1.9527 0.0508561 .
## factor(STATE)53    -1.3877e-01  8.7775e-02 -1.5810 0.1138833
## factor(STATE)54    -1.9348e-01  1.4867e-01 -1.3013 0.1931451
## factor(STATE)55    -1.7410e-01  8.2203e-02 -2.1179 0.0341824 *
## EVEHPURP            2.6482e-05  3.8208e-06  6.9311 4.203e-12 ***
## HISP_REF           -3.9539e-02  3.0687e-02 -1.2885 0.1975840
## factor(HIGH_EDU)10 -1.4020e-01  2.7252e-01 -0.5145 0.6069264
## factor(HIGH_EDU)11  1.2386e-01  2.6519e-01  0.4671 0.6404433
## factor(HIGH_EDU)12  4.0986e-01  2.6233e-01  1.5624 0.1181979
## factor(HIGH_EDU)13  5.7415e-01  2.6247e-01  2.1875 0.0287055 *
## factor(HIGH_EDU)14  6.1421e-01  2.6330e-01  2.3328 0.0196628 *
## factor(HIGH_EDU)15  5.9033e-01  2.6241e-01  2.2497 0.0244725 *
## factor(HIGH_EDU)16  6.5317e-01  2.6268e-01  2.4866 0.0129002 *
## factor(QSQ118)1     1.6082e-02  6.1853e-02  0.2600 0.7948606
## factor(QSQ118)2    -3.1063e-02  6.2433e-02 -0.4975 0.6188057
## factor(QSQ118)3    -1.0314e-01  6.1360e-02 -1.6810 0.0927733 .
## factor(QSQ118)4     3.5237e-02  6.1333e-02  0.5745 0.5656239
## factor(QSQ118)5     4.5020e-02  6.1801e-02  0.7285 0.4663313
## factor(QSQ118)6     6.2848e-03  6.2123e-02  0.1012 0.9194180
## factor(QSQ118)7     6.1980e-02  6.2050e-02  0.9989 0.3178581
## factor(QSQ118)8     2.2741e-01  6.3117e-02  3.6029 0.0003148 ***
## factor(QSQ118)9     2.2382e-01  6.3229e-02  3.5398 0.0004007 ***
## factor(QSQ118)10    2.4191e-01  6.2439e-02  3.8744 0.0001070 ***
## factor(QSQ118)11    1.1153e-01  6.2751e-02  1.7774 0.0755070 .
## factor(QSQ118)12    1.4405e-01  6.2344e-02  2.3105 0.0208618 *
```

```
## factor(QSQ118)13    4.8045e-02  6.2567e-02  0.7679 0.4425534
## factor(QSQ118)14   -6.3322e-02  6.2796e-02 -1.0084 0.3132772
## factor(QSQ118)15   -9.9209e-02  6.2265e-02 -1.5933 0.1110897
## factor(QSQ118)16   -2.0250e-01  6.2870e-02 -3.2209 0.0012783 **
## factor(QSQ118)17   -3.8551e-01  6.5136e-02 -5.9185 3.260e-09 ***
## factor(QSQ118)18   -1.3160e-01  6.4702e-02 -2.0340 0.0419545 *
## factor(QSQ118)19    1.2173e-02  6.3405e-02  0.1920 0.8477556
## factor(QSQ118)20    5.9430e-02  6.3964e-02  0.9291 0.3528382
## factor(QSQ118)21    2.4536e-02  6.4478e-02  0.3805 0.7035484
## factor(QSQ118)22   -9.9884e-02  6.4909e-02 -1.5388 0.1238509
## factor(QSQ118)23    2.9293e-02  6.5351e-02  0.4482 0.6539805
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(lifinsmodel,vcov. = vcovHC(lifinsmodel, type = "HC3"))
```

```
##
## t test of coefficients:
##
##                       Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)         9.8281e-04  1.5490e-01   0.0063 0.9949375
## HOUSB               4.8179e-01  4.1382e-02  11.6426 < 2.2e-16 ***
## AGE_REF             1.4857e-02  6.2516e-04  23.7654 < 2.2e-16 ***
## factor(CUTENURE)2  -2.9727e-01  2.2071e-02 -13.4690 < 2.2e-16 ***
## factor(CUTENURE)4  -4.9246e-01  1.9356e-02 -25.4419 < 2.2e-16 ***
## factor(CUTENURE)5  -5.1260e-01  5.5704e-02  -9.2022 < 2.2e-16 ***
## factor(CUTENURE)6  -3.6209e-01  9.0743e-02  -3.9903 6.605e-05 ***
## FAM_SIZE            6.7156e-02  6.4536e-03  10.4059 < 2.2e-16 ***
## TINCHRS             7.1785e-03  3.5828e-04  20.0361 < 2.2e-16 ***
## factor(MARITAL1)2  -2.6421e-01  3.0413e-02  -8.6876 < 2.2e-16 ***
## factor(MARITAL1)3  -2.5706e-01  2.4546e-02 -10.4728 < 2.2e-16 ***
## factor(MARITAL1)4  -3.4456e-01  4.4652e-02  -7.7165 1.208e-14 ***
## factor(MARITAL1)5  -3.5714e-01  2.3555e-02 -15.1618 < 2.2e-16 ***
## factor(REF_RACE)2   3.8830e-01  2.6253e-02  14.7907 < 2.2e-16 ***
## factor(REF_RACE)3   2.7232e-01  1.0220e-01   2.6645 0.0077110 **
## factor(REF_RACE)4  -1.5422e-01  3.2937e-02  -4.6824 2.840e-06 ***
## factor(REF_RACE)5   5.3456e-02  1.0372e-01   0.5154 0.6062921
## factor(REF_RACE)6   2.1664e-01  5.8799e-02   3.6844 0.0002294 ***
## SEX_REF             1.0808e-01  1.5301e-02   7.0639 1.631e-12 ***
## ALCBEVPQ            8.5478e-04  4.3075e-05  19.8441 < 2.2e-16 ***
## TOBACCPQ            3.5943e-05  4.1101e-05   0.8745 0.3818446
## factor(STATE)2     -6.0715e-01  7.6959e-02  -7.8893 3.072e-15 ***
## factor(STATE)4     -6.7844e-01  7.0565e-02  -9.6144 < 2.2e-16 ***
## factor(STATE)6     -5.8294e-01  5.8319e-02  -9.9957 < 2.2e-16 ***
## factor(STATE)8     -7.6464e-01  7.6145e-02 -10.0419 < 2.2e-16 ***
## factor(STATE)9     -3.3625e-01  7.4989e-02  -4.4840 7.335e-06 ***
## factor(STATE)10    -4.6155e-01  2.0287e-01  -2.2751 0.0229015 *
## factor(STATE)11    -1.4254e-01  1.6071e-01  -0.8869 0.3751328
## factor(STATE)12    -7.7106e-01  6.0417e-02 -12.7622 < 2.2e-16 ***
## factor(STATE)13    -4.2425e-01  7.6781e-02  -5.5254 3.296e-08 ***
```

```
## factor(STATE)15    -7.4165e-01  7.7885e-02  -9.5223 < 2.2e-16 ***
## factor(STATE)17    -4.0711e-01  6.9339e-02  -5.8713 4.339e-09 ***
## factor(STATE)18    -8.9164e-01  1.0570e-01  -8.4356 < 2.2e-16 ***
## factor(STATE)19    -3.8135e-01  2.0031e-01  -1.9037 0.0569461 .
## factor(STATE)20    -7.1193e-02  9.0793e-02  -0.7841 0.4329657
## factor(STATE)21    -2.8033e-01  8.4889e-02  -3.3023 0.0009592 ***
## factor(STATE)22    -9.2709e-01  7.4272e-02 -12.4824 < 2.2e-16 ***
## factor(STATE)24    -4.1873e-01  7.6906e-02  -5.4447 5.202e-08 ***
## factor(STATE)25    -4.5907e-01  6.9246e-02  -6.6295 3.387e-11 ***
## factor(STATE)26    -3.3831e-01  6.9306e-02  -4.8815 1.055e-06 ***
## factor(STATE)27    -2.3857e-01  7.7921e-02  -3.0617 0.0022013 **
## factor(STATE)28     4.9281e-03  1.1683e-01   0.0422 0.9663534
## factor(STATE)29    -2.4285e-02  8.6725e-02  -0.2800 0.7794584
## factor(STATE)31    -2.4811e-01  7.3521e-02  -3.3747 0.0007392 ***
## factor(STATE)32    -3.5092e-01  9.0047e-02  -3.8971 9.742e-05 ***
## factor(STATE)33     1.9469e-01  1.9205e-01   1.0137 0.3107089
## factor(STATE)34    -4.3461e-01  7.5746e-02  -5.7378 9.623e-09 ***
## factor(STATE)36    -6.2481e-01  6.1879e-02 -10.0974 < 2.2e-16 ***
## factor(STATE)37    -4.6816e-01  6.3738e-02  -7.3451 2.073e-13 ***
## factor(STATE)39    -3.8271e-01  6.5187e-02  -5.8710 4.347e-09 ***
## factor(STATE)40    -6.6746e-02  1.0307e-01  -0.6476 0.5172566
## factor(STATE)41    -4.7458e-01  7.5082e-02  -6.3209 2.612e-10 ***
## factor(STATE)42    -4.0446e-01  6.6432e-02  -6.0884 1.145e-09 ***
## factor(STATE)45    -2.3813e-01  8.4902e-02  -2.8047 0.0050366 **
## factor(STATE)46    -7.3381e-01  1.2259e-01  -5.9858 2.161e-09 ***
## factor(STATE)47     1.4806e-01  9.3517e-02   1.5832 0.1133688
## factor(STATE)48    -5.2361e-01  6.1324e-02  -8.5384 < 2.2e-16 ***
## factor(STATE)49    -5.1962e-01  6.8671e-02  -7.5668 3.861e-14 ***
## factor(STATE)51    -2.4046e-01  7.5850e-02  -3.1702 0.0015237 **
## factor(STATE)53    -4.4150e-01  7.3148e-02  -6.0356 1.589e-09 ***
## factor(STATE)54    -1.3309e-01  1.2179e-01  -1.0928 0.2744704
## factor(STATE)55    -2.1059e-01  7.0567e-02  -2.9842 0.0028440 **
## HISP_REF            2.6004e-01  2.1702e-02  11.9825 < 2.2e-16 ***
## factor(HIGH_EDU)10 -1.5827e-02  1.4112e-01  -0.1121 0.9107051
## factor(HIGH_EDU)11  1.3088e-01  1.3611e-01   0.9616 0.3362542
## factor(HIGH_EDU)12  1.8479e-01  1.3366e-01   1.3825 0.1668100
## factor(HIGH_EDU)13  3.2044e-01  1.3391e-01   2.3930 0.0167134 *
## factor(HIGH_EDU)14  4.6671e-01  1.3491e-01   3.4593 0.0005418 ***
## factor(HIGH_EDU)15  4.4888e-01  1.3386e-01   3.3532 0.0007991 ***
## factor(HIGH_EDU)16  6.8557e-01  1.3446e-01   5.0986 3.428e-07 ***
## QSQ118             -7.8368e-04  1.0876e-03  -0.7205 0.4711898
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(perinsmodel,vcov. = vcovHC(perinsmodel, type = "HC3"))
```

```
##
## t test of coefficients:
##
##                      Estimate  Std. Error  t value  Pr(>|t|)
```

```
## (Intercept)         3.10800993  0.22579308  13.7649 < 2.2e-16 ***
## HOUSB              -0.74208211  0.04263030 -17.4074 < 2.2e-16 ***
## AGE_REF            -0.02104819  0.00055404 -37.9906 < 2.2e-16 ***
## factor(CUTENURE)2  -0.41512741  0.01948240 -21.3078 < 2.2e-16 ***
## factor(CUTENURE)4  -0.49485860  0.01600440 -30.9202 < 2.2e-16 ***
## factor(CUTENURE)5  -0.99415646  0.06176110 -16.0968 < 2.2e-16 ***
## factor(CUTENURE)6  -2.32201303  0.15099482 -15.3781 < 2.2e-16 ***
## FAM_SIZE            0.19572304  0.00494307  39.5955 < 2.2e-16 ***
## TINCHRS             0.04820722  0.00031827 151.4660 < 2.2e-16 ***
## factor(MARITAL1)2   0.11520635  0.03401537   3.3869 0.0007072 ***
## factor(MARITAL1)3   0.71166801  0.02277555  31.2470 < 2.2e-16 ***
## factor(MARITAL1)4   0.82157396  0.04488282  18.3049 < 2.2e-16 ***
## factor(MARITAL1)5   0.78712973  0.02122647  37.0825 < 2.2e-16 ***
## factor(REF_RACE)2   0.09439713  0.02238236   4.2175 2.473e-05 ***
## factor(REF_RACE)3  -0.14064223  0.08598383  -1.6357 0.1019095
## factor(REF_RACE)4   0.15842030  0.02473539   6.4046 1.515e-10 ***
## factor(REF_RACE)5   0.22131154  0.07897203   2.8024 0.0050734 **
## factor(REF_RACE)6   0.08948794  0.04700903   1.9036 0.0569611 .
## SEX_REF            -0.08984656  0.01298868  -6.9173 4.633e-12 ***
## factor(STATE)2      0.27559258  0.06349785   4.3402 1.425e-05 ***
## factor(STATE)4     -0.12710196  0.06175252  -2.0582 0.0395692 *
## factor(STATE)6      0.24807250  0.04780748   5.1890 2.119e-07 ***
## factor(STATE)8      0.30315609  0.06126713   4.9481 7.507e-07 ***
## factor(STATE)9      0.18656227  0.06016556   3.1008 0.0019305 **
## factor(STATE)10     0.58270640  0.15789680   3.6904 0.0002240 ***
## factor(STATE)11     0.33779387  0.13719556   2.4621 0.0138131 *
## factor(STATE)12    -0.11964503  0.05130560  -2.3320 0.0197025 *
## factor(STATE)13     0.12873507  0.06233678   2.0652 0.0389112 *
## factor(STATE)15     0.27679579  0.06664571   4.1532 3.281e-05 ***
## factor(STATE)17     0.11253451  0.05593044   2.0120 0.0442182 *
## factor(STATE)18    -0.01998664  0.10209810  -0.1958 0.8447991
## factor(STATE)19     0.04788814  0.18209621   0.2630 0.7925646
## factor(STATE)20    -0.03569879  0.07283256  -0.4901 0.6240298
## factor(STATE)21     0.04418770  0.07288829   0.6062 0.5443577
## factor(STATE)22    -0.20295826  0.06645432  -3.0541 0.0022580 **
## factor(STATE)24     0.39858198  0.06173912   6.4559 1.081e-10 ***
## factor(STATE)25     0.14813014  0.05617085   2.6371 0.0083624 **
## factor(STATE)26     0.00520424  0.05804058   0.0897 0.9285533
## factor(STATE)27     0.12864299  0.06232170   2.0642 0.0390038 *
## factor(STATE)28    -0.18639228  0.10647953  -1.7505 0.0800356 .
## factor(STATE)29     0.13117346  0.06724365   1.9507 0.0510935 .
## factor(STATE)31     0.01343474  0.05906132   0.2275 0.8200580
## factor(STATE)32     0.12742228  0.07454671   1.7093 0.0873998 .
## factor(STATE)33     0.35295530  0.12726255   2.7734 0.0055478 **
## factor(STATE)34     0.24726198  0.06204933   3.9849 6.755e-05 ***
## factor(STATE)36     0.15832587  0.05128129   3.0874 0.0020198 **
## factor(STATE)37    -0.00427553  0.05205973  -0.0821 0.9345456
## factor(STATE)39     0.14262572  0.05349747   2.6660 0.0076767 **
## factor(STATE)40     0.26948782  0.08101958   3.3262 0.0008807 ***
## factor(STATE)41     0.16792555  0.06145511   2.7325 0.0062869 **
```

```
## factor(STATE)42     0.12212069  0.05521494   2.2117 0.0269876 *
## factor(STATE)45    -0.10437863  0.07099842  -1.4702 0.1415234
## factor(STATE)46    -0.27790287  0.09696001  -2.8662 0.0041558 **
## factor(STATE)47     0.13635673  0.08089261   1.6857 0.0918664 .
## factor(STATE)48     0.10615672  0.05002698   2.1220 0.0338413 *
## factor(STATE)49     0.02707683  0.05676208   0.4770 0.6333467
## factor(STATE)51     0.27163453  0.05973321   4.5475 5.437e-06 ***
## factor(STATE)53     0.36268525  0.05952855   6.0926 1.115e-09 ***
## factor(STATE)54     0.08221904  0.10378884   0.7922 0.4282600
## factor(STATE)55     0.11427307  0.05618055   2.0340 0.0419512 *
## HISP_REF           -0.16864965  0.01861253  -9.0611 < 2.2e-16 ***
## factor(HIGH_EDU)10 -0.03048015  0.22551720  -0.1352 0.8924883
## factor(HIGH_EDU)11  0.37978167  0.21893380   1.7347 0.0827996 .
## factor(HIGH_EDU)12  0.84037269  0.21667426   3.8785 0.0001052 ***
## factor(HIGH_EDU)13  1.09544991  0.21671337   5.0548 4.316e-07 ***
## factor(HIGH_EDU)14  1.36661592  0.21713477   6.2939 3.111e-10 ***
## factor(HIGH_EDU)15  1.57008990  0.21669024   7.2458 4.333e-13 ***
## factor(HIGH_EDU)16  1.71330425  0.21692576   7.8981 2.863e-15 ***
## QSQ118              0.00666765  0.00097059   6.8697 6.476e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(hlthinsmodel,vcov. = vcovHC(hlthinsmodel, type = "HC3"))
```

```
##
## t test of coefficients:
##
##                       Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)         1.5376e+00  2.4209e-01   6.3516 2.140e-10 ***
## HOUSB              -5.3777e-01  4.8037e-02 -11.1949 < 2.2e-16 ***
## AGE_REF             5.2092e-02  6.9653e-04  74.7881 < 2.2e-16 ***
## factor(CUTENURE)2  -2.6050e-01  2.0308e-02 -12.8271 < 2.2e-16 ***
## factor(CUTENURE)4  -6.7745e-01  2.2137e-02 -30.6030 < 2.2e-16 ***
## factor(CUTENURE)5  -1.3173e+00  8.2349e-02 -15.9960 < 2.2e-16 ***
## factor(CUTENURE)6  -2.3534e+00  1.5313e-01 -15.3688 < 2.2e-16 ***
## FAM_SIZE           -1.9099e-02  7.7997e-03  -2.4487 0.0143386 *
## TINCHRS             5.0407e-04  3.5215e-04   1.4314 0.1523139
## factor(MARITAL1)2  -3.0064e-01  2.5327e-02 -11.8705 < 2.2e-16 ***
## factor(MARITAL1)3  -5.0446e-01  2.5192e-02 -20.0241 < 2.2e-16 ***
## factor(MARITAL1)4  -8.2544e-01  6.4055e-02 -12.8863 < 2.2e-16 ***
## factor(MARITAL1)5  -4.7734e-01  2.8083e-02 -16.9976 < 2.2e-16 ***
## factor(REF_RACE)2  -1.4134e-01  2.7645e-02  -5.1126 3.183e-07 ***
## factor(REF_RACE)3  -2.3534e-01  1.2454e-01  -1.8897 0.0588001 .
## factor(REF_RACE)4  -3.6028e-02  3.6843e-02  -0.9779 0.3281414
## factor(REF_RACE)5   1.6050e-01  1.1802e-01   1.3599 0.1738766
## factor(REF_RACE)6   8.1338e-03  6.7110e-02   0.1212 0.9035327
## SEX_REF             6.8169e-02  1.6007e-02   4.2587 2.058e-05 ***
## FOODPQ              1.7668e-04  7.2232e-06  24.4605 < 2.2e-16 ***
## ALCBEVPQ            1.8238e-04  4.1236e-05   4.4230 9.746e-06 ***
## MEDSRVPQ            1.0786e-04  1.1731e-05   9.1948 < 2.2e-16 ***
```

```
## PREDRGPQ           -5.3640e-02  1.7071e-01  -0.3142 0.7533537
## MEDSUPPQ            2.4314e-04  4.2382e-05   5.7369 9.675e-09 ***
## TOBACCPQ           -2.4469e-05  4.7683e-05  -0.5132 0.6078453
## factor(STATE)2     -2.1447e-01  7.7391e-02  -2.7712 0.0055865 **
## factor(STATE)4     -2.6873e-01  7.1894e-02  -3.7379 0.0001857 ***
## factor(STATE)6     -4.3136e-01  5.4795e-02  -7.8722 3.524e-15 ***
## factor(STATE)8     -4.1286e-02  7.6128e-02  -0.5423 0.5875931
## factor(STATE)9      1.5622e-01  6.7099e-02   2.3283 0.0199007 *
## factor(STATE)10    -4.6664e-01  2.2253e-01  -2.0970 0.0359968 *
## factor(STATE)11    -6.2789e-02  1.5178e-01  -0.4137 0.6791042
## factor(STATE)12    -4.1336e-02  5.6940e-02  -0.7260 0.4678677
## factor(STATE)13     2.3132e-01  7.1524e-02   3.2342 0.0012204 **
## factor(STATE)15    -6.2675e-01  8.0977e-02  -7.7399 1.005e-14 ***
## factor(STATE)17     1.4377e-01  6.5308e-02   2.2015 0.0277057 *
## factor(STATE)18    -2.7214e-01  1.2132e-01  -2.2432 0.0248854 *
## factor(STATE)19    -2.2700e-01  1.9885e-01  -1.1415 0.2536513
## factor(STATE)20    -6.1478e-02  8.5265e-02  -0.7210 0.4708978
## factor(STATE)21    -1.5706e-01  8.1623e-02  -1.9243 0.0543261 .
## factor(STATE)22    -3.8777e-01  8.3786e-02  -4.6281 3.695e-06 ***
## factor(STATE)24     1.9409e-01  6.7856e-02   2.8603 0.0042333 **
## factor(STATE)25     1.0664e-01  6.3381e-02   1.6825 0.0924694 .
## factor(STATE)26    -3.0302e-01  6.7077e-02  -4.5174 6.267e-06 ***
## factor(STATE)27     1.5904e-01  7.1264e-02   2.2317 0.0256365 *
## factor(STATE)28    -1.1897e-02  9.6741e-02  -0.1230 0.9021256
## factor(STATE)29     1.5464e-01  7.5406e-02   2.0508 0.0402941 *
## factor(STATE)31     1.5001e-01  6.7318e-02   2.2283 0.0258612 *
## factor(STATE)32    -3.5445e-01  9.0210e-02  -3.9291 8.533e-05 ***
## factor(STATE)33     4.0319e-01  1.8074e-01   2.2308 0.0256990 *
## factor(STATE)34    -1.9056e-01  7.4192e-02  -2.5685 0.0102167 *
## factor(STATE)36    -5.2392e-01  5.9262e-02  -8.8408 < 2.2e-16 ***
## factor(STATE)37    -8.5335e-02  5.8822e-02  -1.4507 0.1468541
## factor(STATE)39    -5.1016e-03  5.9138e-02  -0.0863 0.9312559
## factor(STATE)40    -7.8769e-02  1.0583e-01  -0.7443 0.4566910
## factor(STATE)41    -3.1490e-01  7.5595e-02  -4.1656 3.108e-05 ***
## factor(STATE)42    -9.3366e-03  6.0848e-02  -0.1534 0.8780514
## factor(STATE)45     1.9312e-02  7.7551e-02   0.2490 0.8033396
## factor(STATE)46     1.0477e-01  1.1779e-01   0.8894 0.3737714
## factor(STATE)47    -3.9608e-02  8.4744e-02  -0.4674 0.6402270
## factor(STATE)48    -3.6954e-02  5.8472e-02  -0.6320 0.5273919
## factor(STATE)49    -2.7941e-01  6.4658e-02  -4.3214 1.552e-05 ***
## factor(STATE)51     1.2007e-01  6.8794e-02   1.7454 0.0809170 .
## factor(STATE)53    -5.2121e-01  7.3238e-02  -7.1166 1.114e-12 ***
## factor(STATE)54    -2.5596e-01  1.2365e-01  -2.0700 0.0384595 *
## factor(STATE)55     4.3741e-01  6.1919e-02   7.0643 1.626e-12 ***
## HISP_REF            6.5743e-01  2.8505e-02  23.0638 < 2.2e-16 ***
## factor(HIGH_EDU)10 -4.3982e-02  2.3902e-01  -0.1840 0.8540070
## factor(HIGH_EDU)11 -5.1140e-01  2.3229e-01  -2.2016 0.0276972 *
## factor(HIGH_EDU)12  6.0538e-02  2.2937e-01   0.2639 0.7918313
## factor(HIGH_EDU)13  3.0737e-01  2.2942e-01   1.3398 0.1803235
## factor(HIGH_EDU)14  6.0074e-01  2.3002e-01   2.6117 0.0090107 **
```

```
## factor(HIGH_EDU)15  8.3717e-01  2.2931e-01   3.6508 0.0002615 ***
## factor(HIGH_EDU)16  9.8623e-01  2.2944e-01   4.2984 1.722e-05 ***
## QSQ118              8.0429e-04  1.1502e-03   0.6993 0.4843819
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Models with factored Quarters
vehinsmodel1 <- lm(log(AdjVEHINSPQ) ~ HOUSB+AGE_REF+factor(CUTENURE)+FAM_SIZE
+TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+VEHQ+GASMOPQ+VEHFINPQ+MAIN
RPPQ+VRNTLOPQ+PUBTRAPQ+factor(STATE)+EVEHPURP+HISP_REF+factor(HIGH_EDU)+facto
r(QSQ118), data=ces_capstone)
summary(vehinsmodel1)
```

```
##
## Call:
## lm(formula = log(AdjVEHINSPQ) ~ HOUSB + AGE_REF + factor(CUTENURE) +
##     FAM_SIZE + TINCHRS + factor(MARITAL1) + factor(REF_RACE) +
##     SEX_REF + VEHQ + GASMOPQ + VEHFINPQ + MAINRPPQ + VRNTLOPQ +
##     PUBTRAPQ + factor(STATE) + EVEHPURP + HISP_REF + factor(HIGH_EDU) +
##     factor(QSQ118), data = ces_capstone)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.5511  -2.5985  -0.1596   2.7743   7.3512
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.159e+00  3.120e-01   3.713 0.000205 ***
## HOUSB                2.113e+00  5.625e-02  37.564  < 2e-16 ***
## AGE_REF             -2.175e-03  8.377e-04  -2.597 0.009419 **
## factor(CUTENURE)2    2.125e-02  2.667e-02   0.797 0.425644
## factor(CUTENURE)4   -1.970e-01  2.572e-02  -7.662 1.85e-14 ***
## factor(CUTENURE)5   -2.439e-01  8.779e-02  -2.779 0.005457 **
## factor(CUTENURE)6   -1.043e+00  2.299e-01  -4.535 5.76e-06 ***
## FAM_SIZE            -1.497e-03  8.339e-03  -0.179 0.857549
## TINCHRS              4.091e-03  4.320e-04   9.469  < 2e-16 ***
## factor(MARITAL1)2    8.308e-02  3.940e-02   2.108 0.034991 *
## factor(MARITAL1)3    1.611e-01  3.140e-02   5.132 2.87e-07 ***
## factor(MARITAL1)4   -1.312e-01  6.843e-02  -1.917 0.055237 .
## factor(MARITAL1)5    7.510e-04  3.250e-02   0.023 0.981565
## factor(REF_RACE)2   -3.797e-02  3.265e-02  -1.163 0.244775
## factor(REF_RACE)3    5.320e-02  1.331e-01   0.400 0.689465
## factor(REF_RACE)4   -2.059e-01  4.189e-02  -4.914 8.94e-07 ***
## factor(REF_RACE)5    4.851e-02  1.334e-01   0.364 0.716200
## factor(REF_RACE)6   -3.661e-02  7.415e-02  -0.494 0.621446
## SEX_REF             -1.883e-03  1.945e-02  -0.097 0.922902
## VEHQ                 1.794e-01  7.849e-03  22.853  < 2e-16 ***
## GASMOPQ              1.336e-03  2.430e-05  54.973  < 2e-16 ***
## VEHFINPQ             2.113e-03  8.653e-05  24.416  < 2e-16 ***
## MAINRPPQ             2.509e-04  1.886e-05  13.302  < 2e-16 ***
```

```
## VRNTLOPQ             3.428e-04  1.847e-05  18.566  < 2e-16 ***
## PUBTRAPQ             2.532e-06  1.374e-05   0.184 0.853756
## factor(STATE)2      -1.395e-01  9.424e-02  -1.480 0.138751
## factor(STATE)4      -2.178e-02  8.863e-02  -0.246 0.805871
## factor(STATE)6      -1.546e-01  6.855e-02  -2.255 0.024138 *
## factor(STATE)8       3.243e-01  9.532e-02   3.402 0.000669 ***
## factor(STATE)9       1.061e-01  8.866e-02   1.196 0.231553
## factor(STATE)10      7.412e-01  2.514e-01   2.949 0.003191 **
## factor(STATE)11     -3.286e-01  1.993e-01  -1.649 0.099104 .
## factor(STATE)12     -3.678e-02  7.306e-02  -0.503 0.614692
## factor(STATE)13      1.830e-01  9.183e-02   1.993 0.046241 *
## factor(STATE)15     -7.284e-02  9.719e-02  -0.749 0.453574
## factor(STATE)17     -2.223e-01  8.264e-02  -2.690 0.007153 **
## factor(STATE)18     -9.103e-01  1.497e-01  -6.081 1.20e-09 ***
## factor(STATE)19     -5.926e-01  2.473e-01  -2.397 0.016551 *
## factor(STATE)20     -1.073e-01  1.081e-01  -0.992 0.321189
## factor(STATE)21      2.163e-01  1.029e-01   2.103 0.035509 *
## factor(STATE)22      2.593e-01  9.866e-02   2.628 0.008583 **
## factor(STATE)24      1.288e-01  8.919e-02   1.444 0.148667
## factor(STATE)25     -2.012e-01  8.223e-02  -2.447 0.014395 *
## factor(STATE)26     -1.122e-01  8.297e-02  -1.352 0.176257
## factor(STATE)27     -1.367e-01  9.211e-02  -1.484 0.137788
## factor(STATE)28     -2.140e-01  1.381e-01  -1.550 0.121116
## factor(STATE)29     -5.477e-02  9.903e-02  -0.553 0.580204
## factor(STATE)31      1.232e-01  8.825e-02   1.396 0.162765
## factor(STATE)32      1.959e-01  1.070e-01   1.831 0.067141 .
## factor(STATE)33     -9.013e-02  2.207e-01  -0.408 0.682977
## factor(STATE)34     -6.835e-02  8.959e-02  -0.763 0.445485
## factor(STATE)36     -5.416e-01  7.364e-02  -7.355 1.93e-13 ***
## factor(STATE)37      1.015e-02  7.516e-02   0.135 0.892550
## factor(STATE)39     -1.529e-02  7.724e-02  -0.198 0.843063
## factor(STATE)40      2.760e-01  1.223e-01   2.257 0.024023 *
## factor(STATE)41      4.523e-02  9.228e-02   0.490 0.624000
## factor(STATE)42      3.575e-03  7.866e-02   0.045 0.963748
## factor(STATE)45      1.147e-01  1.007e-01   1.138 0.255004
## factor(STATE)46     -1.651e-01  1.616e-01  -1.022 0.306909
## factor(STATE)47     -8.421e-03  1.092e-01  -0.077 0.938550
## factor(STATE)48      1.485e-01  7.290e-02   2.037 0.041669 *
## factor(STATE)49      7.391e-02  8.188e-02   0.903 0.366702
## factor(STATE)51     -1.727e-01  8.807e-02  -1.961 0.049840 *
## factor(STATE)53     -1.388e-01  8.815e-02  -1.574 0.115415
## factor(STATE)54     -1.935e-01  1.524e-01  -1.270 0.204227
## factor(STATE)55     -1.741e-01  8.328e-02  -2.091 0.036570 *
## EVEHPURP             2.648e-05  3.196e-06   8.285  < 2e-16 ***
## HISP_REF            -3.954e-02  3.061e-02  -1.292 0.196441
## factor(HIGH_EDU)10 -1.402e-01  3.080e-01  -0.455 0.649005
## factor(HIGH_EDU)11  1.239e-01  2.981e-01   0.416 0.677750
## factor(HIGH_EDU)12  4.099e-01  2.951e-01   1.389 0.164854
## factor(HIGH_EDU)13  5.742e-01  2.951e-01   1.945 0.051729 .
## factor(HIGH_EDU)14  6.142e-01  2.958e-01   2.076 0.037888 *
```

```
## factor(HIGH_EDU)15  5.903e-01  2.950e-01   2.001 0.045413 *
## factor(HIGH_EDU)16  6.532e-01  2.953e-01   2.212 0.026974 *
## factor(QSQ118)1     1.608e-02  6.256e-02   0.257 0.797119
## factor(QSQ118)2    -3.106e-02  6.324e-02  -0.491 0.623265
## factor(QSQ118)3    -1.031e-01  6.194e-02  -1.665 0.095895 .
## factor(QSQ118)4     3.524e-02  6.233e-02   0.565 0.571860
## factor(QSQ118)5     4.502e-02  6.282e-02   0.717 0.473576
## factor(QSQ118)6     6.285e-03  6.293e-02   0.100 0.920451
## factor(QSQ118)7     6.198e-02  6.276e-02   0.988 0.323342
## factor(QSQ118)8     2.274e-01  6.378e-02   3.566 0.000363 ***
## factor(QSQ118)9     2.238e-01  6.346e-02   3.527 0.000420 ***
## factor(QSQ118)10    2.419e-01  6.312e-02   3.833 0.000127 ***
## factor(QSQ118)11    1.115e-01  6.314e-02   1.766 0.077317 .
## factor(QSQ118)12    1.440e-01  6.308e-02   2.283 0.022410 *
## factor(QSQ118)13    4.804e-02  6.308e-02   0.762 0.446304
## factor(QSQ118)14   -6.332e-02  6.387e-02  -0.991 0.321519
## factor(QSQ118)15   -9.921e-02  6.282e-02  -1.579 0.114266
## factor(QSQ118)16   -2.025e-01  6.304e-02  -3.212 0.001317 **
## factor(QSQ118)17   -3.855e-01  6.501e-02  -5.930 3.04e-09 ***
## factor(QSQ118)18   -1.316e-01  6.471e-02  -2.034 0.041972 *
## factor(QSQ118)19    1.217e-02  6.358e-02   0.191 0.848163
## factor(QSQ118)20    5.943e-02  6.403e-02   0.928 0.353349
## factor(QSQ118)21    2.454e-02  6.409e-02   0.383 0.701841
## factor(QSQ118)22   -9.988e-02  6.444e-02  -1.550 0.121124
## factor(QSQ118)23    2.929e-02  6.431e-02   0.455 0.648752
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.861 on 91893 degrees of freedom
## Multiple R-squared:  0.1285, Adjusted R-squared:  0.1276
## F-statistic: 139.7 on 97 and 91893 DF,  p-value: < 2.2e-16
```

```
lifinsmodel1<- lm(log(AdjLIFINSPQ) ~ HOUSB+AGE_REF+factor(CUTENURE)+FAM_SIZE+
TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+ALCBEVPQ+TOBACCPQ+factor(ST
ATE)+HISP_REF+factor(HIGH_EDU)+factor(QSQ118), data = ces_capstone)
summary(lifinsmodel1)
```

```
##
## Call:
## lm(formula = log(AdjLIFINSPQ) ~ HOUSB + AGE_REF + factor(CUTENURE) +
##     FAM_SIZE + TINCHRS + factor(MARITAL1) + factor(REF_RACE) +
##     SEX_REF + ALCBEVPQ + TOBACCPQ + factor(STATE) + HISP_REF +
##     factor(HIGH_EDU) + factor(QSQ118), data = ces_capstone)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.6203 -1.4821 -0.8975  0.3347  9.1334
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)         9.309e-03  2.443e-01   0.038 0.969610
## HOUSB               4.832e-01  4.354e-02  11.097  < 2e-16 ***
## AGE_REF             1.479e-02  6.548e-04  22.579  < 2e-16 ***
## factor(CUTENURE)2  -2.970e-01  2.084e-02 -14.249  < 2e-16 ***
## factor(CUTENURE)4  -4.923e-01  1.985e-02 -24.794  < 2e-16 ***
## factor(CUTENURE)5  -5.122e-01  6.870e-02  -7.457 8.95e-14 ***
## factor(CUTENURE)6  -3.734e-01  1.801e-01  -2.074 0.038094 *
## FAM_SIZE            6.729e-02  6.395e-03  10.521  < 2e-16 ***
## TINCHRS             7.167e-03  3.348e-04  21.410  < 2e-16 ***
## factor(MARITAL1)2  -2.649e-01  3.074e-02  -8.618  < 2e-16 ***
## factor(MARITAL1)3  -2.564e-01  2.452e-02 -10.459  < 2e-16 ***
## factor(MARITAL1)4  -3.467e-01  5.359e-02  -6.470 9.86e-11 ***
## factor(MARITAL1)5  -3.563e-01  2.540e-02 -14.029  < 2e-16 ***
## factor(REF_RACE)2   3.868e-01  2.558e-02  15.122  < 2e-16 ***
## factor(REF_RACE)3   2.782e-01  1.043e-01   2.668 0.007638 **
## factor(REF_RACE)4  -1.537e-01  3.280e-02  -4.687 2.78e-06 ***
## factor(REF_RACE)5   5.860e-02  1.046e-01   0.560 0.575173
## factor(REF_RACE)6   2.163e-01  5.809e-02   3.724 0.000196 ***
## SEX_REF             1.083e-01  1.523e-02   7.109 1.18e-12 ***
## ALCBEVPQ            8.486e-04  3.337e-05  25.433  < 2e-16 ***
## TOBACCPQ            3.389e-05  3.879e-05   0.874 0.382286
## factor(STATE)2     -6.083e-01  7.378e-02  -8.246  < 2e-16 ***
## factor(STATE)4     -6.729e-01  6.942e-02  -9.692  < 2e-16 ***
## factor(STATE)6     -5.844e-01  5.370e-02 -10.882  < 2e-16 ***
## factor(STATE)8     -7.672e-01  7.467e-02 -10.274  < 2e-16 ***
## factor(STATE)9     -3.326e-01  6.944e-02  -4.790 1.67e-06 ***
## factor(STATE)10    -4.456e-01  1.969e-01  -2.263 0.023652 *
## factor(STATE)11    -1.322e-01  1.560e-01  -0.847 0.396720
## factor(STATE)12    -7.702e-01  5.718e-02 -13.470  < 2e-16 ***
## factor(STATE)13    -4.272e-01  7.194e-02  -5.938 2.90e-09 ***
## factor(STATE)15    -7.414e-01  7.614e-02  -9.738  < 2e-16 ***
## factor(STATE)17    -4.053e-01  6.472e-02  -6.263 3.79e-10 ***
## factor(STATE)18    -8.931e-01  1.173e-01  -7.616 2.65e-14 ***
## factor(STATE)19    -3.832e-01  1.937e-01  -1.978 0.047892 *
## factor(STATE)20    -6.528e-02  8.469e-02  -0.771 0.440817
## factor(STATE)21    -2.837e-01  8.061e-02  -3.520 0.000431 ***
## factor(STATE)22    -9.299e-01  7.724e-02 -12.038  < 2e-16 ***
## factor(STATE)24    -4.171e-01  6.987e-02  -5.970 2.38e-09 ***
## factor(STATE)25    -4.580e-01  6.437e-02  -7.115 1.13e-12 ***
## factor(STATE)26    -3.387e-01  6.491e-02  -5.218 1.81e-07 ***
## factor(STATE)27    -2.378e-01  7.217e-02  -3.295 0.000983 ***
## factor(STATE)28     5.817e-03  1.082e-01   0.054 0.957122
## factor(STATE)29    -2.141e-02  7.758e-02  -0.276 0.782541
## factor(STATE)31    -2.467e-01  6.912e-02  -3.569 0.000358 ***
## factor(STATE)32    -3.502e-01  8.382e-02  -4.178 2.95e-05 ***
## factor(STATE)33     1.963e-01  1.729e-01   1.136 0.256155
## factor(STATE)34    -4.352e-01  7.006e-02  -6.211 5.29e-10 ***
## factor(STATE)36    -6.252e-01  5.747e-02 -10.880  < 2e-16 ***
## factor(STATE)37    -4.701e-01  5.888e-02  -7.984 1.43e-15 ***
## factor(STATE)39    -3.852e-01  6.051e-02  -6.366 1.94e-10 ***
```

```
## factor(STATE)40    -6.987e-02  9.583e-02  -0.729 0.465940
## factor(STATE)41    -4.719e-01  7.230e-02  -6.527 6.76e-11 ***
## factor(STATE)42    -4.048e-01  6.157e-02  -6.575 4.89e-11 ***
## factor(STATE)45    -2.385e-01  7.892e-02  -3.022 0.002513 **
## factor(STATE)46    -7.208e-01  1.266e-01  -5.694 1.25e-08 ***
## factor(STATE)47     1.493e-01  8.558e-02   1.744 0.081135 .
## factor(STATE)48    -5.264e-01  5.709e-02  -9.220  < 2e-16 ***
## factor(STATE)49    -5.185e-01  6.419e-02  -8.078 6.69e-16 ***
## factor(STATE)51    -2.392e-01  6.900e-02  -3.466 0.000528 ***
## factor(STATE)53    -4.388e-01  6.908e-02  -6.352 2.13e-10 ***
## factor(STATE)54    -1.259e-01  1.194e-01  -1.055 0.291631
## factor(STATE)55    -2.101e-01  6.524e-02  -3.220 0.001281 **
## HISP_REF            2.611e-01  2.405e-02  10.859  < 2e-16 ***
## factor(HIGH_EDU)10 -1.581e-03  2.413e-01  -0.007 0.994775
## factor(HIGH_EDU)11  1.403e-01  2.336e-01   0.601 0.547990
## factor(HIGH_EDU)12  1.945e-01  2.312e-01   0.841 0.400123
## factor(HIGH_EDU)13  3.281e-01  2.312e-01   1.419 0.155948
## factor(HIGH_EDU)14  4.742e-01  2.318e-01   2.046 0.040776 *
## factor(HIGH_EDU)15  4.584e-01  2.311e-01   1.983 0.047363 *
## factor(HIGH_EDU)16  6.946e-01  2.313e-01   3.002 0.002679 **
## factor(QSQ118)1     5.579e-02  4.900e-02   1.138 0.254956
## factor(QSQ118)2     9.303e-03  4.954e-02   0.188 0.851040
## factor(QSQ118)3    -9.854e-03  4.853e-02  -0.203 0.839099
## factor(QSQ118)4     5.312e-03  4.883e-02   0.109 0.913379
## factor(QSQ118)5    -6.753e-04  4.921e-02  -0.014 0.989052
## factor(QSQ118)6    -3.699e-02  4.930e-02  -0.750 0.453112
## factor(QSQ118)7     2.824e-02  4.917e-02   0.574 0.565712
## factor(QSQ118)8     6.570e-02  4.993e-02   1.316 0.188225
## factor(QSQ118)9    -7.944e-02  4.966e-02  -1.600 0.109664
## factor(QSQ118)10   -4.365e-02  4.941e-02  -0.883 0.376974
## factor(QSQ118)11   -7.652e-02  4.942e-02  -1.548 0.121577
## factor(QSQ118)12   -1.356e-01  4.940e-02  -2.746 0.006037 **
## factor(QSQ118)13   -9.955e-02  4.942e-02  -2.014 0.043964 *
## factor(QSQ118)14   -8.263e-02  5.003e-02  -1.652 0.098639 .
## factor(QSQ118)15   -1.334e-01  4.920e-02  -2.711 0.006715 **
## factor(QSQ118)16   -1.275e-01  4.930e-02  -2.585 0.009727 **
## factor(QSQ118)17   -3.384e-02  5.069e-02  -0.668 0.504383
## factor(QSQ118)18   -1.386e-02  5.060e-02  -0.274 0.784097
## factor(QSQ118)19    7.159e-03  4.977e-02   0.144 0.885621
## factor(QSQ118)20    8.755e-02  5.014e-02   1.746 0.080774 .
## factor(QSQ118)21    3.732e-02  5.015e-02   0.744 0.456790
## factor(QSQ118)22    2.371e-02  5.043e-02   0.470 0.638262
## factor(QSQ118)23    1.184e-02  5.033e-02   0.235 0.814031
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.241 on 91898 degrees of freedom
## Multiple R-squared:  0.08325,    Adjusted R-squared:  0.08234
## F-statistic: 90.71 on 92 and 91898 DF,  p-value: < 2.2e-16
```

```
perinsmodel1 <- lm(log(AdjPERINSPQ) ~ HOUSB+AGE_REF+factor(CUTENURE)+FAM_SIZE
+TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+factor(STATE)+HISP_REF+fac
tor(HIGH_EDU)+factor(QSQ118), data= ces_capstone)
summary(perinsmodel1)
```

```
##
## Call:
## lm(formula = log(AdjPERINSPQ) ~ HOUSB + AGE_REF + factor(CUTENURE) +
##     FAM_SIZE + TINCHRS + factor(MARITAL1) + factor(REF_RACE) +
##     SEX_REF + factor(STATE) + HISP_REF + factor(HIGH_EDU) + factor(QSQ11
8),
##     data = ces_capstone)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.7440  -1.2419   0.1627   1.2812   7.8492
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          3.0938498  0.2102097  14.718  < 2e-16 ***
## HOUSB               -0.7415390  0.0374214 -19.816  < 2e-16 ***
## AGE_REF             -0.0210589  0.0005631 -37.396  < 2e-16 ***
## factor(CUTENURE)2   -0.4154486  0.0179310 -23.169  < 2e-16 ***
## factor(CUTENURE)4   -0.4953020  0.0170664 -29.022  < 2e-16 ***
## factor(CUTENURE)5   -0.9958021  0.0590955 -16.851  < 2e-16 ***
## factor(CUTENURE)6   -2.3289810  0.1549084 -15.035  < 2e-16 ***
## FAM_SIZE             0.1962193  0.0054902  35.740  < 2e-16 ***
## TINCHRS              0.0482212  0.0002870 168.018  < 2e-16 ***
## factor(MARITAL1)2    0.1159628  0.0264405   4.386 1.16e-05 ***
## factor(MARITAL1)3    0.7129525  0.0210842  33.815  < 2e-16 ***
## factor(MARITAL1)4    0.8205539  0.0460979  17.800  < 2e-16 ***
## factor(MARITAL1)5    0.7882387  0.0218499  36.075  < 2e-16 ***
## factor(REF_RACE)2    0.0928064  0.0219624   4.226 2.38e-05 ***
## factor(REF_RACE)3   -0.1355013  0.0897234  -1.510 0.130993
## factor(REF_RACE)4    0.1594221  0.0281529   5.663 1.49e-08 ***
## factor(REF_RACE)5    0.2272858  0.0899444   2.527 0.011507 *
## factor(REF_RACE)6    0.0899640  0.0499717   1.800 0.071816 .
## SEX_REF             -0.0896588  0.0130858  -6.852 7.35e-12 ***
## factor(STATE)2       0.2753524  0.0634452   4.340 1.43e-05 ***
## factor(STATE)4      -0.1236613  0.0597171  -2.071 0.038382 *
## factor(STATE)6       0.2460819  0.0461653   5.330 9.82e-08 ***
## factor(STATE)8       0.3002048  0.0642040   4.676 2.93e-06 ***
## factor(STATE)9       0.1911741  0.0597332   3.200 0.001373 **
## factor(STATE)10      0.5911742  0.1694122   3.490 0.000484 ***
## factor(STATE)11      0.3468817  0.1341857   2.585 0.009737 **
## factor(STATE)12     -0.1207328  0.0491874  -2.455 0.014108 *
## factor(STATE)13      0.1242093  0.0618833   2.007 0.044736 *
## factor(STATE)15      0.2777765  0.0654847   4.242 2.22e-05 ***
## factor(STATE)17      0.1136857  0.0556631   2.042 0.041116 *
## factor(STATE)18     -0.0192489  0.1008910  -0.191 0.848691
```

```
## factor(STATE)19     0.0454511  0.1666570   0.273 0.785067
## factor(STATE)20    -0.0289679  0.0728561  -0.398 0.690922
## factor(STATE)21     0.0425808  0.0693463   0.614 0.539196
## factor(STATE)22    -0.2071335  0.0664503  -3.117 0.001827 **
## factor(STATE)24     0.3963689  0.0600926   6.596 4.25e-11 ***
## factor(STATE)25     0.1489743  0.0553611   2.691 0.007126 **
## factor(STATE)26     0.0063782  0.0558446   0.114 0.909069
## factor(STATE)27     0.1288422  0.0620600   2.076 0.037888 *
## factor(STATE)28    -0.1849424  0.0930709  -1.987 0.046913 *
## factor(STATE)29     0.1329979  0.0667344   1.993 0.046271 *
## factor(STATE)31     0.0157829  0.0594613   0.265 0.790678
## factor(STATE)32     0.1268940  0.0721047   1.760 0.078435 .
## factor(STATE)33     0.3617192  0.1487455   2.432 0.015026 *
## factor(STATE)34     0.2472829  0.0602635   4.103 4.08e-05 ***
## factor(STATE)36     0.1591441  0.0494087   3.221 0.001278 **
## factor(STATE)37    -0.0057987  0.0506523  -0.114 0.908857
## factor(STATE)39     0.1399958  0.0520459   2.690 0.007150 **
## factor(STATE)40     0.2664586  0.0824453   3.232 0.001230 **
## factor(STATE)41     0.1708220  0.0621718   2.748 0.006005 **
## factor(STATE)42     0.1215885  0.0529633   2.296 0.021695 *
## factor(STATE)45    -0.1038318  0.0678924  -1.529 0.126179
## factor(STATE)46    -0.2755843  0.1089100  -2.530 0.011395 *
## factor(STATE)47     0.1363376  0.0736260   1.852 0.064063 .
## factor(STATE)48     0.1028995  0.0491077   2.095 0.036139 *
## factor(STATE)49     0.0283262  0.0551856   0.513 0.607749
## factor(STATE)51     0.2704443  0.0593512   4.557 5.20e-06 ***
## factor(STATE)53     0.3646087  0.0593854   6.140 8.30e-10 ***
## factor(STATE)54     0.0896276  0.1026905   0.873 0.382778
## factor(STATE)55     0.1138208  0.0561233   2.028 0.042558 *
## HISP_REF           -0.1681965  0.0206027  -8.164 3.29e-16 ***
## factor(HIGH_EDU)10 -0.0177258  0.2076278  -0.085 0.931965
## factor(HIGH_EDU)11  0.3891043  0.2009196   1.937 0.052795 .
## factor(HIGH_EDU)12  0.8515200  0.1988944   4.281 1.86e-05 ***
## factor(HIGH_EDU)13  1.1049255  0.1989129   5.555 2.79e-08 ***
## factor(HIGH_EDU)14  1.3755890  0.1993872   6.899 5.27e-12 ***
## factor(HIGH_EDU)15  1.5810320  0.1988485   7.951 1.87e-15 ***
## factor(HIGH_EDU)16  1.7239731  0.1990146   8.663  < 2e-16 ***
## factor(QSQ118)1     0.2644961  0.0421559   6.274 3.53e-10 ***
## factor(QSQ118)2     0.0171121  0.0426174   0.402 0.688031
## factor(QSQ118)3     0.0030637  0.0417510   0.073 0.941504
## factor(QSQ118)4    -0.0014166  0.0420125  -0.034 0.973101
## factor(QSQ118)5     0.0147753  0.0423384   0.349 0.727104
## factor(QSQ118)6     0.0316910  0.0424158   0.747 0.454974
## factor(QSQ118)7     0.0540623  0.0422969   1.278 0.201196
## factor(QSQ118)8     0.0554520  0.0429538   1.291 0.196718
## factor(QSQ118)9    -0.0243889  0.0427183  -0.571 0.568052
## factor(QSQ118)10    0.0312811  0.0425036   0.736 0.461756
## factor(QSQ118)11    0.0304798  0.0425173   0.717 0.473450
## factor(QSQ118)12    0.0182008  0.0424995   0.428 0.668463
## factor(QSQ118)13    0.0294080  0.0425150   0.692 0.489122
```

```
## factor(QSQ118)14    0.0301644  0.0430398   0.701 0.483399
## factor(QSQ118)15    0.0368559  0.0423247   0.871 0.383872
## factor(QSQ118)16    0.0763577  0.0424066   1.801 0.071768 .
## factor(QSQ118)17    0.0843141  0.0435997   1.934 0.053138 .
## factor(QSQ118)18    0.1331869  0.0435203   3.060 0.002211 **
## factor(QSQ118)19    0.1944352  0.0428087   4.542 5.58e-06 ***
## factor(QSQ118)20    0.2423340  0.0431250   5.619 1.92e-08 ***
## factor(QSQ118)21    0.1799838  0.0431415   4.172 3.02e-05 ***
## factor(QSQ118)22    0.1714057  0.0433750   3.952 7.77e-05 ***
## factor(QSQ118)23    0.2227793  0.0432949   5.146 2.67e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.928 on 91900 degrees of freedom
## Multiple R-squared:  0.543,  Adjusted R-squared:  0.5425
## F-statistic:  1213 on 90 and 91900 DF,  p-value: < 2.2e-16
```

```
hlthinsmodel1 <-lm(log(AdjHLTHINPQ) ~ HOUSB+AGE_REF+factor(CUTENURE)+FAM_SIZE
+TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+FOODPQ+ALCBEVPQ+MEDSRVPQ+P
REDRGPQ+MEDSUPPQ+TOBACCPQ+factor(STATE)+HISP_REF+factor(HIGH_EDU)+factor(QSQ1
18),data=ces_capstone)
summary(hlthinsmodel1)
```

```
##
## Call:
## lm(formula = log(AdjHLTHINPQ) ~ HOUSB + AGE_REF + factor(CUTENURE) +
##     FAM_SIZE + TINCHRS + factor(MARITAL1) + factor(REF_RACE) +
##     SEX_REF + FOODPQ + ALCBEVPQ + MEDSRVPQ + PREDRGPQ + MEDSUPPQ +
##     TOBACCPQ + factor(STATE) + HISP_REF + factor(HIGH_EDU) +
##     factor(QSQ118), data = ces_capstone)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.2432 -0.8872  0.5636  1.5549  7.0639
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)         1.531e+00  2.548e-01   6.008 1.89e-09 ***
## HOUSB              -5.382e-01  4.592e-02 -11.719  < 2e-16 ***
## AGE_REF             5.206e-02  6.837e-04  76.141  < 2e-16 ***
## factor(CUTENURE)2  -2.601e-01  2.175e-02 -11.960  < 2e-16 ***
## factor(CUTENURE)4  -6.770e-01  2.074e-02 -32.645  < 2e-16 ***
## factor(CUTENURE)5  -1.317e+00  7.164e-02 -18.377  < 2e-16 ***
## factor(CUTENURE)6  -2.350e+00  1.878e-01 -12.512  < 2e-16 ***
## FAM_SIZE           -1.933e-02  6.846e-03  -2.824 0.004751 **
## TINCHRS             4.916e-04  3.510e-04   1.401 0.161346
## factor(MARITAL1)2  -3.007e-01  3.208e-02  -9.373  < 2e-16 ***
## factor(MARITAL1)3  -5.047e-01  2.558e-02 -19.729  < 2e-16 ***
## factor(MARITAL1)4  -8.270e-01  5.589e-02 -14.797  < 2e-16 ***
## factor(MARITAL1)5  -4.783e-01  2.650e-02 -18.049  < 2e-16 ***
```

```
## factor(REF_RACE)2  -1.405e-01  2.669e-02  -5.263 1.42e-07 ***
## factor(REF_RACE)3  -2.316e-01  1.088e-01  -2.130 0.033214 *
## factor(REF_RACE)4  -3.565e-02  3.422e-02  -1.042 0.297492
## factor(REF_RACE)5   1.643e-01  1.090e-01   1.506 0.131979
## factor(REF_RACE)6   9.961e-03  6.058e-02   0.164 0.869397
## SEX_REF             6.857e-02  1.589e-02   4.316 1.59e-05 ***
## FOODPQ              1.770e-04  6.142e-06  28.809  < 2e-16 ***
## ALCBEVPQ            1.830e-04  3.741e-05   4.891 1.00e-06 ***
## MEDSRVPQ            1.079e-04  9.731e-06  11.090  < 2e-16 ***
## PREDRGPQ           -5.435e-02  1.875e-01  -0.290 0.771929
## MEDSUPPQ            2.429e-04  4.211e-05   5.768 8.05e-09 ***
## TOBACCPQ           -2.368e-05  4.049e-05  -0.585 0.558596
## factor(STATE)2     -2.157e-01  7.695e-02  -2.804 0.005053 **
## factor(STATE)4     -2.682e-01  7.241e-02  -3.704 0.000213 ***
## factor(STATE)6     -4.301e-01  5.600e-02  -7.680 1.61e-14 ***
## factor(STATE)8     -4.034e-02  7.788e-02  -0.518 0.604453
## factor(STATE)9      1.567e-01  7.243e-02   2.164 0.030482 *
## factor(STATE)10    -4.630e-01  2.054e-01  -2.254 0.024185 *
## factor(STATE)11    -5.671e-02  1.627e-01  -0.349 0.727457
## factor(STATE)12    -4.004e-02  5.964e-02  -0.671 0.502038
## factor(STATE)13     2.332e-01  7.503e-02   3.108 0.001881 **
## factor(STATE)15    -6.265e-01  7.942e-02  -7.888 3.11e-15 ***
## factor(STATE)17     1.457e-01  6.751e-02   2.159 0.030874 *
## factor(STATE)18    -2.741e-01  1.223e-01  -2.241 0.025056 *
## factor(STATE)19    -2.287e-01  2.021e-01  -1.132 0.257785
## factor(STATE)20    -6.045e-02  8.833e-02  -0.684 0.493735
## factor(STATE)21    -1.576e-01  8.406e-02  -1.875 0.060849 .
## factor(STATE)22    -3.864e-01  8.057e-02  -4.796 1.62e-06 ***
## factor(STATE)24     1.978e-01  7.287e-02   2.715 0.006630 **
## factor(STATE)25     1.078e-01  6.713e-02   1.606 0.108355
## factor(STATE)26    -3.021e-01  6.771e-02  -4.461 8.15e-06 ***
## factor(STATE)27     1.609e-01  7.528e-02   2.137 0.032592 *
## factor(STATE)28    -1.183e-02  1.128e-01  -0.105 0.916478
## factor(STATE)29     1.568e-01  8.092e-02   1.938 0.052632 .
## factor(STATE)31     1.508e-01  7.212e-02   2.091 0.036503 *
## factor(STATE)32    -3.518e-01  8.742e-02  -4.024 5.72e-05 ***
## factor(STATE)33     3.979e-01  1.803e-01   2.207 0.027336 *
## factor(STATE)34    -1.888e-01  7.307e-02  -2.584 0.009779 **
## factor(STATE)36    -5.233e-01  5.993e-02  -8.732  < 2e-16 ***
## factor(STATE)37    -8.533e-02  6.141e-02  -1.390 0.164641
## factor(STATE)39    -4.025e-03  6.311e-02  -0.064 0.949150
## factor(STATE)40    -7.775e-02  9.995e-02  -0.778 0.436631
## factor(STATE)41    -3.145e-01  7.541e-02  -4.171 3.04e-05 ***
## factor(STATE)42    -8.030e-03  6.423e-02  -0.125 0.900509
## factor(STATE)45     1.920e-02  8.231e-02   0.233 0.815570
## factor(STATE)46     1.148e-01  1.321e-01   0.869 0.384790
## factor(STATE)47    -3.834e-02  8.926e-02  -0.429 0.667562
## factor(STATE)48    -3.501e-02  5.955e-02  -0.588 0.556557
## factor(STATE)49    -2.784e-01  6.697e-02  -4.158 3.22e-05 ***
## factor(STATE)51     1.214e-01  7.197e-02   1.686 0.091729 .
```

```
## factor(STATE)53    -5.199e-01  7.205e-02  -7.216 5.39e-13 ***
## factor(STATE)54    -2.552e-01  1.245e-01  -2.049 0.040443 *
## factor(STATE)55     4.385e-01  6.806e-02   6.443 1.18e-10 ***
## HISP_REF            6.577e-01  2.508e-02  26.220  < 2e-16 ***
## factor(HIGH_EDU)10 -4.444e-02  2.517e-01  -0.177 0.859847
## factor(HIGH_EDU)11 -5.139e-01  2.436e-01  -2.110 0.034887 *
## factor(HIGH_EDU)12  5.859e-02  2.411e-01   0.243 0.808006
## factor(HIGH_EDU)13  3.048e-01  2.412e-01   1.264 0.206200
## factor(HIGH_EDU)14  5.976e-01  2.417e-01   2.472 0.013423 *
## factor(HIGH_EDU)15  8.347e-01  2.411e-01   3.462 0.000536 ***
## factor(HIGH_EDU)16  9.833e-01  2.413e-01   4.075 4.61e-05 ***
## factor(QSQ118)1    -4.991e-03  5.111e-02  -0.098 0.922204
## factor(QSQ118)2    -2.819e-02  5.167e-02  -0.546 0.585287
## factor(QSQ118)3    -1.311e-02  5.061e-02  -0.259 0.795547
## factor(QSQ118)4     4.945e-02  5.093e-02   0.971 0.331623
## factor(QSQ118)5     2.208e-02  5.133e-02   0.430 0.667067
## factor(QSQ118)6     4.456e-02  5.142e-02   0.866 0.386220
## factor(QSQ118)7     6.664e-02  5.128e-02   1.300 0.193739
## factor(QSQ118)8     4.360e-02  5.207e-02   0.837 0.402434
## factor(QSQ118)9     1.831e-02  5.179e-02   0.354 0.723696
## factor(QSQ118)10    4.107e-02  5.153e-02   0.797 0.425417
## factor(QSQ118)11    4.841e-02  5.155e-02   0.939 0.347608
## factor(QSQ118)12    1.892e-02  5.153e-02   0.367 0.713434
## factor(QSQ118)13   -1.879e-02  5.156e-02  -0.364 0.715571
## factor(QSQ118)14   -5.532e-02  5.220e-02  -1.060 0.289284
## factor(QSQ118)15   -2.024e-02  5.133e-02  -0.394 0.693314
## factor(QSQ118)16    1.596e-02  5.145e-02   0.310 0.756396
## factor(QSQ118)17    5.895e-03  5.294e-02   0.111 0.911337
## factor(QSQ118)18    6.662e-02  5.284e-02   1.261 0.207409
## factor(QSQ118)19    7.419e-02  5.197e-02   1.428 0.153409
## factor(QSQ118)20    2.884e-02  5.238e-02   0.551 0.581929
## factor(QSQ118)21   -6.972e-03  5.241e-02  -0.133 0.894186
## factor(QSQ118)22   -1.568e-02  5.271e-02  -0.297 0.766101
## factor(QSQ118)23    8.104e-02  5.263e-02   1.540 0.123602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.338 on 91894 degrees of freedom
## Multiple R-squared:  0.2482, Adjusted R-squared:  0.2474
## F-statistic:   316 on 96 and 91894 DF,  p-value: < 2.2e-16

#Robust Standard Errors of factored Quarter models
coeftest(vehinsmodel1, vcov. = vcovHC(vehinsmodel1, type = "HC3"))

##
## t test of coefficients:
##
##                      Estimate  Std. Error t value  Pr(>|t|)
## (Intercept)         1.1586e+00  2.8148e-01  4.1161 3.857e-05 ***
## HOUSB               2.1129e+00  5.7216e-02 36.9290 < 2.2e-16 ***
```

```
## AGE_REF             -2.1750e-03  8.3021e-04 -2.6199 0.0087979 **
## factor(CUTENURE)2    2.1246e-02  2.7226e-02  0.7803 0.4351897
## factor(CUTENURE)4   -1.9704e-01  2.5764e-02 -7.6479 2.062e-14 ***
## factor(CUTENURE)5   -2.4394e-01  8.4168e-02 -2.8983 0.0037528 **
## factor(CUTENURE)6   -1.0428e+00  1.5092e-01 -6.9095 4.897e-12 ***
## FAM_SIZE            -1.4968e-03  8.6095e-03 -0.1738 0.8619851
## TINCHRS              4.0907e-03  4.4628e-04  9.1662 < 2.2e-16 ***
## factor(MARITAL1)2    8.3078e-02  3.8669e-02  2.1484 0.0316809 *
## factor(MARITAL1)3    1.6114e-01  3.1226e-02  5.1605 2.468e-07 ***
## factor(MARITAL1)4   -1.3118e-01  6.6771e-02 -1.9646 0.0494661 *
## factor(MARITAL1)5    7.5095e-04  3.2299e-02  0.0232 0.9814511
## factor(REF_RACE)2   -3.7974e-02  3.2484e-02 -1.1690 0.2423967
## factor(REF_RACE)3    5.3197e-02  1.2979e-01  0.4099 0.6819158
## factor(REF_RACE)4   -2.0586e-01  4.1753e-02 -4.9306 8.214e-07 ***
## factor(REF_RACE)5    4.8513e-02  1.3765e-01  0.3524 0.7245016
## factor(REF_RACE)6   -3.6614e-02  7.2057e-02 -0.5081 0.6113654
## SEX_REF             -1.8826e-03  1.9529e-02 -0.0964 0.9232020
## VEHQ                 1.7936e-01  8.8711e-03 20.2186 < 2.2e-16 ***
## GASMOPQ              1.3359e-03  4.2913e-05 31.1313 < 2.2e-16 ***
## VEHFINPQ             2.1127e-03  9.5501e-05 22.1222 < 2.2e-16 ***
## MAINRPPQ             2.5091e-04  2.8362e-05  8.8469 < 2.2e-16 ***
## VRNTLOPQ             3.4284e-04  4.3383e-05  7.9026 2.761e-15 ***
## PUBTRAPQ             2.5319e-06  1.4750e-05  0.1717 0.8637106
## factor(STATE)2      -1.3951e-01  9.4069e-02 -1.4831 0.1380510
## factor(STATE)4      -2.1780e-02  8.8856e-02 -0.2451 0.8063635
## factor(STATE)6      -1.5459e-01  6.8645e-02 -2.2520 0.0243254 *
## factor(STATE)8       3.2429e-01  9.8181e-02  3.3030 0.0009570 ***
## factor(STATE)9       1.0607e-01  8.8871e-02  1.1935 0.2326621
## factor(STATE)10      7.4122e-01  2.5512e-01  2.9054 0.0036686 **
## factor(STATE)11     -3.2862e-01  1.8594e-01 -1.7673 0.0771750 .
## factor(STATE)12     -3.6779e-02  7.3668e-02 -0.4992 0.6176048
## factor(STATE)13      1.8304e-01  9.4394e-02  1.9391 0.0524910 .
## factor(STATE)15     -7.2843e-02  9.4936e-02 -0.7673 0.4429182
## factor(STATE)17     -2.2229e-01  8.1434e-02 -2.7297 0.0063409 **
## factor(STATE)18     -9.1035e-01  1.4297e-01 -6.3674 1.932e-10 ***
## factor(STATE)19     -5.9258e-01  2.4512e-01 -2.4175 0.0156279 *
## factor(STATE)20     -1.0726e-01  1.0776e-01 -0.9954 0.3195633
## factor(STATE)21      2.1633e-01  1.0306e-01  2.0990 0.0358189 *
## factor(STATE)22      2.5931e-01  1.0152e-01  2.5544 0.0106390 *
## factor(STATE)24      1.2882e-01  9.0088e-02  1.4299 0.1527423
## factor(STATE)25     -2.0124e-01  8.1538e-02 -2.4681 0.0135864 *
## factor(STATE)26     -1.1221e-01  8.5140e-02 -1.3179 0.1875389
## factor(STATE)27     -1.3670e-01  9.3277e-02 -1.4655 0.1427940
## factor(STATE)28     -2.1405e-01  1.3528e-01 -1.5823 0.1135876
## factor(STATE)29     -5.4770e-02  9.8997e-02 -0.5533 0.5800929
## factor(STATE)31      1.2318e-01  8.7453e-02  1.4085 0.1589735
## factor(STATE)32      1.9592e-01  1.0886e-01  1.7998 0.0718954 .
## factor(STATE)33     -9.0133e-02  2.1407e-01 -0.4210 0.6737296
## factor(STATE)34     -6.8353e-02  9.1152e-02 -0.7499 0.4533260
## factor(STATE)36     -5.4164e-01  7.3293e-02 -7.3901 1.479e-13 ***
```

```
## factor(STATE)37     1.0152e-02  7.4749e-02  0.1358 0.8919655
## factor(STATE)39    -1.5292e-02  7.6336e-02 -0.2003 0.8412252
## factor(STATE)40     2.7605e-01  1.2090e-01  2.2833 0.0224148 *
## factor(STATE)41     4.5233e-02  9.1936e-02  0.4920 0.6227158
## factor(STATE)42     3.5752e-03  7.8211e-02  0.0457 0.9635397
## factor(STATE)45     1.1467e-01  1.0115e-01  1.1336 0.2569668
## factor(STATE)46    -1.6512e-01  1.6036e-01 -1.0297 0.3031715
## factor(STATE)47    -8.4211e-03  1.0906e-01 -0.0772 0.9384519
## factor(STATE)48     1.4849e-01  7.3465e-02  2.0212 0.0432627 *
## factor(STATE)49     7.3913e-02  8.1692e-02  0.9048 0.3655816
## factor(STATE)51    -1.7274e-01  8.8461e-02 -1.9527 0.0508561 .
## factor(STATE)53    -1.3877e-01  8.7775e-02 -1.5810 0.1138833
## factor(STATE)54    -1.9348e-01  1.4867e-01 -1.3013 0.1931451
## factor(STATE)55    -1.7410e-01  8.2203e-02 -2.1179 0.0341824 *
## EVEHPURP            2.6482e-05  3.8208e-06  6.9311 4.203e-12 ***
## HISP_REF           -3.9539e-02  3.0687e-02 -1.2885 0.1975840
## factor(HIGH_EDU)10 -1.4020e-01  2.7252e-01 -0.5145 0.6069264
## factor(HIGH_EDU)11  1.2386e-01  2.6519e-01  0.4671 0.6404433
## factor(HIGH_EDU)12  4.0986e-01  2.6233e-01  1.5624 0.1181979
## factor(HIGH_EDU)13  5.7415e-01  2.6247e-01  2.1875 0.0287055 *
## factor(HIGH_EDU)14  6.1421e-01  2.6330e-01  2.3328 0.0196628 *
## factor(HIGH_EDU)15  5.9033e-01  2.6241e-01  2.2497 0.0244725 *
## factor(HIGH_EDU)16  6.5317e-01  2.6268e-01  2.4866 0.0129002 *
## factor(QSQ118)1     1.6082e-02  6.1853e-02  0.2600 0.7948606
## factor(QSQ118)2    -3.1063e-02  6.2433e-02 -0.4975 0.6188057
## factor(QSQ118)3    -1.0314e-01  6.1360e-02 -1.6810 0.0927733 .
## factor(QSQ118)4     3.5237e-02  6.1333e-02  0.5745 0.5656239
## factor(QSQ118)5     4.5020e-02  6.1801e-02  0.7285 0.4663313
## factor(QSQ118)6     6.2848e-03  6.2123e-02  0.1012 0.9194180
## factor(QSQ118)7     6.1980e-02  6.2050e-02  0.9989 0.3178581
## factor(QSQ118)8     2.2741e-01  6.3117e-02  3.6029 0.0003148 ***
## factor(QSQ118)9     2.2382e-01  6.3229e-02  3.5398 0.0004007 ***
## factor(QSQ118)10    2.4191e-01  6.2439e-02  3.8744 0.0001070 ***
## factor(QSQ118)11    1.1153e-01  6.2751e-02  1.7774 0.0755070 .
## factor(QSQ118)12    1.4405e-01  6.2344e-02  2.3105 0.0208618 *
## factor(QSQ118)13    4.8045e-02  6.2567e-02  0.7679 0.4425534
## factor(QSQ118)14   -6.3322e-02  6.2796e-02 -1.0084 0.3132772
## factor(QSQ118)15   -9.9209e-02  6.2265e-02 -1.5933 0.1110897
## factor(QSQ118)16   -2.0250e-01  6.2870e-02 -3.2209 0.0012783 **
## factor(QSQ118)17   -3.8551e-01  6.5136e-02 -5.9185 3.260e-09 ***
## factor(QSQ118)18   -1.3160e-01  6.4702e-02 -2.0340 0.0419545 *
## factor(QSQ118)19    1.2173e-02  6.3405e-02  0.1920 0.8477556
## factor(QSQ118)20    5.9430e-02  6.3964e-02  0.9291 0.3528382
## factor(QSQ118)21    2.4536e-02  6.4478e-02  0.3805 0.7035484
## factor(QSQ118)22   -9.9884e-02  6.4909e-02 -1.5388 0.1238509
## factor(QSQ118)23    2.9293e-02  6.5351e-02  0.4482 0.6539805
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(lifinsmodel1,vcov. = vcovHC(lifinsmodel1, type = "HC3"))
```

```
##
## t test of coefficients:
##
##                        Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)         9.3090e-03  1.5764e-01   0.0591 0.9529096
## HOUSB               4.8317e-01  4.1404e-02  11.6696 < 2.2e-16 ***
## AGE_REF             1.4786e-02  6.2529e-04  23.6460 < 2.2e-16 ***
## factor(CUTENURE)2  -2.9699e-01  2.2067e-02 -13.4586 < 2.2e-16 ***
## factor(CUTENURE)4  -4.9226e-01  1.9356e-02 -25.4326 < 2.2e-16 ***
## factor(CUTENURE)5  -5.1224e-01  5.5745e-02  -9.1890 < 2.2e-16 ***
## factor(CUTENURE)6  -3.7345e-01  9.0231e-02  -4.1388 3.494e-05 ***
## FAM_SIZE            6.7290e-02  6.4531e-03  10.4275 < 2.2e-16 ***
## TINCHRS             7.1671e-03  3.5825e-04  20.0059 < 2.2e-16 ***
## factor(MARITAL1)2  -2.6489e-01  3.0404e-02  -8.7126 < 2.2e-16 ***
## factor(MARITAL1)3  -2.5642e-01  2.4553e-02 -10.4434 < 2.2e-16 ***
## factor(MARITAL1)4  -3.4671e-01  4.4731e-02  -7.7509 9.217e-15 ***
## factor(MARITAL1)5  -3.5635e-01  2.3553e-02 -15.1296 < 2.2e-16 ***
## factor(REF_RACE)2   3.8681e-01  2.6259e-02  14.7305 < 2.2e-16 ***
## factor(REF_RACE)3   2.7822e-01  1.0202e-01   2.7272 0.0063882 **
## factor(REF_RACE)4  -1.5373e-01  3.2913e-02  -4.6707 3.006e-06 ***
## factor(REF_RACE)5   5.8596e-02  1.0378e-01   0.5646 0.5723237
## factor(REF_RACE)6   2.1633e-01  5.8672e-02   3.6871 0.0002270 ***
## SEX_REF             1.0825e-01  1.5302e-02   7.0744 1.512e-12 ***
## ALCBEVPQ            8.4863e-04  4.3134e-05  19.6744 < 2.2e-16 ***
## TOBACCPQ            3.3893e-05  4.1161e-05   0.8234 0.4102641
## factor(STATE)2     -6.0834e-01  7.6965e-02  -7.9041 2.727e-15 ***
## factor(STATE)4     -6.7288e-01  7.0554e-02  -9.5370 < 2.2e-16 ***
## factor(STATE)6     -5.8438e-01  5.8271e-02 -10.0288 < 2.2e-16 ***
## factor(STATE)8     -7.6721e-01  7.6121e-02 -10.0788 < 2.2e-16 ***
## factor(STATE)9     -3.3264e-01  7.4951e-02  -4.4380 9.089e-06 ***
## factor(STATE)10    -4.4562e-01  2.0218e-01  -2.2040 0.0275251 *
## factor(STATE)11    -1.3223e-01  1.6083e-01  -0.8222 0.4109907
## factor(STATE)12    -7.7022e-01  6.0375e-02 -12.7574 < 2.2e-16 ***
## factor(STATE)13    -4.2717e-01  7.6687e-02  -5.5703 2.550e-08 ***
## factor(STATE)15    -7.4143e-01  7.7886e-02  -9.5195 < 2.2e-16 ***
## factor(STATE)17    -4.0532e-01  6.9306e-02  -5.8482 4.985e-09 ***
## factor(STATE)18    -8.9313e-01  1.0573e-01  -8.4475 < 2.2e-16 ***
## factor(STATE)19    -3.8324e-01  1.9989e-01  -1.9173 0.0552051 .
## factor(STATE)20    -6.5277e-02  9.0733e-02  -0.7194 0.4718709
## factor(STATE)21    -2.8375e-01  8.4884e-02  -3.3428 0.0008298 ***
## factor(STATE)22    -9.2989e-01  7.4308e-02 -12.5139 < 2.2e-16 ***
## factor(STATE)24    -4.1712e-01  7.6855e-02  -5.4273 5.736e-08 ***
## factor(STATE)25    -4.5798e-01  6.9184e-02  -6.6197 3.618e-11 ***
## factor(STATE)26    -3.3872e-01  6.9263e-02  -4.8903 1.009e-06 ***
## factor(STATE)27    -2.3784e-01  7.7940e-02  -3.0516 0.0022772 **
## factor(STATE)28     5.8165e-03  1.1676e-01   0.0498 0.9602693
## factor(STATE)29    -2.1414e-02  8.6695e-02  -0.2470 0.8049074
## factor(STATE)31    -2.4672e-01  7.3491e-02  -3.3571 0.0007880 ***
## factor(STATE)32    -3.5019e-01  8.9931e-02  -3.8940 9.869e-05 ***
## factor(STATE)33     1.9634e-01  1.9218e-01   1.0217 0.3069451
```

```
## factor(STATE)34    -4.3515e-01  7.5694e-02  -5.7489 9.012e-09 ***
## factor(STATE)36    -6.2520e-01  6.1840e-02 -10.1100 < 2.2e-16 ***
## factor(STATE)37    -4.7007e-01  6.3716e-02  -7.3777 1.624e-13 ***
## factor(STATE)39    -3.8521e-01  6.5148e-02  -5.9129 3.373e-09 ***
## factor(STATE)40    -6.9873e-02  1.0301e-01  -0.6783 0.4975765
## factor(STATE)41    -4.7188e-01  7.5040e-02  -6.2883 3.223e-10 ***
## factor(STATE)42    -4.0485e-01  6.6401e-02  -6.0970 1.085e-09 ***
## factor(STATE)45    -2.3848e-01  8.4892e-02  -2.8093 0.0049666 **
## factor(STATE)46    -7.2077e-01  1.2251e-01  -5.8833 4.036e-09 ***
## factor(STATE)47     1.4927e-01  9.3616e-02   1.5945 0.1108343
## factor(STATE)48    -5.2638e-01  6.1290e-02  -8.5884 < 2.2e-16 ***
## factor(STATE)49    -5.1854e-01  6.8617e-02  -7.5570 4.162e-14 ***
## factor(STATE)51    -2.3916e-01  7.5791e-02  -3.1555 0.0016025 **
## factor(STATE)53    -4.3881e-01  7.3082e-02  -6.0043 1.928e-09 ***
## factor(STATE)54    -1.2590e-01  1.2188e-01  -1.0329 0.3016435
## factor(STATE)55    -2.1009e-01  7.0505e-02  -2.9798 0.0028849 **
## HISP_REF            2.6114e-01  2.1697e-02  12.0356 < 2.2e-16 ***
## factor(HIGH_EDU)10 -1.5805e-03  1.4136e-01  -0.0112 0.9910789
## factor(HIGH_EDU)11  1.4031e-01  1.3633e-01   1.0292 0.3034021
## factor(HIGH_EDU)12  1.9453e-01  1.3389e-01   1.4529 0.1462674
## factor(HIGH_EDU)13  3.2806e-01  1.3413e-01   2.4459 0.0144522 *
## factor(HIGH_EDU)14  4.7415e-01  1.3514e-01   3.5086 0.0004507 ***
## factor(HIGH_EDU)15  4.5837e-01  1.3409e-01   3.4183 0.0006304 ***
## factor(HIGH_EDU)16  6.9459e-01  1.3469e-01   5.1570 2.515e-07 ***
## factor(QSQ118)1     5.5786e-02  5.0347e-02   1.1080 0.2678502
## factor(QSQ118)2     9.3030e-03  4.9467e-02   0.1881 0.8508275
## factor(QSQ118)3    -9.8538e-03  4.8310e-02  -0.2040 0.8383784
## factor(QSQ118)4     5.3120e-03  4.8597e-02   0.1093 0.9129574
## factor(QSQ118)5    -6.7532e-04  4.8916e-02  -0.0138 0.9889851
## factor(QSQ118)6    -3.6990e-02  4.8953e-02  -0.7556 0.4498752
## factor(QSQ118)7     2.8240e-02  4.9478e-02   0.5708 0.5681629
## factor(QSQ118)8     6.5698e-02  5.0686e-02   1.2962 0.1949215
## factor(QSQ118)9    -7.9436e-02  4.9169e-02  -1.6156 0.1061892
## factor(QSQ118)10   -4.3649e-02  4.9181e-02  -0.8875 0.3748013
## factor(QSQ118)11   -7.6518e-02  4.8566e-02  -1.5756 0.1151296
## factor(QSQ118)12   -1.3564e-01  4.8066e-02  -2.8220 0.0047734 **
## factor(QSQ118)13   -9.9554e-02  4.8572e-02  -2.0496 0.0404045 *
## factor(QSQ118)14   -8.2626e-02  4.9259e-02  -1.6774 0.0934703 .
## factor(QSQ118)15   -1.3336e-01  4.8370e-02  -2.7571 0.0058331 **
## factor(QSQ118)16   -1.2745e-01  4.8495e-02  -2.6282 0.0085856 **
## factor(QSQ118)17   -3.3842e-02  5.1307e-02  -0.6596 0.5095116
## factor(QSQ118)18   -1.3864e-02  5.0703e-02  -0.2734 0.7845204
## factor(QSQ118)19    7.1587e-03  5.0129e-02   0.1428 0.8864442
## factor(QSQ118)20    8.7549e-02  5.1034e-02   1.7155 0.0862574 .
## factor(QSQ118)21    3.7322e-02  5.0582e-02   0.7379 0.4605967
## factor(QSQ118)22    2.3708e-02  5.0958e-02   0.4652 0.6417586
## factor(QSQ118)23    1.1839e-02  5.0608e-02   0.2339 0.8150370
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(perinsmodel1,vcov. = vcovHC(perinsmodel1, type = "HC3"))

##
## t test of coefficients:
##
##                        Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)         3.09384984  0.22813524  13.5615 < 2.2e-16 ***
## HOUSB              -0.74153905  0.04269664 -17.3676 < 2.2e-16 ***
## AGE_REF            -0.02105891  0.00055342 -38.0527 < 2.2e-16 ***
## factor(CUTENURE)2  -0.41544859  0.01946783 -21.3403 < 2.2e-16 ***
## factor(CUTENURE)4  -0.49530203  0.01601946 -30.9188 < 2.2e-16 ***
## factor(CUTENURE)5  -0.99580208  0.06162758 -16.1584 < 2.2e-16 ***
## factor(CUTENURE)6  -2.32898102  0.15060971 -15.4637 < 2.2e-16 ***
## FAM_SIZE            0.19621931  0.00495942  39.5650 < 2.2e-16 ***
## TINCHRS             0.04822116  0.00031832 151.4843 < 2.2e-16 ***
## factor(MARITAL1)2   0.11596280  0.03387555   3.4232 0.0006192 ***
## factor(MARITAL1)3   0.71295252  0.02276494  31.3180 < 2.2e-16 ***
## factor(MARITAL1)4   0.82055392  0.04479815  18.3167 < 2.2e-16 ***
## factor(MARITAL1)5   0.78823875  0.02122440  37.1383 < 2.2e-16 ***
## factor(REF_RACE)2   0.09280640  0.02237961   4.1469 3.373e-05 ***
## factor(REF_RACE)3  -0.13550130  0.08603563  -1.5749 0.1152728
## factor(REF_RACE)4   0.15942214  0.02471787   6.4497 1.126e-10 ***
## factor(REF_RACE)5   0.22728583  0.07911583   2.8728 0.0040691 **
## factor(REF_RACE)6   0.08996405  0.04692005   1.9174 0.0551914 .
## SEX_REF            -0.08965878  0.01298627  -6.9041 5.084e-12 ***
## factor(STATE)2      0.27535244  0.06348855   4.3370 1.446e-05 ***
## factor(STATE)4     -0.12366130  0.06186036  -1.9990 0.0456070 *
## factor(STATE)6      0.24608186  0.04781906   5.1461 2.665e-07 ***
## factor(STATE)8      0.30020485  0.06122136   4.9036 9.426e-07 ***
## factor(STATE)9      0.19117409  0.06017800   3.1768 0.0014895 **
## factor(STATE)10     0.59117423  0.15725206   3.7594 0.0001704 ***
## factor(STATE)11     0.34688168  0.13710814   2.5300 0.0114084 *
## factor(STATE)12    -0.12073278  0.05131914  -2.3526 0.0186454 *
## factor(STATE)13     0.12420928  0.06226624   1.9948 0.0460666 *
## factor(STATE)15     0.27777647  0.06673964   4.1621 3.156e-05 ***
## factor(STATE)17     0.11368573  0.05592647   2.0328 0.0420785 *
## factor(STATE)18    -0.01924890  0.10205432  -0.1886 0.8503956
## factor(STATE)19     0.04545112  0.17975007   0.2529 0.8003790
## factor(STATE)20    -0.02896795  0.07287214  -0.3975 0.6909868
## factor(STATE)21     0.04258082  0.07282175   0.5847 0.5587330
## factor(STATE)22    -0.20713348  0.06649471  -3.1150 0.0018398 **
## factor(STATE)24     0.39636892  0.06169939   6.4242 1.332e-10 ***
## factor(STATE)25     0.14897434  0.05620448   2.6506 0.0080368 **
## factor(STATE)26     0.00637818  0.05805172   0.1099 0.9125122
## factor(STATE)27     0.12884218  0.06235642   2.0662 0.0388104 *
## factor(STATE)28    -0.18494245  0.10653604  -1.7360 0.0825740 .
## factor(STATE)29     0.13299786  0.06713900   1.9809 0.0476018 *
## factor(STATE)31     0.01578288  0.05906296   0.2672 0.7892994
## factor(STATE)32     0.12689403  0.07443356   1.7048 0.0882359 .
## factor(STATE)33     0.36171921  0.12717264   2.8443 0.0044517 **
```

```
## factor(STATE)34     0.24728294  0.06209418   3.9824 6.828e-05 ***
## factor(STATE)36     0.15914413  0.05130869   3.1017 0.0019247 **
## factor(STATE)37    -0.00579870  0.05207446  -0.1114 0.9113359
## factor(STATE)39     0.13999575  0.05349946   2.6168 0.0088781 **
## factor(STATE)40     0.26645856  0.08089454   3.2939 0.0009884 ***
## factor(STATE)41     0.17082204  0.06148014   2.7785 0.0054623 **
## factor(STATE)42     0.12158853  0.05523696   2.2012 0.0277232 *
## factor(STATE)45    -0.10383178  0.07112530  -1.4598 0.1443366
## factor(STATE)46    -0.27558433  0.09723178  -2.8343 0.0045936 **
## factor(STATE)47     0.13633759  0.08107964   1.6815 0.0926640 .
## factor(STATE)48     0.10289955  0.05002630   2.0569 0.0396978 *
## factor(STATE)49     0.02832625  0.05674418   0.4992 0.6176453
## factor(STATE)51     0.27044434  0.05978419   4.5237 6.085e-06 ***
## factor(STATE)53     0.36460865  0.05941381   6.1368 8.456e-10 ***
## factor(STATE)54     0.08962760  0.10382032   0.8633 0.3879774
## factor(STATE)55     0.11382080  0.05614076   2.0274 0.0426226 *
## HISP_REF           -0.16819646  0.01857917  -9.0530 < 2.2e-16 ***
## factor(HIGH_EDU)10 -0.01772581  0.22688107  -0.0781 0.9377262
## factor(HIGH_EDU)11  0.38910431  0.22030280   1.7662 0.0773614 .
## factor(HIGH_EDU)12  0.85152001  0.21804628   3.9052 9.420e-05 ***
## factor(HIGH_EDU)13  1.10492552  0.21808079   5.0666 4.058e-07 ***
## factor(HIGH_EDU)14  1.37558896  0.21850251   6.2955 3.077e-10 ***
## factor(HIGH_EDU)15  1.58103198  0.21806423   7.2503 4.191e-13 ***
## factor(HIGH_EDU)16  1.72397313  0.21830264   7.8972 2.884e-15 ***
## factor(QSQ118)1     0.26449614  0.05829940   4.5369 5.717e-06 ***
## factor(QSQ118)2     0.01711214  0.04042069   0.4234 0.6720401
## factor(QSQ118)3     0.00306368  0.03934592   0.0779 0.9379353
## factor(QSQ118)4    -0.00141661  0.03948503  -0.0359 0.9713804
## factor(QSQ118)5     0.01477529  0.03994195   0.3699 0.7114437
## factor(QSQ118)6     0.03169099  0.03975889   0.7971 0.4254069
## factor(QSQ118)7     0.05406228  0.03960040   1.3652 0.1721949
## factor(QSQ118)8     0.05545195  0.04076724   1.3602 0.1737672
## factor(QSQ118)9    -0.02438893  0.04070605  -0.5991 0.5490760
## factor(QSQ118)10    0.03128106  0.03974194   0.7871 0.4312227
## factor(QSQ118)11    0.03047982  0.03992551   0.7634 0.4452166
## factor(QSQ118)12    0.01820084  0.04047093   0.4497 0.6529090
## factor(QSQ118)13    0.02940805  0.04065025   0.7234 0.4694110
## factor(QSQ118)14    0.03016441  0.04113661   0.7333 0.4633932
## factor(QSQ118)15    0.03685589  0.04035909   0.9132 0.3611402
## factor(QSQ118)16    0.07635775  0.04064293   1.8787 0.0602823 .
## factor(QSQ118)17    0.08431412  0.04221123   1.9974 0.0457810 *
## factor(QSQ118)18    0.13318691  0.04207184   3.1657 0.0015476 **
## factor(QSQ118)19    0.19443523  0.04138666   4.6980 2.631e-06 ***
## factor(QSQ118)20    0.24233400  0.04173567   5.8064 6.404e-09 ***
## factor(QSQ118)21    0.17998380  0.04135494   4.3522 1.349e-05 ***
## factor(QSQ118)22    0.17140568  0.04204882   4.0763 4.579e-05 ***
## factor(QSQ118)23    0.22277935  0.04167938   5.3451 9.060e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(hlthinsmodel1,vcov. = vcovHC(hlthinsmodel1, type = "HC3"))

##
## t test of coefficients:
##
##                        Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)         1.5310e+00  2.4460e-01   6.2593 3.885e-10 ***
## HOUSB              -5.3817e-01  4.8049e-02 -11.2004 < 2.2e-16 ***
## AGE_REF             5.2056e-02  6.9691e-04  74.6952 < 2.2e-16 ***
## factor(CUTENURE)2  -2.6007e-01  2.0311e-02 -12.8043 < 2.2e-16 ***
## factor(CUTENURE)4  -6.7700e-01  2.2147e-02 -30.5679 < 2.2e-16 ***
## factor(CUTENURE)5  -1.3166e+00  8.2428e-02 -15.9732 < 2.2e-16 ***
## factor(CUTENURE)6  -2.3497e+00  1.5340e-01 -15.3173 < 2.2e-16 ***
## FAM_SIZE           -1.9331e-02  7.8040e-03  -2.4770 0.0132500 *
## TINCHRS             4.9156e-04  3.5227e-04   1.3954 0.1629012
## factor(MARITAL1)2  -3.0066e-01  2.5337e-02 -11.8666 < 2.2e-16 ***
## factor(MARITAL1)3  -5.0473e-01  2.5195e-02 -20.0329 < 2.2e-16 ***
## factor(MARITAL1)4  -8.2705e-01  6.4077e-02 -12.9071 < 2.2e-16 ***
## factor(MARITAL1)5  -4.7833e-01  2.8091e-02 -17.0278 < 2.2e-16 ***
## factor(REF_RACE)2  -1.4049e-01  2.7655e-02  -5.0800 3.782e-07 ***
## factor(REF_RACE)3  -2.3163e-01  1.2448e-01  -1.8608 0.0627768 .
## factor(REF_RACE)4  -3.5648e-02  3.6852e-02  -0.9673 0.3333902
## factor(REF_RACE)5   1.6425e-01  1.1805e-01   1.3914 0.1641172
## factor(REF_RACE)6   9.9609e-03  6.7115e-02   0.1484 0.8820154
## SEX_REF             6.8573e-02  1.6009e-02   4.2836 1.841e-05 ***
## FOODPQ              1.7696e-04  7.2480e-06  24.4146 < 2.2e-16 ***
## ALCBEVPQ            1.8297e-04  4.1257e-05   4.4350 9.220e-06 ***
## MEDSRVPQ            1.0792e-04  1.1741e-05   9.1910 < 2.2e-16 ***
## PREDRGPQ           -5.4345e-02  1.7029e-01  -0.3191 0.7496276
## MEDSUPPQ            2.4289e-04  4.2435e-05   5.7238 1.045e-08 ***
## TOBACCPQ           -2.3684e-05  4.7737e-05  -0.4961 0.6197921
## factor(STATE)2     -2.1574e-01  7.7428e-02  -2.7863 0.0053321 **
## factor(STATE)4     -2.6817e-01  7.1945e-02  -3.7274 0.0001936 ***
## factor(STATE)6     -4.3012e-01  5.4811e-02  -7.8474 4.292e-15 ***
## factor(STATE)8     -4.0342e-02  7.6146e-02  -0.5298 0.5962523
## factor(STATE)9      1.5672e-01  6.7135e-02   2.3345 0.0195725 *
## factor(STATE)10    -4.6298e-01  2.2264e-01  -2.0795 0.0375767 *
## factor(STATE)11    -5.6709e-02  1.5193e-01  -0.3733 0.7089594
## factor(STATE)12    -4.0036e-02  5.6956e-02  -0.7029 0.4821021
## factor(STATE)13     2.3323e-01  7.1534e-02   3.2604 0.0011130 **
## factor(STATE)15    -6.2648e-01  8.0990e-02  -7.7353 1.042e-14 ***
## factor(STATE)17     1.4573e-01  6.5315e-02   2.2311 0.0256749 *
## factor(STATE)18    -2.7405e-01  1.2136e-01  -2.2582 0.0239384 *
## factor(STATE)19    -2.2866e-01  1.9916e-01  -1.1481 0.2509231
## factor(STATE)20    -6.0453e-02  8.5308e-02  -0.7086 0.4785440
## factor(STATE)21    -1.5759e-01  8.1669e-02  -1.9296 0.0536578 .
## factor(STATE)22    -3.8636e-01  8.3854e-02  -4.6075 4.080e-06 ***
## factor(STATE)24     1.9783e-01  6.7891e-02   2.9140 0.0035688 **
## factor(STATE)25     1.0779e-01  6.3408e-02   1.6999 0.0891452 .
## factor(STATE)26    -3.0206e-01  6.7070e-02  -4.5037 6.685e-06 ***
```

```
## factor(STATE)27     1.6089e-01  7.1264e-02   2.2576 0.0239731 *
## factor(STATE)28    -1.1833e-02  9.6704e-02  -0.1224 0.9026156
## factor(STATE)29     1.5682e-01  7.5416e-02   2.0794 0.0375858 *
## factor(STATE)31     1.5082e-01  6.7320e-02   2.2403 0.0250709 *
## factor(STATE)32    -3.5180e-01  9.0212e-02  -3.8997 9.637e-05 ***
## factor(STATE)33     3.9793e-01  1.8095e-01   2.1991 0.0278716 *
## factor(STATE)34    -1.8878e-01  7.4214e-02  -2.5438 0.0109676 *
## factor(STATE)36    -5.2333e-01  5.9283e-02  -8.8277 < 2.2e-16 ***
## factor(STATE)37    -8.5332e-02  5.8834e-02  -1.4504 0.1469535
## factor(STATE)39    -4.0250e-03  5.9151e-02  -0.0680 0.9457497
## factor(STATE)40    -7.7749e-02  1.0580e-01  -0.7349 0.4624054
## factor(STATE)41    -3.1450e-01  7.5626e-02  -4.1586 3.205e-05 ***
## factor(STATE)42    -8.0305e-03  6.0868e-02  -0.1319 0.8950381
## factor(STATE)45     1.9199e-02  7.7584e-02   0.2475 0.8045504
## factor(STATE)46     1.1477e-01  1.1781e-01   0.9742 0.3299406
## factor(STATE)47    -3.8335e-02  8.4760e-02  -0.4523 0.6510687
## factor(STATE)48    -3.5012e-02  5.8478e-02  -0.5987 0.5493660
## factor(STATE)49    -2.7844e-01  6.4688e-02  -4.3043 1.677e-05 ***
## factor(STATE)51     1.2136e-01  6.8847e-02   1.7627 0.0779475 .
## factor(STATE)53    -5.1992e-01  7.3250e-02  -7.0979 1.276e-12 ***
## factor(STATE)54    -2.5517e-01  1.2366e-01  -2.0636 0.0390600 *
## factor(STATE)55     4.3851e-01  6.1938e-02   7.0799 1.453e-12 ***
## HISP_REF            6.5767e-01  2.8505e-02  23.0721 < 2.2e-16 ***
## factor(HIGH_EDU)10 -4.4443e-02  2.3964e-01  -0.1855 0.8528747
## factor(HIGH_EDU)11 -5.1388e-01  2.3295e-01  -2.2060 0.0273879 *
## factor(HIGH_EDU)12  5.8593e-02  2.3004e-01   0.2547 0.7989500
## factor(HIGH_EDU)13  3.0484e-01  2.3009e-01   1.3249 0.1852167
## factor(HIGH_EDU)14  5.9764e-01  2.3069e-01   2.5907 0.0095808 **
## factor(HIGH_EDU)15  8.3466e-01  2.2999e-01   3.6292 0.0002844 ***
## factor(HIGH_EDU)16  9.8334e-01  2.3012e-01   4.2731 1.930e-05 ***
## factor(QSQ118)1    -4.9913e-03  5.7216e-02  -0.0872 0.9304835
## factor(QSQ118)2    -2.8193e-02  5.1605e-02  -0.5463 0.5848442
## factor(QSQ118)3    -1.3115e-02  5.0268e-02  -0.2609 0.7941723
## factor(QSQ118)4     4.9446e-02  5.0603e-02   0.9771 0.3285088
## factor(QSQ118)5     2.2080e-02  5.0774e-02   0.4349 0.6636623
## factor(QSQ118)6     4.4558e-02  5.0819e-02   0.8768 0.3805951
## factor(QSQ118)7     6.6644e-02  5.0514e-02   1.3193 0.1870642
## factor(QSQ118)8     4.3598e-02  5.0893e-02   0.8567 0.3916401
## factor(QSQ118)9     1.8308e-02  5.0979e-02   0.3591 0.7194990
## factor(QSQ118)10    4.1072e-02  5.1092e-02   0.8039 0.4214591
## factor(QSQ118)11    4.8414e-02  5.1114e-02   0.9472 0.3435463
## factor(QSQ118)12    1.8923e-02  5.1431e-02   0.3679 0.7129341
## factor(QSQ118)13   -1.8787e-02  5.1373e-02  -0.3657 0.7145887
## factor(QSQ118)14   -5.5316e-02  5.1765e-02  -1.0686 0.2852581
## factor(QSQ118)15   -2.0243e-02  5.0973e-02  -0.3971 0.6912630
## factor(QSQ118)16    1.5961e-02  5.0757e-02   0.3145 0.7531720
## factor(QSQ118)17    5.8951e-03  5.2379e-02   0.1125 0.9103904
## factor(QSQ118)18    6.6621e-02  5.1941e-02   1.2826 0.1996253
## factor(QSQ118)19    7.4186e-02  5.1738e-02   1.4339 0.1516129
## factor(QSQ118)20    2.8840e-02  5.1854e-02   0.5562 0.5780873
```

```
## factor(QSQ118)21   -6.9716e-03  5.2113e-02  -0.1338 0.8935784
## factor(QSQ118)22   -1.5680e-02  5.2636e-02  -0.2979 0.7657846
## factor(QSQ118)23    8.1037e-02  5.2629e-02   1.5398 0.1236209
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Extra Analysis
vehinsmodel2 <- lm(log(AdjVEHINSPQ) ~ HOUSB+I(HOUSB^2)+AGE_REF+factor(CUTENURE)+FAM_SIZE+TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+VEHQ+log(GASMOPQ+1)+log(VEHFINPQ+1)+log(MAINRPPQ+1)+log(VRNTLOPQ+1)+log(PUBTRAPQ+1)+factor(STATE)+log(EVEHPURP+1)+HISP_REF+factor(HIGH_EDU)+factor(QSQ118), data=ces_capstone)
summary(vehinsmodel2)
```

```
##
## Call:
## lm(formula = log(AdjVEHINSPQ) ~ HOUSB + I(HOUSB^2) + AGE_REF +
##     factor(CUTENURE) + FAM_SIZE + TINCHRS + factor(MARITAL1) +
##     factor(REF_RACE) + SEX_REF + VEHQ + log(GASMOPQ + 1) + log(VEHFINPQ +
##     1) + log(MAINRPPQ + 1) + log(VRNTLOPQ + 1) + log(PUBTRAPQ +
##     1) + factor(STATE) + log(EVEHPURP + 1) + HISP_REF + factor(HIGH_EDU) +

##     factor(QSQ118), data = ces_capstone)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.8457 -2.6505  0.5684  2.4170  9.8516
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -1.4749408  0.3030114  -4.868 1.13e-06 ***
## HOUSB               7.9786825  0.1552740  51.385  < 2e-16 ***
## I(HOUSB^2)         -7.6423944  0.1929932 -39.599  < 2e-16 ***
## AGE_REF             0.0028415  0.0008146   3.488 0.000486 ***
## factor(CUTENURE)2   0.1943763  0.0261433   7.435 1.05e-13 ***
## factor(CUTENURE)4  -0.0457879  0.0250177  -1.830 0.067220 .
## factor(CUTENURE)5   0.2726735  0.0851776   3.201 0.001369 **
## factor(CUTENURE)6   0.3198293  0.2227737   1.436 0.151100
## FAM_SIZE            0.0169310  0.0080183   2.112 0.034728 *
## TINCHRS             0.0048400  0.0004195  11.538  < 2e-16 ***
## factor(MARITAL1)2   0.1964214  0.0381283   5.152 2.59e-07 ***
## factor(MARITAL1)3   0.1722838  0.0303546   5.676 1.39e-08 ***
## factor(MARITAL1)4  -0.0794833  0.0661474  -1.202 0.229518
## factor(MARITAL1)5   0.1301335  0.0314618   4.136 3.53e-05 ***
## factor(REF_RACE)2   0.0160383  0.0315994   0.508 0.611770
## factor(REF_RACE)3   0.0669102  0.1286890   0.520 0.603109
## factor(REF_RACE)4  -0.1475927  0.0405143  -3.643 0.000270 ***
## factor(REF_RACE)5   0.0468719  0.1289932   0.363 0.716331
## factor(REF_RACE)6  -0.0508594  0.0716775  -0.710 0.477979
## SEX_REF             0.0087534  0.0188115   0.465 0.641702
```

```
## VEHQ                 0.1051022  0.0077104  13.631  < 2e-16 ***
## log(GASMOPQ + 1)     0.4284715  0.0063129  67.872  < 2e-16 ***
## log(VEHFINPQ + 1)    0.0669207  0.0094530   7.079 1.46e-12 ***
## log(MAINRPPQ + 1)    0.1021348  0.0038855  26.286  < 2e-16 ***
## log(VRNTLOPQ + 1)    0.0867964  0.0042740  20.308  < 2e-16 ***
## log(PUBTRAPQ + 1)    0.0084018  0.0043169   1.946 0.051627 .
## factor(STATE)2      -0.0970332  0.0911730  -1.064 0.287207
## factor(STATE)4      -0.0232202  0.0856767  -0.271 0.786376
## factor(STATE)6      -0.1960328  0.0664088  -2.952 0.003159 **
## factor(STATE)8       0.2221444  0.0923078   2.407 0.016105 *
## factor(STATE)9       0.0330881  0.0857817   0.386 0.699701
## factor(STATE)10      0.5440118  0.2430705   2.238 0.025218 *
## factor(STATE)11      0.0079609  0.1931018   0.041 0.967115
## factor(STATE)12     -0.0943442  0.0707050  -1.334 0.182097
## factor(STATE)13      0.0892637  0.0887788   1.005 0.314677
## factor(STATE)15     -0.0312273  0.0940414  -0.332 0.739845
## factor(STATE)17     -0.2904312  0.0800805  -3.627 0.000287 ***
## factor(STATE)18     -0.9335904  0.1447208  -6.451 1.12e-10 ***
## factor(STATE)19     -0.6821249  0.2390078  -2.854 0.004318 **
## factor(STATE)20     -0.1395336  0.1045441  -1.335 0.181982
## factor(STATE)21      0.1991290  0.0994635   2.002 0.045284 *
## factor(STATE)22      0.4300250  0.0954337   4.506 6.61e-06 ***
## factor(STATE)24      0.0542516  0.0863201   0.628 0.529682
## factor(STATE)25     -0.2279916  0.0797298  -2.860 0.004243 **
## factor(STATE)26     -0.1155696  0.0803129  -1.439 0.150156
## factor(STATE)27     -0.1584650  0.0891640  -1.777 0.075534 .
## factor(STATE)28     -0.1770108  0.1334745  -1.326 0.184784
## factor(STATE)29     -0.1272654  0.0957656  -1.329 0.183876
## factor(STATE)31     -0.0009977  0.0853213  -0.012 0.990671
## factor(STATE)32      0.1481629  0.1035096   1.431 0.152321
## factor(STATE)33     -0.3172280  0.2134673  -1.486 0.137263
## factor(STATE)34     -0.1630033  0.0869010  -1.876 0.060695 .
## factor(STATE)36     -0.3165920  0.0718722  -4.405 1.06e-05 ***
## factor(STATE)37     -0.0282565  0.0726650  -0.389 0.697381
## factor(STATE)39     -0.0825257  0.0747080  -1.105 0.269317
## factor(STATE)40      0.1529208  0.1182797   1.293 0.196058
## factor(STATE)41     -0.0106562  0.0893267  -0.119 0.905042
## factor(STATE)42     -0.0564406  0.0762879  -0.740 0.459401
## factor(STATE)45      0.0884782  0.0973785   0.909 0.363563
## factor(STATE)46     -0.2278229  0.1562066  -1.458 0.144714
## factor(STATE)47     -0.1132957  0.1056025  -1.073 0.283341
## factor(STATE)48      0.0160134  0.0705265   0.227 0.820381
## factor(STATE)49      0.0280578  0.0792084   0.354 0.723169
## factor(STATE)51     -0.2933098  0.0852555  -3.440 0.000581 ***
## factor(STATE)53     -0.1586674  0.0853996  -1.858 0.063181 .
## factor(STATE)54     -0.1484141  0.1473148  -1.007 0.313715
## factor(STATE)55     -0.2546520  0.0805378  -3.162 0.001568 **
## log(EVEHPURP + 1)    0.0402994  0.0061953   6.505 7.82e-11 ***
## HISP_REF            -0.0072148  0.0296030  -0.244 0.807449
## factor(HIGH_EDU)10  -0.0854617  0.2977532  -0.287 0.774096
```

```
## factor(HIGH_EDU)11 -0.0053642  0.2881338  -0.019 0.985147
## factor(HIGH_EDU)12  0.0708385  0.2852654   0.248 0.803884
## factor(HIGH_EDU)13  0.1335341  0.2853350   0.468 0.639792
## factor(HIGH_EDU)14  0.1510007  0.2860370   0.528 0.597566
## factor(HIGH_EDU)15  0.1206603  0.2852872   0.423 0.672337
## factor(HIGH_EDU)16  0.1742641  0.2855633   0.610 0.541700
## factor(QSQ118)1    -0.0013823  0.0604663  -0.023 0.981761
## factor(QSQ118)2    -0.0380435  0.0611201  -0.622 0.533655
## factor(QSQ118)3    -0.1214524  0.0598761  -2.028 0.042523 *
## factor(QSQ118)4     0.0104769  0.0602502   0.174 0.861952
## factor(QSQ118)5     0.0231381  0.0607203   0.381 0.703159
## factor(QSQ118)6    -0.0139678  0.0608328  -0.230 0.818395
## factor(QSQ118)7     0.0502792  0.0606611   0.829 0.407190
## factor(QSQ118)8     0.3050545  0.0616834   4.945 7.61e-07 ***
## factor(QSQ118)9     0.3013573  0.0614341   4.905 9.34e-07 ***
## factor(QSQ118)10    0.2883135  0.0610727   4.721 2.35e-06 ***
## factor(QSQ118)11    0.1415999  0.0610858   2.318 0.020449 *
## factor(QSQ118)12    0.1857268  0.0610086   3.044 0.002333 **
## factor(QSQ118)13    0.0857613  0.0609792   1.406 0.159608
## factor(QSQ118)14   -0.0281941  0.0617320  -0.457 0.647876
## factor(QSQ118)15   -0.0508788  0.0607139  -0.838 0.402028
## factor(QSQ118)16   -0.0977805  0.0608604  -1.607 0.108138
## factor(QSQ118)17   -0.2026338  0.0625857  -3.238 0.001205 **
## factor(QSQ118)18   -0.0321283  0.0624472  -0.514 0.606913
## factor(QSQ118)19    0.0745163  0.0614146   1.213 0.225006
## factor(QSQ118)20    0.1442342  0.0618707   2.331 0.019744 *
## factor(QSQ118)21    0.1064851  0.0619047   1.720 0.085409 .
## factor(QSQ118)22    0.0120960  0.0622305   0.194 0.845884
## factor(QSQ118)23    0.1303180  0.0621164   2.098 0.035911 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.765 on 91892 degrees of freedom
## Multiple R-squared:  0.1858, Adjusted R-squared:  0.1849
## F-statistic:   214 on 98 and 91892 DF,  p-value: < 2.2e-16
```

```
lifinsmodel2<- lm(log(AdjLIFINSPQ) ~ HOUSB+I(HOUSB^2)+AGE_REF+factor(CUTENUR
E)+FAM_SIZE+TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+ALCBEVPQ+log(TO
BACCPQ+1)+factor(STATE)+HISP_REF+factor(HIGH_EDU)+factor(QSQ118), data = ces_
capstone)
summary(lifinsmodel2)
```

```
##
## Call:
## lm(formula = log(AdjLIFINSPQ) ~ HOUSB + I(HOUSB^2) + AGE_REF +
##     factor(CUTENURE) + FAM_SIZE + TINCHRS + factor(MARITAL1) +
##     factor(REF_RACE) + SEX_REF + ALCBEVPQ + log(TOBACCPQ + 1) +
##     factor(STATE) + HISP_REF + factor(HIGH_EDU) + factor(QSQ118),
##     data = ces_capstone)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.4617 -1.4832 -0.8929  0.4198  9.0809
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -2.352e-01  2.443e-01  -0.962 0.335832
## HOUSB               2.512e+00  1.238e-01  20.286  < 2e-16 ***
## I(HOUSB^2)         -2.704e+00  1.547e-01 -17.483  < 2e-16 ***
## AGE_REF             1.488e-02  6.538e-04  22.763  < 2e-16 ***
## factor(CUTENURE)2  -2.464e-01  2.101e-02 -11.726  < 2e-16 ***
## factor(CUTENURE)4  -4.989e-01  1.984e-02 -25.152  < 2e-16 ***
## factor(CUTENURE)5  -4.133e-01  6.881e-02  -6.007 1.90e-09 ***
## factor(CUTENURE)6  -2.055e-01  1.801e-01  -1.141 0.253666
## FAM_SIZE            6.845e-02  6.388e-03  10.715  < 2e-16 ***
## TINCHRS             7.698e-03  3.354e-04  22.949  < 2e-16 ***
## factor(MARITAL1)2  -2.738e-01  3.069e-02  -8.922  < 2e-16 ***
## factor(MARITAL1)3  -2.555e-01  2.449e-02 -10.430  < 2e-16 ***
## factor(MARITAL1)4  -3.363e-01  5.352e-02  -6.284 3.31e-10 ***
## factor(MARITAL1)5  -3.513e-01  2.536e-02 -13.849  < 2e-16 ***
## factor(REF_RACE)2   3.768e-01  2.554e-02  14.752  < 2e-16 ***
## factor(REF_RACE)3   2.907e-01  1.041e-01   2.791 0.005250 **
## factor(REF_RACE)4  -1.647e-01  3.276e-02  -5.027 5.00e-07 ***
## factor(REF_RACE)5   5.025e-02  1.044e-01   0.481 0.630213
## factor(REF_RACE)6   2.183e-01  5.799e-02   3.764 0.000167 ***
## SEX_REF             1.026e-01  1.521e-02   6.749 1.50e-11 ***
## ALCBEVPQ            8.301e-04  3.331e-05  24.922  < 2e-16 ***
## log(TOBACCPQ + 1)  -7.682e-03  3.905e-03  -1.967 0.049133 *
## factor(STATE)2     -6.215e-01  7.366e-02  -8.437  < 2e-16 ***
## factor(STATE)4     -6.854e-01  6.931e-02  -9.888  < 2e-16 ***
## factor(STATE)6     -6.032e-01  5.364e-02 -11.246  < 2e-16 ***
## factor(STATE)8     -7.857e-01  7.456e-02 -10.538  < 2e-16 ***
## factor(STATE)9     -3.505e-01  6.934e-02  -5.054 4.32e-07 ***
## factor(STATE)10    -4.482e-01  1.966e-01  -2.280 0.022622 *
## factor(STATE)11    -1.605e-01  1.558e-01  -1.030 0.302896
## factor(STATE)12    -7.815e-01  5.709e-02 -13.688  < 2e-16 ***
## factor(STATE)13    -4.401e-01  7.183e-02  -6.127 9.00e-10 ***
## factor(STATE)15    -7.621e-01  7.604e-02 -10.023  < 2e-16 ***
## factor(STATE)17    -4.242e-01  6.462e-02  -6.565 5.23e-11 ***
## factor(STATE)18    -8.912e-01  1.171e-01  -7.612 2.72e-14 ***
## factor(STATE)19    -3.822e-01  1.934e-01  -1.976 0.048114 *
## factor(STATE)20    -6.557e-02  8.455e-02  -0.776 0.438007
## factor(STATE)21    -2.828e-01  8.047e-02  -3.514 0.000441 ***
## factor(STATE)22    -9.299e-01  7.712e-02 -12.057  < 2e-16 ***
## factor(STATE)24    -4.212e-01  6.976e-02  -6.039 1.56e-09 ***
## factor(STATE)25    -4.721e-01  6.427e-02  -7.345 2.07e-13 ***
## factor(STATE)26    -3.406e-01  6.481e-02  -5.256 1.47e-07 ***
## factor(STATE)27    -2.440e-01  7.206e-02  -3.386 0.000708 ***
## factor(STATE)28     1.256e-02  1.080e-01   0.116 0.907409
## factor(STATE)29    -2.467e-02  7.746e-02  -0.319 0.750057
```

```
## factor(STATE)31    -2.586e-01  6.901e-02  -3.747 0.000179 ***
## factor(STATE)32    -3.513e-01  8.368e-02  -4.198 2.70e-05 ***
## factor(STATE)33     1.956e-01  1.726e-01   1.133 0.257034
## factor(STATE)34    -4.606e-01  6.998e-02  -6.582 4.67e-11 ***
## factor(STATE)36    -6.420e-01  5.739e-02 -11.187  < 2e-16 ***
## factor(STATE)37    -4.778e-01  5.878e-02  -8.128 4.43e-16 ***
## factor(STATE)39    -3.950e-01  6.041e-02  -6.539 6.23e-11 ***
## factor(STATE)40    -7.732e-02  9.567e-02  -0.808 0.419016
## factor(STATE)41    -4.896e-01  7.219e-02  -6.782 1.19e-11 ***
## factor(STATE)42    -4.170e-01  6.147e-02  -6.783 1.18e-11 ***
## factor(STATE)45    -2.445e-01  7.879e-02  -3.104 0.001912 **
## factor(STATE)46    -7.288e-01  1.264e-01  -5.766 8.13e-09 ***
## factor(STATE)47     1.520e-01  8.545e-02   1.778 0.075358 .
## factor(STATE)48    -5.384e-01  5.700e-02  -9.444  < 2e-16 ***
## factor(STATE)49    -5.278e-01  6.413e-02  -8.231  < 2e-16 ***
## factor(STATE)51    -2.480e-01  6.889e-02  -3.599 0.000319 ***
## factor(STATE)53    -4.476e-01  6.897e-02  -6.490 8.63e-11 ***
## factor(STATE)54    -1.186e-01  1.192e-01  -0.995 0.319826
## factor(STATE)55    -2.263e-01  6.514e-02  -3.475 0.000511 ***
## HISP_REF            2.786e-01  2.406e-02  11.575  < 2e-16 ***
## factor(HIGH_EDU)10  7.582e-03  2.409e-01   0.031 0.974896
## factor(HIGH_EDU)11  1.456e-01  2.332e-01   0.624 0.532387
## factor(HIGH_EDU)12  1.938e-01  2.308e-01   0.840 0.401156
## factor(HIGH_EDU)13  3.314e-01  2.308e-01   1.436 0.151104
## factor(HIGH_EDU)14  4.767e-01  2.314e-01   2.060 0.039396 *
## factor(HIGH_EDU)15  4.655e-01  2.308e-01   2.017 0.043682 *
## factor(HIGH_EDU)16  7.079e-01  2.310e-01   3.065 0.002176 **
## factor(QSQ118)1     5.125e-02  4.892e-02   1.047 0.294881
## factor(QSQ118)2     3.335e-03  4.946e-02   0.067 0.946230
## factor(QSQ118)3    -1.389e-02  4.845e-02  -0.287 0.774429
## factor(QSQ118)4    -9.591e-04  4.875e-02  -0.020 0.984305
## factor(QSQ118)5    -3.950e-03  4.913e-02  -0.080 0.935917
## factor(QSQ118)6    -4.271e-02  4.922e-02  -0.868 0.385526
## factor(QSQ118)7     2.357e-02  4.909e-02   0.480 0.631082
## factor(QSQ118)8     6.069e-02  4.985e-02   1.218 0.223405
## factor(QSQ118)9    -8.068e-02  4.957e-02  -1.628 0.103633
## factor(QSQ118)10   -4.859e-02  4.932e-02  -0.985 0.324585
## factor(QSQ118)11   -7.969e-02  4.934e-02  -1.615 0.106318
## factor(QSQ118)12   -1.373e-01  4.932e-02  -2.784 0.005376 **
## factor(QSQ118)13   -1.014e-01  4.934e-02  -2.055 0.039897 *
## factor(QSQ118)14   -8.549e-02  4.995e-02  -1.712 0.086971 .
## factor(QSQ118)15   -1.353e-01  4.912e-02  -2.755 0.005872 **
## factor(QSQ118)16   -1.295e-01  4.921e-02  -2.632 0.008491 **
## factor(QSQ118)17   -3.688e-02  5.061e-02  -0.729 0.466165
## factor(QSQ118)18   -1.721e-02  5.052e-02  -0.341 0.733349
## factor(QSQ118)19    2.439e-03  4.968e-02   0.049 0.960842
## factor(QSQ118)20    8.411e-02  5.005e-02   1.680 0.092866 .
## factor(QSQ118)21    3.282e-02  5.007e-02   0.655 0.512187
## factor(QSQ118)22    2.310e-02  5.034e-02   0.459 0.646405
## factor(QSQ118)23    1.107e-02  5.025e-02   0.220 0.825580
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.238 on 91897 degrees of freedom
## Multiple R-squared:  0.08632,    Adjusted R-squared:  0.08539
## F-statistic: 93.35 on 93 and 91897 DF,  p-value: < 2.2e-16
```

```
perinsmodel2 <- lm(log(AdjPERINSPQ) ~ HOUSB+I(HOUSB^2)+AGE_REF+factor(CUTENUR
E)+FAM_SIZE+TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+factor(STATE)+H
ISP_REF+factor(HIGH_EDU)+factor(QSQ118), data= ces_capstone)
summary(perinsmodel2)
```

```
##
## Call:
## lm(formula = log(AdjPERINSPQ) ~ HOUSB + I(HOUSB^2) + AGE_REF +
##     factor(CUTENURE) + FAM_SIZE + TINCHRS + factor(MARITAL1) +
##     factor(REF_RACE) + SEX_REF + factor(STATE) + HISP_REF + factor(HIGH_ED
U) +
##     factor(QSQ118), data = ces_capstone)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -10.633  -1.229   0.159   1.277   7.878
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          2.9729576  0.2104347  14.128  < 2e-16 ***
## HOUSB                0.2658447  0.1064979   2.496 0.012553 *
## I(HOUSB^2)          -1.3443476  0.1330684 -10.103  < 2e-16 ***
## AGE_REF             -0.0209988  0.0005629 -37.308  < 2e-16 ***
## factor(CUTENURE)2   -0.3901340  0.0180955 -21.560  < 2e-16 ***
## factor(CUTENURE)4   -0.4992958  0.0170616 -29.264  < 2e-16 ***
## factor(CUTENURE)5   -0.9476814  0.0592548 -15.993  < 2e-16 ***
## factor(CUTENURE)6   -2.2419424  0.1550628 -14.458  < 2e-16 ***
## FAM_SIZE             0.1962805  0.0054872  35.771  < 2e-16 ***
## TINCHRS              0.0484640  0.0002878 168.366  < 2e-16 ***
## factor(MARITAL1)2    0.1115616  0.0264295   4.221 2.43e-05 ***
## factor(MARITAL1)3    0.7121425  0.0210727  33.795  < 2e-16 ***
## factor(MARITAL1)4    0.8239254  0.0460738  17.883  < 2e-16 ***
## factor(MARITAL1)5    0.7899897  0.0218386  36.174  < 2e-16 ***
## factor(REF_RACE)2    0.0899117  0.0219522   4.096 4.21e-05 ***
## factor(REF_RACE)3   -0.1306655  0.0896754  -1.457 0.145094
## factor(REF_RACE)4    0.1560932  0.0281394   5.547 2.91e-08 ***
## factor(REF_RACE)5    0.2234978  0.0898958   2.486 0.012914 *
## factor(REF_RACE)6    0.0903661  0.0499442   1.809 0.070401 .
## SEX_REF             -0.0918440  0.0130804  -7.022 2.21e-12 ***
## factor(STATE)2       0.2693651  0.0634131   4.248 2.16e-05 ***
## factor(STATE)4      -0.1287713  0.0596864  -2.157 0.030972 *
## factor(STATE)6       0.2386215  0.0461459   5.171 2.33e-07 ***
## factor(STATE)8       0.2920646  0.0641738   4.551 5.34e-06 ***
```

```
## factor(STATE)9      0.1826816  0.0597063   3.060 0.002216 **
## factor(STATE)10     0.5877321  0.1693195   3.471 0.000518 ***
## factor(STATE)11     0.3330403  0.1341189   2.483 0.013024 *
## factor(STATE)12    -0.1249331  0.0491622  -2.541 0.011047 *
## factor(STATE)13     0.1196735  0.0618509   1.935 0.053009 .
## factor(STATE)15     0.2700104  0.0654532   4.125 3.71e-05 ***
## factor(STATE)17     0.1050071  0.0556392   1.887 0.059125 .
## factor(STATE)18    -0.0189531  0.1008356  -0.188 0.850908
## factor(STATE)19     0.0453510  0.1665654   0.272 0.785414
## factor(STATE)20    -0.0298677  0.0728161  -0.410 0.681675
## factor(STATE)21     0.0424535  0.0693082   0.613 0.540187
## factor(STATE)22    -0.2052755  0.0664140  -3.091 0.001996 **
## factor(STATE)24     0.3961273  0.0600596   6.596 4.26e-11 ***
## factor(STATE)25     0.1429953  0.0553338   2.584 0.009761 **
## factor(STATE)26     0.0051130  0.0558141   0.092 0.927010
## factor(STATE)27     0.1255754  0.0620268   2.025 0.042918 *
## factor(STATE)28    -0.1823818  0.0930201  -1.961 0.049920 *
## factor(STATE)29     0.1303819  0.0666983   1.955 0.050609 .
## factor(STATE)31     0.0106491  0.0594308   0.179 0.857793
## factor(STATE)32     0.1262907  0.0720651   1.752 0.079699 .
## factor(STATE)33     0.3611887  0.1486638   2.430 0.015119 *
## factor(STATE)34     0.2366378  0.0602396   3.928 8.56e-05 ***
## factor(STATE)36     0.1519710  0.0493867   3.077 0.002090 **
## factor(STATE)37    -0.0089743  0.0506254  -0.177 0.859298
## factor(STATE)39     0.1351972  0.0520194   2.599 0.009352 **
## factor(STATE)40     0.2621442  0.0824011   3.181 0.001467 **
## factor(STATE)41     0.1629986  0.0621424   2.623 0.008718 **
## factor(STATE)42     0.1158656  0.0529372   2.189 0.028619 *
## factor(STATE)45    -0.1061038  0.0678555  -1.564 0.117898
## factor(STATE)46    -0.2781673  0.1088505  -2.555 0.010605 *
## factor(STATE)47     0.1360326  0.0735855   1.849 0.064514 .
## factor(STATE)48     0.0979742  0.0490831   1.996 0.045927 *
## factor(STATE)49     0.0279777  0.0551553   0.507 0.611979
## factor(STATE)51     0.2662298  0.0593201   4.488 7.20e-06 ***
## factor(STATE)53     0.3605511  0.0593541   6.075 1.25e-09 ***
## factor(STATE)54     0.0913405  0.1026342   0.890 0.373489
## factor(STATE)55     0.1064650  0.0560972   1.898 0.057717 .
## HISP_REF           -0.1631512  0.0205975  -7.921 2.38e-15 ***
## factor(HIGH_EDU)10 -0.0153122  0.2075139  -0.074 0.941179
## factor(HIGH_EDU)11  0.3877181  0.2008093   1.931 0.053514 .
## factor(HIGH_EDU)12  0.8475949  0.1987855   4.264 2.01e-05 ***
## factor(HIGH_EDU)13  1.1033161  0.1988037   5.550 2.87e-08 ***
## factor(HIGH_EDU)14  1.3741254  0.1992777   6.896 5.40e-12 ***
## factor(HIGH_EDU)15  1.5840574  0.1987395   7.971 1.60e-15 ***
## factor(HIGH_EDU)16  1.7312346  0.1989065   8.704  < 2e-16 ***
## factor(QSQ118)1     0.2618062  0.0421336   6.214 5.20e-10 ***
## factor(QSQ118)2     0.0140418  0.0425951   0.330 0.741659
## factor(QSQ118)3     0.0009941  0.0417286   0.024 0.980994
## factor(QSQ118)4    -0.0044242  0.0419905  -0.105 0.916088
## factor(QSQ118)5     0.0133002  0.0423153   0.314 0.753286
```

```
## factor(QSQ118)6     0.0287835  0.0423935   0.679 0.497165
## factor(QSQ118)7     0.0518209  0.0422742   1.226 0.220267
## factor(QSQ118)8     0.0533188  0.0429307   1.242 0.214249
## factor(QSQ118)9    -0.0246117  0.0426949  -0.576 0.564309
## factor(QSQ118)10    0.0289874  0.0424809   0.682 0.495011
## factor(QSQ118)11    0.0290345  0.0424942   0.683 0.494445
## factor(QSQ118)12    0.0176375  0.0424762   0.415 0.677973
## factor(QSQ118)13    0.0283106  0.0424918   0.666 0.505247
## factor(QSQ118)14    0.0286241  0.0430164   0.665 0.505782
## factor(QSQ118)15    0.0358975  0.0423016   0.849 0.396102
## factor(QSQ118)16    0.0753202  0.0423834   1.777 0.075553 .
## factor(QSQ118)17    0.0824578  0.0435761   1.892 0.058458 .
## factor(QSQ118)18    0.1309784  0.0434969   3.011 0.002603 **
## factor(QSQ118)19    0.1919395  0.0427859   4.486 7.26e-06 ***
## factor(QSQ118)20    0.2399816  0.0431019   5.568 2.59e-08 ***
## factor(QSQ118)21    0.1774600  0.0431185   4.116 3.86e-05 ***
## factor(QSQ118)22    0.1704935  0.0433513   3.933 8.40e-05 ***
## factor(QSQ118)23    0.2223363  0.0432711   5.138 2.78e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.927 on 91899 degrees of freedom
## Multiple R-squared:  0.5435, Adjusted R-squared:  0.543
## F-statistic:  1202 on 91 and 91899 DF,  p-value: < 2.2e-16
```

```
hlthinsmodel2 <-lm(log(AdjHLTHINPQ) ~ HOUSB+I(HOUSB^2)+AGE_REF+factor(CUTENUR
E)+FAM_SIZE+TINCHRS+factor(MARITAL1)+factor(REF_RACE)+SEX_REF+log(FOODPQ+1)+l
og(ALCBEVPQ+1)+log(MEDSRVPQ+1)+log(PREDRGPQ+1)+log(MEDSUPPQ+1)+log(TOBACCPQ+
1)+factor(STATE)+HISP_REF+factor(HIGH_EDU)+factor(QSQ118),data=ces_capstone)
summary(hlthinsmodel2)
```

```
##
## Call:
## lm(formula = log(AdjHLTHINPQ) ~ HOUSB + I(HOUSB^2) + AGE_REF +
##     factor(CUTENURE) + FAM_SIZE + TINCHRS + factor(MARITAL1) +
##     factor(REF_RACE) + SEX_REF + log(FOODPQ + 1) + log(ALCBEVPQ +
##     1) + log(MEDSRVPQ + 1) + log(PREDRGPQ + 1) + log(MEDSUPPQ +
##     1) + log(TOBACCPQ + 1) + factor(STATE) + HISP_REF + factor(HIGH_EDU) +

##     factor(QSQ118), data = ces_capstone)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.0031 -0.8724  0.5424  1.5417  8.4952
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -4.389e-01  2.589e-01  -1.695 0.090082 .
## HOUSB               1.421e-01  1.315e-01   1.080 0.279929
## I(HOUSB^2)         -1.019e+00  1.627e-01  -6.266 3.73e-10 ***
```

```
## AGE_REF              5.214e-02  6.802e-04  76.645  < 2e-16 ***
## factor(CUTENURE)2   -2.432e-01  2.180e-02 -11.154  < 2e-16 ***
## factor(CUTENURE)4   -6.195e-01  2.066e-02 -29.990  < 2e-16 ***
## factor(CUTENURE)5   -1.185e+00  7.127e-02 -16.631  < 2e-16 ***
## factor(CUTENURE)6   -2.046e+00  1.865e-01 -10.967  < 2e-16 ***
## FAM_SIZE            -1.834e-02  6.788e-03  -2.702 0.006898 **
## TINCHRS              6.766e-05  3.505e-04   0.193 0.846920
## factor(MARITAL1)2   -2.180e-01  3.188e-02  -6.838 8.06e-12 ***
## factor(MARITAL1)3   -4.543e-01  2.540e-02 -17.887  < 2e-16 ***
## factor(MARITAL1)4   -7.740e-01  5.543e-02 -13.963  < 2e-16 ***
## factor(MARITAL1)5   -4.231e-01  2.630e-02 -16.091  < 2e-16 ***
## factor(REF_RACE)2   -1.057e-01  2.651e-02  -3.986 6.72e-05 ***
## factor(REF_RACE)3   -2.013e-01  1.078e-01  -1.867 0.061896 .
## factor(REF_RACE)4    1.500e-02  3.406e-02   0.440 0.659616
## factor(REF_RACE)5    2.054e-01  1.081e-01   1.900 0.057385 .
## factor(REF_RACE)6    1.080e-02  6.006e-02   0.180 0.857278
## SEX_REF              5.255e-02  1.579e-02   3.327 0.000878 ***
## log(FOODPQ + 1)      3.088e-01  9.977e-03  30.951  < 2e-16 ***
## log(ALCBEVPQ + 1)    2.712e-02  3.424e-03   7.919 2.41e-15 ***
## log(MEDSRVPQ + 1)    9.905e-02  3.150e-03  31.444  < 2e-16 ***
## log(PREDRGPQ + 1)   -3.274e-02  2.681e-01  -0.122 0.902796
## log(MEDSUPPQ + 1)    3.704e-02  5.689e-03   6.512 7.47e-11 ***
## log(TOBACCPQ + 1)   -2.427e-02  4.051e-03  -5.992 2.08e-09 ***
## factor(STATE)2      -2.000e-01  7.629e-02  -2.621 0.008767 **
## factor(STATE)4      -2.701e-01  7.179e-02  -3.762 0.000169 ***
## factor(STATE)6      -4.339e-01  5.558e-02  -7.806 5.94e-15 ***
## factor(STATE)8      -9.088e-02  7.723e-02  -1.177 0.239270
## factor(STATE)9       1.145e-01  7.185e-02   1.594 0.110880
## factor(STATE)10     -4.773e-01  2.036e-01  -2.345 0.019054 *
## factor(STATE)11     -1.346e-01  1.613e-01  -0.835 0.403919
## factor(STATE)12     -7.973e-03  5.913e-02  -0.135 0.892741
## factor(STATE)13      2.214e-01  7.438e-02   2.977 0.002915 **
## factor(STATE)15     -6.271e-01  7.876e-02  -7.962 1.71e-15 ***
## factor(STATE)17      1.133e-01  6.696e-02   1.692 0.090605 .
## factor(STATE)18     -2.598e-01  1.212e-01  -2.143 0.032102 *
## factor(STATE)19     -2.372e-01  2.003e-01  -1.184 0.236247
## factor(STATE)20     -9.290e-02  8.760e-02  -1.061 0.288884
## factor(STATE)21     -1.611e-01  8.332e-02  -1.933 0.053202 .
## factor(STATE)22     -3.187e-01  7.992e-02  -3.988 6.66e-05 ***
## factor(STATE)24      2.026e-01  7.226e-02   2.803 0.005058 **
## factor(STATE)25      1.049e-01  6.658e-02   1.576 0.114984
## factor(STATE)26     -3.237e-01  6.712e-02  -4.823 1.42e-06 ***
## factor(STATE)27      1.335e-01  7.468e-02   1.788 0.073773 .
## factor(STATE)28      3.685e-02  1.118e-01   0.329 0.741791
## factor(STATE)29      1.328e-01  8.023e-02   1.655 0.097921 .
## factor(STATE)31      1.268e-01  7.151e-02   1.773 0.076183 .
## factor(STATE)32     -3.530e-01  8.667e-02  -4.073 4.64e-05 ***
## factor(STATE)33      3.910e-01  1.787e-01   2.187 0.028723 *
## factor(STATE)34     -2.150e-01  7.246e-02  -2.967 0.003013 **
## factor(STATE)36     -5.229e-01  5.944e-02  -8.796  < 2e-16 ***
```

```
## factor(STATE)37    -7.866e-02  6.088e-02  -1.292 0.196295
## factor(STATE)39    -1.414e-02  6.258e-02  -0.226 0.821299
## factor(STATE)40    -1.061e-01  9.906e-02  -1.071 0.284188
## factor(STATE)41    -3.496e-01  7.479e-02  -4.675 2.95e-06 ***
## factor(STATE)42    -4.157e-02  6.372e-02  -0.652 0.514116
## factor(STATE)45     3.711e-02  8.158e-02   0.455 0.649168
## factor(STATE)46     1.517e-01  1.309e-01   1.159 0.246488
## factor(STATE)47    -5.165e-02  8.848e-02  -0.584 0.559355
## factor(STATE)48    -3.375e-02  5.904e-02  -0.572 0.567546
## factor(STATE)49    -3.092e-01  6.647e-02  -4.652 3.29e-06 ***
## factor(STATE)51     1.027e-01  7.135e-02   1.439 0.150129
## factor(STATE)53    -5.727e-01  7.146e-02  -8.015 1.12e-15 ***
## factor(STATE)54    -1.925e-01  1.234e-01  -1.560 0.118877
## factor(STATE)55     4.102e-01  6.748e-02   6.078 1.22e-09 ***
## HISP_REF            6.589e-01  2.494e-02  26.424  < 2e-16 ***
## factor(HIGH_EDU)10 -1.181e-01  2.495e-01  -0.473 0.635897
## factor(HIGH_EDU)11 -6.023e-01  2.415e-01  -2.494 0.012618 *
## factor(HIGH_EDU)12 -8.563e-02  2.390e-01  -0.358 0.720182
## factor(HIGH_EDU)13  1.181e-01  2.391e-01   0.494 0.621197
## factor(HIGH_EDU)14  3.912e-01  2.397e-01   1.632 0.102673
## factor(HIGH_EDU)15  5.863e-01  2.391e-01   2.452 0.014190 *
## factor(HIGH_EDU)16  7.136e-01  2.393e-01   2.982 0.002868 **
## factor(QSQ118)1    -6.190e-03  5.066e-02  -0.122 0.902755
## factor(QSQ118)2    -1.074e-02  5.121e-02  -0.210 0.833826
## factor(QSQ118)3    -1.618e-04  5.017e-02  -0.003 0.997427
## factor(QSQ118)4     5.229e-02  5.048e-02   1.036 0.300301
## factor(QSQ118)5     3.446e-02  5.087e-02   0.677 0.498121
## factor(QSQ118)6     5.630e-02  5.097e-02   1.105 0.269335
## factor(QSQ118)7     8.440e-02  5.083e-02   1.661 0.096816 .
## factor(QSQ118)8     5.997e-02  5.161e-02   1.162 0.245265
## factor(QSQ118)9     6.373e-02  5.135e-02   1.241 0.214559
## factor(QSQ118)10    7.170e-02  5.108e-02   1.404 0.160433
## factor(QSQ118)11    8.292e-02  5.110e-02   1.623 0.104659
## factor(QSQ118)12    3.420e-02  5.108e-02   0.670 0.503064
## factor(QSQ118)13    6.189e-03  5.110e-02   0.121 0.903601
## factor(QSQ118)14   -4.113e-02  5.174e-02  -0.795 0.426642
## factor(QSQ118)15   -3.423e-03  5.089e-02  -0.067 0.946374
## factor(QSQ118)16    2.633e-02  5.100e-02   0.516 0.605710
## factor(QSQ118)17    1.321e-02  5.246e-02   0.252 0.801207
## factor(QSQ118)18    7.607e-02  5.236e-02   1.453 0.146284
## factor(QSQ118)19    7.684e-02  5.150e-02   1.492 0.135668
## factor(QSQ118)20    4.192e-02  5.192e-02   0.807 0.419486
## factor(QSQ118)21    6.064e-03  5.196e-02   0.117 0.907092
## factor(QSQ118)22    4.178e-03  5.223e-02   0.080 0.936241
## factor(QSQ118)23    1.009e-01  5.214e-02   1.935 0.053027 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.317 on 91893 degrees of freedom
```

```
## Multiple R-squared:  0.2614, Adjusted R-squared:  0.2606
## F-statistic: 335.3 on 97 and 91893 DF,  p-value: < 2.2e-16

#Robust Standard Errors for the Extra Analysis Models
coeftest(vehinsmodel2, vcov. = vcovHC(vehinsmodel2, type = "HC3"))

##
## t test of coefficients:
##
##                        Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)        -1.47494076  0.24728060  -5.9646 2.461e-09 ***
## HOUSB               7.97868252  0.15270661  52.2484 < 2.2e-16 ***
## I(HOUSB^2)         -7.64239443  0.18854396 -40.5338 < 2.2e-16 ***
## AGE_REF             0.00284149  0.00080323   3.5376 0.0004040 ***
## factor(CUTENURE)2   0.19437631  0.02689774   7.2265 4.995e-13 ***
## factor(CUTENURE)4  -0.04578793  0.02507711  -1.8259 0.0678707 .
## factor(CUTENURE)5   0.27267352  0.08176751   3.3347 0.0008541 ***
## factor(CUTENURE)6   0.31982935  0.14376380   2.2247 0.0261047 *
## FAM_SIZE            0.01693102  0.00824281   2.0540 0.0399751 *
## TINCHRS             0.00484003  0.00043024  11.2495 < 2.2e-16 ***
## factor(MARITAL1)2   0.19642138  0.03712845   5.2903 1.224e-07 ***
## factor(MARITAL1)3   0.17228378  0.03019128   5.7064 1.157e-08 ***
## factor(MARITAL1)4  -0.07948329  0.06399465  -1.2420 0.2142286
## factor(MARITAL1)5   0.13013352  0.03103351   4.1933 2.752e-05 ***
## factor(REF_RACE)2   0.01603829  0.03095449   0.5181 0.6043724
## factor(REF_RACE)3   0.06691019  0.12091716   0.5534 0.5800213
## factor(REF_RACE)4  -0.14759265  0.04074551  -3.6223 0.0002921 ***
## factor(REF_RACE)5   0.04687195  0.13257205   0.3536 0.7236707
## factor(REF_RACE)6  -0.05085944  0.06955628  -0.7312 0.4646598
## SEX_REF             0.00875338  0.01889516   0.4633 0.6431788
## VEHQ                0.10510222  0.00828265  12.6894 < 2.2e-16 ***
## log(GASMOPQ + 1)    0.42847147  0.00594465  72.0768 < 2.2e-16 ***
## log(VEHFINPQ + 1)   0.06692072  0.00962069   6.9559 3.526e-12 ***
## log(MAINRPPQ + 1)   0.10213483  0.00403447  25.3156 < 2.2e-16 ***
## log(VRNTLOPQ + 1)   0.08679640  0.00442102  19.6326 < 2.2e-16 ***
## log(PUBTRAPQ + 1)   0.00840183  0.00429434   1.9565 0.0504107 .
## factor(STATE)2     -0.09703319  0.09104618  -1.0658 0.2865359
## factor(STATE)4     -0.02322017  0.08574086  -0.2708 0.7865316
## factor(STATE)6     -0.19603278  0.06645837  -2.9497 0.0031815 **
## factor(STATE)8      0.22214439  0.09472517   2.3451 0.0190217 *
## factor(STATE)9      0.03308812  0.08571032   0.3860 0.6994636
## factor(STATE)10     0.54401177  0.23988350   2.2678 0.0233427 *
## factor(STATE)11     0.00796093  0.17299765   0.0460 0.9632963
## factor(STATE)12    -0.09434419  0.07131106  -1.3230 0.1858403
## factor(STATE)13     0.08926369  0.09131303   0.9776 0.3282960
## factor(STATE)15    -0.03122730  0.09164570  -0.3407 0.7333006
## factor(STATE)17    -0.29043117  0.07885989  -3.6829 0.0002307 ***
## factor(STATE)18    -0.93359044  0.14164290  -6.5912 4.388e-11 ***
## factor(STATE)19    -0.68212493  0.24392344  -2.7965 0.0051675 **
## factor(STATE)20    -0.13953361  0.10420008  -1.3391 0.1805437
```

```
## factor(STATE)21     0.19912899  0.09940078   2.0033 0.0451487 *
## factor(STATE)22     0.43002500  0.09783706   4.3953 1.107e-05 ***
## factor(STATE)24     0.05425162  0.08675866   0.6253 0.5317649
## factor(STATE)25    -0.22799160  0.07853294  -2.9031 0.0036954 **
## factor(STATE)26    -0.11556965  0.08204168  -1.4087 0.1589362
## factor(STATE)27    -0.15846500  0.08992983  -1.7621 0.0780564 .
## factor(STATE)28    -0.17701080  0.13016645  -1.3599 0.1738711
## factor(STATE)29    -0.12726541  0.09565727  -1.3304 0.1833796
## factor(STATE)31    -0.00099766  0.08422052  -0.0118 0.9905487
## factor(STATE)32     0.14816288  0.10449715   1.4179 0.1562335
## factor(STATE)33    -0.31722799  0.21561470  -1.4713 0.1412209
## factor(STATE)34    -0.16300328  0.08790497  -1.8543 0.0636977 .
## factor(STATE)36    -0.31659203  0.07082750  -4.4699 7.835e-06 ***
## factor(STATE)37    -0.02825651  0.07217671  -0.3915 0.6954355
## factor(STATE)39    -0.08252566  0.07386885  -1.1172 0.2639153
## factor(STATE)40     0.15292076  0.11889773   1.2862 0.1983927
## factor(STATE)41    -0.01065619  0.08913572  -0.1196 0.9048398
## factor(STATE)42    -0.05644059  0.07572287  -0.7454 0.4560577
## factor(STATE)45     0.08847823  0.09787018   0.9040 0.3659784
## factor(STATE)46    -0.22782290  0.15613636  -1.4591 0.1445334
## factor(STATE)47    -0.11329569  0.10470634  -1.0820 0.2792408
## factor(STATE)48     0.01601342  0.07114038   0.2251 0.8219051
## factor(STATE)49     0.02805778  0.07902356   0.3551 0.7225486
## factor(STATE)51    -0.29330978  0.08582216  -3.4176 0.0006319 ***
## factor(STATE)53    -0.15866739  0.08536159  -1.8588 0.0630632 .
## factor(STATE)54    -0.14841413  0.14421270  -1.0291 0.3034196
## factor(STATE)55    -0.25465204  0.07951797  -3.2024 0.0013631 **
## log(EVEHPURP + 1)   0.04029943  0.00631418   6.3824 1.752e-10 ***
## HISP_REF           -0.00721481  0.02955106  -0.2441 0.8071173
## factor(HIGH_EDU)10 -0.08546171  0.23562092  -0.3627 0.7168235
## factor(HIGH_EDU)11 -0.00536423  0.22826551  -0.0235 0.9812515
## factor(HIGH_EDU)12  0.07083846  0.22553254   0.3141 0.7534502
## factor(HIGH_EDU)13  0.13353409  0.22574122   0.5915 0.5541627
## factor(HIGH_EDU)14  0.15100066  0.22666515   0.6662 0.5052953
## factor(HIGH_EDU)15  0.12066035  0.22573062   0.5345 0.5929745
## factor(HIGH_EDU)16  0.17426413  0.22608387   0.7708 0.4408309
## factor(QSQ118)1    -0.00138230  0.05976948  -0.0231 0.9815488
## factor(QSQ118)2    -0.03804351  0.06033290  -0.6306 0.5283298
## factor(QSQ118)3    -0.12145239  0.05920373  -2.0514 0.0402278 *
## factor(QSQ118)4     0.01047689  0.05934438   0.1765 0.8598670
## factor(QSQ118)5     0.02313813  0.06008831   0.3851 0.7001874
## factor(QSQ118)6    -0.01396784  0.06022710  -0.2319 0.8166011
## factor(QSQ118)7     0.05027916  0.06011843   0.8363 0.4029685
## factor(QSQ118)8     0.30505455  0.06073233   5.0229 5.098e-07 ***
## factor(QSQ118)9     0.30135730  0.06137913   4.9098 9.134e-07 ***
## factor(QSQ118)10    0.28831353  0.06002449   4.8033 1.563e-06 ***
## factor(QSQ118)11    0.14159994  0.06065404   2.3346 0.0195690 *
## factor(QSQ118)12    0.18572678  0.06037405   3.0763 0.0020967 **
## factor(QSQ118)13    0.08576133  0.06036557   1.4207 0.1554076
## factor(QSQ118)14   -0.02819405  0.06075805  -0.4640 0.6426215
```

```
## factor(QSQ118)15   -0.05087876  0.06007593  -0.8469 0.3970488
## factor(QSQ118)16   -0.09778047  0.06061996  -1.6130 0.1067462
## factor(QSQ118)17   -0.20263376  0.06220745  -3.2574 0.0011248 **
## factor(QSQ118)18   -0.03212826  0.06239363  -0.5149 0.6066042
## factor(QSQ118)19    0.07451633  0.06131153   1.2154 0.2242273
## factor(QSQ118)20    0.14423423  0.06172690   2.3367 0.0194595 *
## factor(QSQ118)21    0.10648511  0.06245110   1.7051 0.0881799 .
## factor(QSQ118)22    0.01209595  0.06272100   0.1929 0.8470743
## factor(QSQ118)23    0.13031804  0.06323042   2.0610 0.0393056 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(lifinsmodel2, vcov. = vcovHC(lifinsmodel2, type = "HC3"))
```

```
##
## t test of coefficients:
##
##                       Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)        -2.3515e-01  1.5739e-01  -1.4941 0.1351622
## HOUSB               2.5121e+00  1.2074e-01  20.8057 < 2.2e-16 ***
## I(HOUSB^2)         -2.7041e+00  1.4918e-01 -18.1261 < 2.2e-16 ***
## AGE_REF             1.4882e-02  6.2427e-04  23.8391 < 2.2e-16 ***
## factor(CUTENURE)2  -2.4636e-01  2.2220e-02 -11.0873 < 2.2e-16 ***
## factor(CUTENURE)4  -4.9895e-01  1.9343e-02 -25.7943 < 2.2e-16 ***
## factor(CUTENURE)5  -4.1330e-01  5.5981e-02  -7.3828 1.563e-13 ***
## factor(CUTENURE)6  -2.0554e-01  9.0024e-02  -2.2831 0.0224253 *
## FAM_SIZE            6.8446e-02  6.4421e-03  10.6248 < 2.2e-16 ***
## TINCHRS             7.6977e-03  3.5848e-04  21.4734 < 2.2e-16 ***
## factor(MARITAL1)2  -2.7383e-01  3.0419e-02  -9.0020 < 2.2e-16 ***
## factor(MARITAL1)3  -2.5545e-01  2.4532e-02 -10.4130 < 2.2e-16 ***
## factor(MARITAL1)4  -3.3634e-01  4.4788e-02  -7.5095 5.988e-14 ***
## factor(MARITAL1)5  -3.5126e-01  2.3511e-02 -14.9404 < 2.2e-16 ***
## factor(REF_RACE)2   3.7680e-01  2.6223e-02  14.3689 < 2.2e-16 ***
## factor(REF_RACE)3   2.9065e-01  1.0235e-01   2.8399 0.0045133 **
## factor(REF_RACE)4  -1.6465e-01  3.2804e-02  -5.0193 5.195e-07 ***
## factor(REF_RACE)5   5.0251e-02  1.0384e-01   0.4839 0.6284562
## factor(REF_RACE)6   2.1828e-01  5.8420e-02   3.7365 0.0001867 ***
## SEX_REF             1.0263e-01  1.5284e-02   6.7147 1.896e-11 ***
## ALCBEVPQ            8.3011e-04  4.2979e-05  19.3145 < 2.2e-16 ***
## log(TOBACCPQ + 1)  -7.6824e-03  3.8789e-03  -1.9805 0.0476460 *
## factor(STATE)2     -6.2148e-01  7.6717e-02  -8.1009 5.521e-16 ***
## factor(STATE)4     -6.8540e-01  7.0446e-02  -9.7295 < 2.2e-16 ***
## factor(STATE)6     -6.0320e-01  5.8177e-02 -10.3683 < 2.2e-16 ***
## factor(STATE)8     -7.8570e-01  7.6083e-02 -10.3269 < 2.2e-16 ***
## factor(STATE)9     -3.5046e-01  7.4922e-02  -4.6776 2.907e-06 ***
## factor(STATE)10    -4.4822e-01  2.0013e-01  -2.2396 0.0251201 *
## factor(STATE)11    -1.6048e-01  1.6045e-01  -1.0002 0.3172309
## factor(STATE)12    -7.8148e-01  6.0284e-02 -12.9632 < 2.2e-16 ***
## factor(STATE)13    -4.4012e-01  7.6545e-02  -5.7498 8.962e-09 ***
## factor(STATE)15    -7.6212e-01  7.7734e-02  -9.8041 < 2.2e-16 ***
```

```
## factor(STATE)17    -4.2424e-01  6.9218e-02  -6.1290 8.881e-10 ***
## factor(STATE)18    -8.9120e-01  1.0563e-01  -8.4366 < 2.2e-16 ***
## factor(STATE)19    -3.8222e-01  1.9895e-01  -1.9212 0.0547078 .
## factor(STATE)20    -6.5571e-02  9.0617e-02  -0.7236 0.4693083
## factor(STATE)21    -2.8279e-01  8.4725e-02  -3.3377 0.0008450 ***
## factor(STATE)22    -9.2988e-01  7.4268e-02 -12.5206 < 2.2e-16 ***
## factor(STATE)24    -4.2124e-01  7.6720e-02  -5.4906 4.015e-08 ***
## factor(STATE)25    -4.7210e-01  6.9065e-02  -6.8356 8.215e-12 ***
## factor(STATE)26    -3.4064e-01  6.9112e-02  -4.9288 8.288e-07 ***
## factor(STATE)27    -2.4401e-01  7.7698e-02  -3.1405 0.0016869 **
## factor(STATE)28     1.2562e-02  1.1597e-01   0.1083 0.9137432
## factor(STATE)29    -2.4675e-02  8.6331e-02  -0.2858 0.7750215
## factor(STATE)31    -2.5856e-01  7.3364e-02  -3.5243 0.0004249 ***
## factor(STATE)32    -3.5128e-01  8.9825e-02  -3.9107 9.208e-05 ***
## factor(STATE)33     1.9565e-01  1.9136e-01   1.0224 0.3065905
## factor(STATE)34    -4.6058e-01  7.5548e-02  -6.0965 1.089e-09 ***
## factor(STATE)36    -6.4199e-01  6.1744e-02 -10.3976 < 2.2e-16 ***
## factor(STATE)37    -4.7777e-01  6.3628e-02  -7.5087 6.023e-14 ***
## factor(STATE)39    -3.9500e-01  6.5069e-02  -6.0705 1.280e-09 ***
## factor(STATE)40    -7.7317e-02  1.0287e-01  -0.7516 0.4522799
## factor(STATE)41    -4.8958e-01  7.4960e-02  -6.5313 6.556e-11 ***
## factor(STATE)42    -4.1699e-01  6.6304e-02  -6.2891 3.208e-10 ***
## factor(STATE)45    -2.4453e-01  8.4894e-02  -2.8804 0.0039723 **
## factor(STATE)46    -7.2878e-01  1.2235e-01  -5.9567 2.583e-09 ***
## factor(STATE)47     1.5195e-01  9.3436e-02   1.6263 0.1038928
## factor(STATE)48    -5.3835e-01  6.1206e-02  -8.7957 < 2.2e-16 ***
## factor(STATE)49    -5.2785e-01  6.8497e-02  -7.7062 1.310e-14 ***
## factor(STATE)51    -2.4795e-01  7.5665e-02  -3.2770 0.0010496 **
## factor(STATE)53    -4.4759e-01  7.2964e-02  -6.1344 8.582e-10 ***
## factor(STATE)54    -1.1856e-01  1.2148e-01  -0.9760 0.3290840
## factor(STATE)55    -2.2635e-01  7.0379e-02  -3.2161 0.0012998 **
## HISP_REF            2.7855e-01  2.1780e-02  12.7895 < 2.2e-16 ***
## factor(HIGH_EDU)10  7.5821e-03  1.4066e-01   0.0539 0.9570113
## factor(HIGH_EDU)11  1.4558e-01  1.3558e-01   1.0738 0.2829339
## factor(HIGH_EDU)12  1.9379e-01  1.3313e-01   1.4556 0.1454948
## factor(HIGH_EDU)13  3.3140e-01  1.3336e-01   2.4851 0.0129547 *
## factor(HIGH_EDU)14  4.7666e-01  1.3436e-01   3.5477 0.0003888 ***
## factor(HIGH_EDU)15  4.6548e-01  1.3331e-01   3.4916 0.0004804 ***
## factor(HIGH_EDU)16  7.0792e-01  1.3391e-01   5.2865 1.250e-07 ***
## factor(QSQ118)1     5.1246e-02  5.0251e-02   1.0198 0.3078247
## factor(QSQ118)2     3.3355e-03  4.9341e-02   0.0676 0.9461037
## factor(QSQ118)3    -1.3885e-02  4.8230e-02  -0.2879 0.7734299
## factor(QSQ118)4    -9.5910e-04  4.8533e-02  -0.0198 0.9842334
## factor(QSQ118)5    -3.9503e-03  4.8848e-02  -0.0809 0.9355453
## factor(QSQ118)6    -4.2715e-02  4.8898e-02  -0.8735 0.3823728
## factor(QSQ118)7     2.3571e-02  4.9398e-02   0.4772 0.6332426
## factor(QSQ118)8     6.0689e-02  5.0590e-02   1.1996 0.2302869
## factor(QSQ118)9    -8.0681e-02  4.9037e-02  -1.6453 0.0999117 .
## factor(QSQ118)10   -4.8588e-02  4.9094e-02  -0.9897 0.3223287
## factor(QSQ118)11   -7.9686e-02  4.8490e-02  -1.6433 0.1003156
```

```
## factor(QSQ118)12   -1.3728e-01  4.7999e-02  -2.8601 0.0042354 **
## factor(QSQ118)13   -1.0138e-01  4.8428e-02  -2.0934 0.0363166 *
## factor(QSQ118)14   -8.5492e-02  4.9119e-02  -1.7405 0.0817758 .
## factor(QSQ118)15   -1.3531e-01  4.8279e-02  -2.8027 0.0050694 **
## factor(QSQ118)16   -1.2953e-01  4.8408e-02  -2.6758 0.0074567 **
## factor(QSQ118)17   -3.6880e-02  5.1206e-02  -0.7202 0.4713933
## factor(QSQ118)18   -1.7209e-02  5.0562e-02  -0.3404 0.7335882
## factor(QSQ118)19    2.4393e-03  4.9979e-02   0.0488 0.9610728
## factor(QSQ118)20    8.4108e-02  5.0898e-02   1.6525 0.0984398 .
## factor(QSQ118)21    3.2819e-02  5.0448e-02   0.6505 0.5153410
## factor(QSQ118)22    2.3095e-02  5.0816e-02   0.4545 0.6494772
## factor(QSQ118)23    1.1073e-02  5.0480e-02   0.2194 0.8263755
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

coeftest(perinsmodel2, vcov. = vcovHC(perinsmodel2, type = "HC3"))

##
## t test of coefficients:
##
##                       Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)         2.97295757  0.22914949  12.9739 < 2.2e-16 ***
## HOUSB               0.26584466  0.11327753   2.3468 0.0189353 *
## I(HOUSB^2)         -1.34434756  0.14788233  -9.0907 < 2.2e-16 ***
## AGE_REF            -0.02099876  0.00055300 -37.9722 < 2.2e-16 ***
## factor(CUTENURE)2  -0.39013397  0.01969835 -19.8054 < 2.2e-16 ***
## factor(CUTENURE)4  -0.49929579  0.01601252 -31.1816 < 2.2e-16 ***
## factor(CUTENURE)5  -0.94768140  0.06176887 -15.3424 < 2.2e-16 ***
## factor(CUTENURE)6  -2.24194236  0.15000018 -14.9463 < 2.2e-16 ***
## FAM_SIZE            0.19628051  0.00495721  39.5950 < 2.2e-16 ***
## TINCHRS             0.04846399  0.00031896 151.9457 < 2.2e-16 ***
## factor(MARITAL1)2   0.11156163  0.03391822   3.2891 0.0010053 **
## factor(MARITAL1)3   0.71214251  0.02275600  31.2947 < 2.2e-16 ***
## factor(MARITAL1)4   0.82392537  0.04478814  18.3961 < 2.2e-16 ***
## factor(MARITAL1)5   0.78998967  0.02120602  37.2531 < 2.2e-16 ***
## factor(REF_RACE)2   0.08991173  0.02237555   4.0183 5.867e-05 ***
## factor(REF_RACE)3  -0.13066550  0.08598362  -1.5197 0.1286010
## factor(REF_RACE)4   0.15609316  0.02468648   6.3230 2.577e-10 ***
## factor(REF_RACE)5   0.22349777  0.07909208   2.8258 0.0047174 **
## factor(REF_RACE)6   0.09036609  0.04689324   1.9271 0.0539753 .
## SEX_REF            -0.09184398  0.01298159  -7.0749 1.506e-12 ***
## factor(STATE)2      0.26936513  0.06341664   4.2475 2.163e-05 ***
## factor(STATE)4     -0.12877131  0.06182078  -2.0830 0.0372560 *
## factor(STATE)6      0.23862146  0.04778269   4.9939 5.928e-07 ***
## factor(STATE)8      0.29206462  0.06112397   4.7782 1.771e-06 ***
## factor(STATE)9      0.18268163  0.06016361   3.0364 0.0023948 **
## factor(STATE)10     0.58773205  0.15667123   3.7514 0.0001760 ***
## factor(STATE)11     0.33304027  0.13673652   2.4356 0.0148676 *
## factor(STATE)12    -0.12493314  0.05129485  -2.4356 0.0148695 *
## factor(STATE)13     0.11967352  0.06221492   1.9236 0.0544141 .
```

```
## factor(STATE)15     0.27001042  0.06664767   4.0513 5.097e-05 ***
## factor(STATE)17     0.10500710  0.05586752   1.8796 0.0601694 .
## factor(STATE)18    -0.01895305  0.10204624  -0.1857 0.8526569
## factor(STATE)19     0.04535100  0.17978949   0.2522 0.8008523
## factor(STATE)20    -0.02986769  0.07286690  -0.4099 0.6818848
## factor(STATE)21     0.04245348  0.07275115   0.5835 0.5595287
## factor(STATE)22    -0.20527552  0.06644046  -3.0896 0.0020048 **
## factor(STATE)24     0.39612728  0.06164989   6.4254 1.321e-10 ***
## factor(STATE)25     0.14299533  0.05615152   2.5466 0.0108795 *
## factor(STATE)26     0.00511298  0.05800264   0.0882 0.9297570
## factor(STATE)27     0.12557537  0.06225537   2.0171 0.0436879 *
## factor(STATE)28    -0.18238182  0.10602228  -1.7202 0.0853955 .
## factor(STATE)29     0.13038193  0.06703852   1.9449 0.0517924 .
## factor(STATE)31     0.01064913  0.05897188   0.1806 0.8566978
## factor(STATE)32     0.12629066  0.07434472   1.6987 0.0893759 .
## factor(STATE)33     0.36118874  0.12800903   2.8216 0.0047797 **
## factor(STATE)34     0.23663783  0.06203058   3.8149 0.0001364 ***
## factor(STATE)36     0.15197095  0.05127030   2.9641 0.0030363 **
## factor(STATE)37    -0.00897429  0.05203883  -0.1725 0.8630812
## factor(STATE)39     0.13519723  0.05347291   2.5283 0.0114623 *
## factor(STATE)40     0.26214416  0.08079341   3.2446 0.0011765 **
## factor(STATE)41     0.16299865  0.06141597   2.6540 0.0079555 **
## factor(STATE)42     0.11586563  0.05520893   2.0987 0.0358482 *
## factor(STATE)45    -0.10610375  0.07115536  -1.4912 0.1359239
## factor(STATE)46    -0.27816729  0.09724541  -2.8605 0.0042311 **
## factor(STATE)47     0.13603263  0.08107739   1.6778 0.0933872 .
## factor(STATE)48     0.09797422  0.04997544   1.9604 0.0499465 *
## factor(STATE)49     0.02797768  0.05670130   0.4934 0.6217155
## factor(STATE)51     0.26622978  0.05968169   4.4608 8.174e-06 ***
## factor(STATE)53     0.36055108  0.05932570   6.0775 1.226e-09 ***
## factor(STATE)54     0.09134053  0.10394329   0.8788 0.3795373
## factor(STATE)55     0.10646496  0.05610699   1.8975 0.0577605 .
## HISP_REF           -0.16315115  0.01856925  -8.7861 < 2.2e-16 ***
## factor(HIGH_EDU)10 -0.01531221  0.22758558  -0.0673 0.9463581
## factor(HIGH_EDU)11  0.38771813  0.22105321   1.7540 0.0794410 .
## factor(HIGH_EDU)12  0.84759493  0.21881000   3.8737 0.0001073 ***
## factor(HIGH_EDU)13  1.10331612  0.21884230   5.0416 4.625e-07 ***
## factor(HIGH_EDU)14  1.37412541  0.21926270   6.2670 3.696e-10 ***
## factor(HIGH_EDU)15  1.58405743  0.21882631   7.2389 4.559e-13 ***
## factor(HIGH_EDU)16  1.73123460  0.21906595   7.9028 2.757e-15 ***
## factor(QSQ118)1     0.26180623  0.05829923   4.4907 7.106e-06 ***
## factor(QSQ118)2     0.01404184  0.04035269   0.3480 0.7278577
## factor(QSQ118)3     0.00099408  0.03930139   0.0253 0.9798207
## factor(QSQ118)4    -0.00442422  0.03943455  -0.1122 0.9106718
## factor(QSQ118)5     0.01330017  0.03988631   0.3335 0.7387940
## factor(QSQ118)6     0.02878346  0.03971030   0.7248 0.4685544
## factor(QSQ118)7     0.05182086  0.03955223   1.3102 0.1901355
## factor(QSQ118)8     0.05331881  0.04070856   1.3098 0.1902773
## factor(QSQ118)9    -0.02461166  0.04064974  -0.6055 0.5448770
## factor(QSQ118)10    0.02898736  0.03968536   0.7304 0.4651295
```

```
## factor(QSQ118)11    0.02903451  0.03986736   0.7283 0.4664455
## factor(QSQ118)12    0.01763747  0.04043316   0.4362 0.6626832
## factor(QSQ118)13    0.02831056  0.04058054   0.6976 0.4854048
## factor(QSQ118)14    0.02862411  0.04108056   0.6968 0.4859423
## factor(QSQ118)15    0.03589746  0.04033060   0.8901 0.3734253
## factor(QSQ118)16    0.07532016  0.04060009   1.8552 0.0635749 .
## factor(QSQ118)17    0.08245785  0.04218403   1.9547 0.0506195 .
## factor(QSQ118)18    0.13097836  0.04204191   3.1154 0.0018374 **
## factor(QSQ118)19    0.19193955  0.04132835   4.6443 3.418e-06 ***
## factor(QSQ118)20    0.23998156  0.04166573   5.7597 8.454e-09 ***
## factor(QSQ118)21    0.17745999  0.04130509   4.2963 1.738e-05 ***
## factor(QSQ118)22    0.17049354  0.04200355   4.0590 4.932e-05 ***
## factor(QSQ118)23    0.22233627  0.04162117   5.3419 9.219e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(hlthinsmodel2, vcov. = vcovHC(hlthinsmodel2, type = "HC3"))
```

```
##
## t test of coefficients:
##
##                      Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)       -4.3889e-01  2.6278e-01  -1.6702 0.0948806 .
## HOUSB              1.4208e-01  1.3583e-01   1.0460 0.2955785
## I(HOUSB^2)        -1.0193e+00  1.7102e-01  -5.9601 2.529e-09 ***
## AGE_REF            5.2137e-02  6.9356e-04  75.1733 < 2.2e-16 ***
## factor(CUTENURE)2 -2.4319e-01  2.0436e-02 -11.9004 < 2.2e-16 ***
## factor(CUTENURE)4 -6.1947e-01  2.2055e-02 -28.0876 < 2.2e-16 ***
## factor(CUTENURE)5 -1.1853e+00  8.2498e-02 -14.3677 < 2.2e-16 ***
## factor(CUTENURE)6 -2.0458e+00  1.5157e-01 -13.4971 < 2.2e-16 ***
## FAM_SIZE          -1.8340e-02  7.7327e-03  -2.3717 0.0177092 *
## TINCHRS            6.7664e-05  3.5170e-04   0.1924 0.8474356
## factor(MARITAL1)2 -2.1801e-01  2.5291e-02  -8.6202 < 2.2e-16 ***
## factor(MARITAL1)3 -4.5429e-01  2.4989e-02 -18.1792 < 2.2e-16 ***
## factor(MARITAL1)4 -7.7396e-01  6.3893e-02 -12.1134 < 2.2e-16 ***
## factor(MARITAL1)5 -4.2314e-01  2.7897e-02 -15.1678 < 2.2e-16 ***
## factor(REF_RACE)2 -1.0566e-01  2.7411e-02  -3.8545 0.0001160 ***
## factor(REF_RACE)3 -2.0131e-01  1.2343e-01  -1.6309 0.1029076
## factor(REF_RACE)4  1.5001e-02  3.6707e-02   0.4087 0.6827906
## factor(REF_RACE)5  2.0544e-01  1.1703e-01   1.7555 0.0791805 .
## factor(REF_RACE)6  1.0800e-02  6.6545e-02   0.1623 0.8710689
## SEX_REF            5.2547e-02  1.5927e-02   3.2992 0.0009700 ***
## log(FOODPQ + 1)    3.0880e-01  1.0742e-02  28.7467 < 2.2e-16 ***
## log(ALCBEVPQ + 1)  2.7116e-02  3.5334e-03   7.6742 1.680e-14 ***
## log(MEDSRVPQ + 1)  9.9048e-02  3.0250e-03  32.7431 < 2.2e-16 ***
## log(PREDRGPQ + 1) -3.2744e-02  2.4274e-01  -0.1349 0.8926962
## log(MEDSUPPQ + 1)  3.7043e-02  5.2169e-03   7.1005 1.252e-12 ***
## log(TOBACCPQ + 1) -2.4273e-02  4.3713e-03  -5.5529 2.818e-08 ***
## factor(STATE)2    -1.9997e-01  7.6897e-02  -2.6005 0.0093090 **
## factor(STATE)4    -2.7007e-01  7.1095e-02  -3.7987 0.0001456 ***
```

```
## factor(STATE)6     -4.3389e-01  5.4503e-02  -7.9608 1.729e-15 ***
## factor(STATE)8     -9.0885e-02  7.5783e-02  -1.1993 0.2304225
## factor(STATE)9      1.1455e-01  6.6666e-02   1.7182 0.0857621 .
## factor(STATE)10    -4.7730e-01  2.2348e-01  -2.1358 0.0326967 *
## factor(STATE)11    -1.3464e-01  1.5005e-01  -0.8973 0.3695617
## factor(STATE)12    -7.9733e-03  5.6575e-02  -0.1409 0.8879237
## factor(STATE)13     2.2140e-01  7.1228e-02   3.1084 0.0018819 **
## factor(STATE)15    -6.2711e-01  8.0413e-02  -7.7987 6.322e-15 ***
## factor(STATE)17     1.1331e-01  6.4841e-02   1.7475 0.0805563 .
## factor(STATE)18    -2.5982e-01  1.2057e-01  -2.1549 0.0311711 *
## factor(STATE)19    -2.3719e-01  1.9552e-01  -1.2131 0.2250900
## factor(STATE)20    -9.2903e-02  8.5083e-02  -1.0919 0.2748721
## factor(STATE)21    -1.6109e-01  8.1100e-02  -1.9863 0.0470052 *
## factor(STATE)22    -3.1873e-01  8.3720e-02  -3.8071 0.0001407 ***
## factor(STATE)24     2.0256e-01  6.7375e-02   3.0065 0.0026437 **
## factor(STATE)25     1.0494e-01  6.2991e-02   1.6659 0.0957281 .
## factor(STATE)26    -3.2372e-01  6.6741e-02  -4.8503 1.235e-06 ***
## factor(STATE)27     1.3353e-01  7.0488e-02   1.8943 0.0581817 .
## factor(STATE)28     3.6850e-02  9.7188e-02   0.3792 0.7045696
## factor(STATE)29     1.3278e-01  7.5172e-02   1.7663 0.0773469 .
## factor(STATE)31     1.2681e-01  6.6854e-02   1.8968 0.0578640 .
## factor(STATE)32    -3.5304e-01  8.9732e-02  -3.9344 8.346e-05 ***
## factor(STATE)33     3.9095e-01  1.8090e-01   2.1611 0.0306889 *
## factor(STATE)34    -2.1496e-01  7.3626e-02  -2.9197 0.0035050 **
## factor(STATE)36    -5.2286e-01  5.8899e-02  -8.8773 < 2.2e-16 ***
## factor(STATE)37    -7.8664e-02  5.8414e-02  -1.3467 0.1780922
## factor(STATE)39    -1.4136e-02  5.8735e-02  -0.2407 0.8098044
## factor(STATE)40    -1.0609e-01  1.0521e-01  -1.0084 0.3132794
## factor(STATE)41    -3.4964e-01  7.5205e-02  -4.6491 3.338e-06 ***
## factor(STATE)42    -4.1572e-02  6.0561e-02  -0.6864 0.4924315
## factor(STATE)45     3.7114e-02  7.6872e-02   0.4828 0.6292349
## factor(STATE)46     1.5173e-01  1.1602e-01   1.3079 0.1909191
## factor(STATE)47    -5.1654e-02  8.3878e-02  -0.6158 0.5380176
## factor(STATE)48    -3.3749e-02  5.8052e-02  -0.5814 0.5609948
## factor(STATE)49    -3.0922e-01  6.4300e-02  -4.8091 1.519e-06 ***
## factor(STATE)51     1.0268e-01  6.8116e-02   1.5075 0.1316936
## factor(STATE)53    -5.7275e-01  7.2653e-02  -7.8833 3.224e-15 ***
## factor(STATE)54    -1.9248e-01  1.2239e-01  -1.5726 0.1158046
## factor(STATE)55     4.1018e-01  6.1552e-02   6.6639 2.681e-11 ***
## HISP_REF            6.5893e-01  2.8302e-02  23.2821 < 2.2e-16 ***
## factor(HIGH_EDU)10 -1.1812e-01  2.5070e-01  -0.4712 0.6375151
## factor(HIGH_EDU)11 -6.0230e-01  2.4420e-01  -2.4664 0.0136481 *
## factor(HIGH_EDU)12 -8.5629e-02  2.4140e-01  -0.3547 0.7228055
## factor(HIGH_EDU)13  1.1815e-01  2.4145e-01   0.4893 0.6246081
## factor(HIGH_EDU)14  3.9115e-01  2.4201e-01   1.6163 0.1060374
## factor(HIGH_EDU)15  5.8632e-01  2.4137e-01   2.4291 0.0151390 *
## factor(HIGH_EDU)16  7.1356e-01  2.4152e-01   2.9545 0.0031329 **
## factor(QSQ118)1    -6.1896e-03  5.6354e-02  -0.1098 0.9125418
## factor(QSQ118)2    -1.0743e-02  5.1029e-02  -0.2105 0.8332480
## factor(QSQ118)3    -1.6176e-04  4.9773e-02  -0.0033 0.9974069
```

```
## factor(QSQ118)4     5.2287e-02  5.0084e-02   1.0440 0.2964978
## factor(QSQ118)5     3.4465e-02  5.0244e-02   0.6860 0.4927447
## factor(QSQ118)6     5.6300e-02  5.0377e-02   1.1176 0.2637498
## factor(QSQ118)7     8.4400e-02  4.9985e-02   1.6885 0.0913193 .
## factor(QSQ118)8     5.9971e-02  5.0410e-02   1.1897 0.2341833
## factor(QSQ118)9     6.3730e-02  5.0485e-02   1.2624 0.2068205
## factor(QSQ118)10    7.1697e-02  5.0667e-02   1.4151 0.1570517
## factor(QSQ118)11    8.2922e-02  5.0645e-02   1.6373 0.1015646
## factor(QSQ118)12    3.4204e-02  5.1014e-02   0.6705 0.5025469
## factor(QSQ118)13    6.1895e-03  5.0793e-02   0.1219 0.9030124
## factor(QSQ118)14   -4.1132e-02  5.1293e-02  -0.8019 0.4226097
## factor(QSQ118)15   -3.4226e-03  5.0461e-02  -0.0678 0.9459246
## factor(QSQ118)16    2.6329e-02  5.0447e-02   0.5219 0.6017297
## factor(QSQ118)17    1.3208e-02  5.1882e-02   0.2546 0.7990528
## factor(QSQ118)18    7.6072e-02  5.1345e-02   1.4816 0.1384494
## factor(QSQ118)19    7.6844e-02  5.1198e-02   1.5009 0.1333762
## factor(QSQ118)20    4.1916e-02  5.1275e-02   0.8175 0.4136610
## factor(QSQ118)21    6.0636e-03  5.1579e-02   0.1176 0.9064168
## factor(QSQ118)22    4.1785e-03  5.2033e-02   0.0803 0.9359950
## factor(QSQ118)23    1.0089e-01  5.2062e-02   1.9378 0.0526494 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Breusch-Pagan test results
bp_results <- list(Vehicle = bptest(vehinsmodel), Life = bptest(lifinsmodel),
 Personal = bptest(perinsmodel), Health = bptest(hlthinsmodel))
bp_results
```

```
## $Vehicle
##
##  studentized Breusch-Pagan test
##
## data:  vehinsmodel
## BP = 4996.7, df = 97, p-value < 2.2e-16
##
##
## $Life
##
##  studentized Breusch-Pagan test
##
## data:  lifinsmodel
## BP = 7583.8, df = 70, p-value < 2.2e-16
##
##
## $Personal
##
##  studentized Breusch-Pagan test
##
## data:  perinsmodel
## BP = 19899, df = 68, p-value < 2.2e-16
```

```
##
##
## $Health
##
##  studentized Breusch-Pagan test
##
## data:  hlthinsmodel
## BP = 8989.8, df = 74, p-value < 2.2e-16
```

```
bp_results1 <- list(Vehicle = bptest(vehinsmodel1), Life = bptest(lifinsmodel
1), Personal = bptest(perinsmodel1), Health = bptest(hlthinsmodel1))
bp_results1
```

```
## $Vehicle
##
##  studentized Breusch-Pagan test
##
## data:  vehinsmodel1
## BP = 4996.7, df = 97, p-value < 2.2e-16
##
##
## $Life
##
##  studentized Breusch-Pagan test
##
## data:  lifinsmodel1
## BP = 7684.7, df = 92, p-value < 2.2e-16
##
##
## $Personal
##
##  studentized Breusch-Pagan test
##
## data:  perinsmodel1
## BP = 24487, df = 90, p-value < 2.2e-16
##
##
## $Health
##
##  studentized Breusch-Pagan test
##
## data:  hlthinsmodel1
## BP = 9423.4, df = 96, p-value < 2.2e-16
```

```
bp_results2 <- list(Vehicle = bptest(vehinsmodel2), Life = bptest(lifinsmodel
2), Personal = bptest(perinsmodel2), Health = bptest(hlthinsmodel2))
bp_results2
```

```
## $Vehicle
##
##  studentized Breusch-Pagan test
```

```
##
## data:  vehinsmodel2
## BP = 6124.2, df = 98, p-value < 2.2e-16
##
##
## $Life
##
##  studentized Breusch-Pagan test
##
## data:  lifinsmodel2
## BP = 7721.1, df = 93, p-value < 2.2e-16
##
##
## $Personal
##
##  studentized Breusch-Pagan test
##
## data:  perinsmodel2
## BP = 24520, df = 91, p-value < 2.2e-16
##
##
## $Health
##
##  studentized Breusch-Pagan test
##
## data:  hlthinsmodel2
## BP = 9240.5, df = 97, p-value < 2.2e-16
```